

Fully Anonymous Secret Sharing

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Abstract

In a secret sharing scheme for a monotone access structure $\mathcal{A} : \{0, 1\}^n \rightarrow \{0, 1\}$, a dealer can share a secret s to n parties such that any authorized subset of parties $A \in \mathcal{A}$ can recover s while all other subsets learn nothing about s . In this work, we study *fully anonymous secret sharing* (FASS), which strengthens standard secret sharing by requiring the following properties:

- **Share Anonymity.** The shares belonging to any unauthorized set of parties not only hide the secret, but also all identifiable information such as party identities and whether or not the shares were generated together. In particular, it suffices that such shares be uniform and independent.
- **Anonymous Reconstruction.** The reconstruction algorithm does not need to know the reconstructing set of parties.

Efficient FASS exists for threshold access structures. For general access structures, the only known construction relies on a monotone DNF representation of $\mathcal{A}$ and has per-party share size $\Omega(\ell n)$ where ℓ is the number of minterms of $\mathcal{A}$. This leaves an exponential gap between standard secret sharing and FASS even for simple access structures. Moreover, even in the threshold case, known schemes could not achieve optimal robust reconstruction when mixing shares of multiple secrets.

Motivated by a recent work of Eldridge et al. [USENIX'24], who demonstrated an application of FASS to stalker detection, we initiate a systematic study of FASS, obtaining the following main results.

- **Near-Optimal Information-Theoretic FASS.** We obtain strong lower bounds, showing that the dependence on the DNF size is generally inherent. In particular, the share size can be exponential in the number of parties or even in the minimum CNF size. This stands in sharp contrast to standard secret sharing, where no super-polynomial lower bounds are known, and where the share size is upper bounded by the CNF size. For DNF with ℓ small minterms, we improve the previous upper bound to $\tilde{O}(\ell)$, matching our lower bound up to a polylogarithmic factor.
- **Computational FASS.** We show that the above negative results can be circumvented in the computational setting, obtaining FASS schemes with succinct shares. Under the learning with errors (LWE) assumption, we present a general compiler from standard secret sharing to FASS that preserves the share size of the underlying scheme. For natural graph access structures, we directly construct succinct FASS from either one-way functions or bilinear maps.
- **Robust FASS.** We show that simple modifications of our computational FASS schemes can allow for robust reconstruction of a polynomially unbounded number of secrets from any mixture of their authorized shares.

1 Introduction

In a secret sharing scheme [Sha79, Bla79, ISN87] for a monotone access structure $\mathcal{A} : \{0, 1\}^n \rightarrow \{0, 1\}$, a dealer can share a secret s to n parties such that the following two properties are satisfied:

- *Correctness*: Any subset of parties that satisfy $\mathcal{A}$ can recover s from their shares.
- *Secrecy*: Any subset of parties that are unauthorized w.r.t. $\mathcal{A}$ learn nothing about s .

In addition to being a fundamental tool for data privacy, secret sharing schemes are a key ingredient in the design of advanced cryptographic primitives such as secure multiparty computation [Yao86, GMW87, BGW88, CCD88], threshold cryptosystems [Des88, DF90] and attribute-based encryption [SW05, GPSW06].

Fully Anonymous Secret Sharing (FASS). In this work, we study secret sharing schemes that satisfy a significantly stronger notion of security than secrecy. We say that a secret sharing scheme is *fully anonymous* if it satisfies the following two properties:

- **Share Anonymity**: The shares belonging to any unauthorized set of parties hide identifiable information such as party identities and whether or not shares were generated together.
- **Anonymous Reconstruction**: The reconstruction algorithm does not require the identities of the reconstructing set of parties.

It is not difficult to see that the standard notion of secrecy does *not* imply anonymity. Indeed, a secret sharing scheme where each share contains a party identity and all shares contain a common identifier can still achieve secrecy; however, it does not achieve share anonymity. Similarly, a secret sharing scheme where shares are of distinct sizes (and the sizes are correlated with party identities) can achieve secrecy but not share anonymity.

Motivation and Prior Work. A preliminary study of FASS schemes began with the works of [GMO03, PO20] and was continued in the recent work of Eldridge et al. [EBG⁺24].¹ In particular, [EBG⁺24] demonstrate a consequential real-world application of FASS: addressing the threat of stalker detection in the ecosystem of bluetooth tracking tags such as Apple AirTags [Nay22].

Inspired by their work, we initiate a systematic study of FASS, providing answers to some of the main open questions regarding the feasibility and efficiency of FASS.

1. **Share size**: Eldridge et al. constructed efficient FASS for *threshold* access structures via a simple modification of Shamir’s scheme, but left open the problem of constructing such schemes for richer access structures – a task that would directly benefit their application (see Section 1.1 for a discussion). Indeed, the only known construction for general access structures—due to the work of [PO20]—relies on a monotone DNF representation² of the access structure and has a per-party share size $\Omega(\ell n)$ where ℓ is the number of minterms and n is the number of parties. This means that even for simple access structures represented by small Boolean formulas, the share size can be *exponentially bigger* than the share size of the best non-anonymous schemes. Therefore, we ask:

Is the exponential gap between anonymous and non-anonymous share size inherent?

¹Their work refers to the share anonymity property as *unlinkability*.

²A monotone DNF is a disjunction of minterms, where each minterm is a conjunction of input variables. For example, $(x_1 \wedge x_2) \vee x_3$ has 2 minterms. Every monotone $\mathcal{A}$ can be represented by a monotone DNF in which the minterms correspond to minimal authorized sets.

2. **Robust reconstruction:** The work of [EBG⁺24] also investigated *robust* FASS schemes where the reconstruction algorithm is required to reconstruct *multiple* secrets given any mixture of their authorized shares. This “multi-dealer correctness” feature is crucial to their application where multiple secrets are simultaneously and independently shared by different dealers, and the reconstructing entity receives an arbitrary mixture of these shares. Note that while reconstruction of multiple secrets can be easily achieved in standard secret sharing by simply “tagging” shares of each secret, this is *not* the case for FASS as such tags violate share anonymity. Using list-decoding, [EBG⁺24] constructed ramp FASS schemes for threshold access structures that support robust reconstruction of a *constant* number of secrets.³ Currently, this is the only known result for robust FASS. We ask:

Do there exist FASS schemes that support robust reconstruction of an unbounded number of secrets?

Before we describe our results addressing these questions, we explain the conceptual challenges in constructing FASS, and briefly touch on connections between FASS and other notions in cryptography.

On the Difficulty of Achieving Anonymity. Achieving share anonymity in secret sharing turns out to be quite challenging. In fact, *most standard techniques for constructing secret sharing schemes completely break down when we require share anonymity.*

Let us start by describing an important exception: threshold secret sharing. In this case, one can achieve full anonymity by attaching to each share a random “identity” [GMO03, EBG⁺24]. Applying this to the polynomial-based Shamir’s scheme, the share of party i is of the form $(\alpha_i, p(\alpha_i))$ for a random interpolation point α_i . However, trying to go beyond the threshold case gives rise to complications.

For example, a ubiquitous technique for building secret sharing schemes is *composition*: if an access structure (say) $\mathcal{A}$ can be decomposed as $\mathcal{A}_0 \wedge \mathcal{A}_1$, then we can share s w.r.t. $\mathcal{A}$ as follows. First, we additively share s into s_0 and s_1 and then share s_0 (resp., s_1) w.r.t. $\mathcal{A}_0$ (resp., $\mathcal{A}_1$). This approach, however, does not achieve share anonymity *even if we use FASS* for sharing s_0 and s_1 . The reason is that if an unauthorized set U is super-authorized with respect to $\mathcal{A}_0$, this is enough to violate anonymity. Indeed, a distinguisher can check if the same value of s_0 can be reconstructed by two distinct subsets of U .

This example shows a fundamental challenge in constructing FASS. Note that share anonymity is inherently a “bounded” security notion in that, in general, we cannot expect to achieve it for an authorized set. Since an unauthorized set for an access structure $\mathcal{A}$ may be authorized for a restricted access structure $\mathcal{A}'$ of $\mathcal{A}$, as in the case of the above decomposition, a FASS scheme must ensure that any attempts to partial reconstruction w.r.t. a “weaker” access structure such as $\mathcal{A}'$ do not yield any identifiable information. See further discussion in Section 2.

Connections Within Cryptography. Fully anonymous secret sharing can be viewed as a secret sharing analogue of *covert computation* [vHL05, CGOS07]. Informally speaking, the notion of covert computation requires that a set of participants can jointly compute a function over their inputs such that no one (including the parties) learns about the existence of the computation unless all parties are participating in the protocol (and the outcome is “favorable” to everyone). All prior work on covert computation only considers two-party computation and dishonest majority secure multiparty computation. Such schemes imply additive (n -out-of- n) secret sharing, which is already fully anonymous.

Secret sharing schemes for partite graph access structures are related to the well-studied notion of *conditional disclosure of secrets* (CDS) [GIKM98], which has found applications in different areas of cryptography (see [AARV17, AV19] and references therein). Leveraging this connection, FASS for such partite structures readily implies a new notion of *anonymous* CDS.

³In a ramp secret sharing scheme, there is a large gap between secrecy and reconstruction thresholds.

1.1 Our Results

We conduct a systematic study of fully anonymous secret sharing. Our results come in two flavors:

- **Near-Optimal Information-Theoretic FASS:** In the information-theoretic setting, we present both positive and negative results. We first present strong lower bounds on share size for FASS schemes, showing the existence of explicit n -party access structures with a monotone CNF representation of size $O(n)$, for which the FASS share size is $2^{\Omega(n)}$. This stands in sharp contrast to standard secret sharing, where the share size is upper bounded by the monotone CNF size (more generally, monotone formula size) of the access structure [BL90], and the strongest known lower bound for *any* access structure is polynomial in n [Csi97]. On the positive side, we construct FASS schemes for every access structure $\mathcal{A}$, where the per-party share size is roughly equal to the size of the monotone DNF representation of $\mathcal{A}$, matching our lower bound for general DNFs up to a polylogarithmic factor. This improves upon the previous construction of [PO20] where the share size has an additional multiplicative dependence on n . The price we pay for the improved share size is inefficient reconstruction. For long secrets, say of length $\Omega(n)$, when the minimal authorized sets are small, we can achieve efficient anonymous reconstruction while preserving near-optimal share size (see Corollary 5.11).
- **Computational FASS:** In the setting of computational security, we circumvent the information-theoretic lower bounds by presenting two kinds of results. First, under the standard learning with errors (LWE) assumption [Reg05], we present a *general compiler* from standard (information-theoretic or computational) secret sharing to FASS with a comparable share size. We also present direct constructions of FASS schemes with succinct shares for specific graph access structures from other standard assumptions, including the Decisional Linear assumptions over bilinear groups or even just one-way functions.
- **Robust FASS:** Our computational FASS schemes can be extended to support efficient robust reconstruction with an arbitrary polynomial number of dealers, also tolerating random insertion noise.

We now elaborate upon our results, starting with a definitional contribution that distinguishes between three useful kinds of FASS.

Notions of FASS. We consider three distinct notions of FASS, where each notion may be either information-theoretic or computational. All three notions require anonymous reconstruction, but vary in the requirements on share anonymity. The strongest and simplest notion is U-FASS, which requires *uniform* anonymity, namely, that any unauthorized set of shares are indistinguishable from the uniform distribution over strings of an appropriate length. Beyond being a natural property to consider, uniformity is qualitatively a useful anonymity property for use cases such as keyless steganography [ACI⁺20] where hiding a share in a stream of (pseudo)random bits allows one plausibly denyability of whether the share value was intentionally hidden or naturally occurring. The weakest notion is S-FASS which requires *single-dealer* anonymity, namely, that the shares of any two unauthorized sets of the same number of parties are indistinguishable. An intermediate notion is M-FASS, which requires *multi-dealer* anonymity. This is a strengthening of single-dealer anonymity, in that it additionally requires that an adversary cannot decide if a set of unauthorized shares were generated together.⁴

It is not difficult to see that U-FASS implies M-FASS, which implies S-FASS (see Section 4 for details). We will consider U-FASS to be our main notion of FASS because it is the strongest. Most of our positive

⁴Our definition of multi-dealer anonymity can be viewed as an extension of the unlinkability property – as defined in [EBG⁺24] for threshold access structures – to general monotone access structures.

results apply to the strongest notion of U-FASS and others to M-FASS, whereas our negative results apply also to M-FASS. We view S-FASS as being a less useful notion, and show a separation between S-FASS and M-FASS share size (see Remark 6.5).

Information-Theoretic FASS: Upper Bounds. We present an improved variant of the U-FASS scheme from [PO20] based on the monotone DNF representation of the access structure. For any monotone DNF formula with ℓ minterms, the per-party share size scales linearly with ℓ . The scheme achieves perfect secrecy, perfect uniform anonymity, and negligible correctness error.

Theorem 1.1 (Informal, see Theorem 5.6). *For any monotone DNF formula Φ with ℓ minterms of maximal size t and statistical correctness parameter λ , there exists a U-FASS scheme realizing Φ that satisfies:*

1. *Perfect U-anonymity (implying also perfect secrecy) and $2^{-\lambda}$ correctness error.*
2. *A per-party share size of $\tilde{O}(\lambda \ell t)$ for sharing a 1-bit secret.*

This should be compared with a previous construction from [PO20] with perfect correctness, in which the per-party share size is $\tilde{O}(\lambda \ell n)$ and where unauthorized shares are $2^{-\lambda}$ -close to uniform. Interestingly, our improved upper bound requires the computational cost of reconstruction to scale exponentially with t . We show how to achieve efficient reconstruction for secrets of length $|s| = \Omega(n/t)$, while maintaining the same (essentially optimal) improved information rate of Theorem 1.1. We briefly remark that the same construction idea achieving Theorem 1.1 can be straightforwardly modified to obtain perfect correctness, perfect secrecy, $2^{-\lambda}$ U-anonymity, and near-optimal share sizes by using a computationally inefficient sharing algorithm.

Information-Theoretic FASS: Lower Bounds. Our main contribution in the information-theoretic setting is a surprisingly strong lower bound, showing that the above upper bound is close to optimal. Specifically, we present an explicit family of DNFs that have a much smaller equivalent CNF form, for which we can prove that the U-FASS share size must grow linearly with the number of minterms.

Theorem 1.2 (Informal, see Corollary 6.19). *For every $m, d \in \mathbb{N}$, there is an n -party access structure $\mathcal{A}_{m,d}$ with $n = md + 1$, represented by a monotone DNF with $\ell = d^m$ minterms of size $t = O(m)$, such that:*

- *Every U-FASS scheme for $\mathcal{A}_{m,d}$ has per-party share size $\Omega(\ell)$.*
- *$\mathcal{A}_{m,d}$ has a monotone CNF representation with $O(m)$ clauses of size $O(d)$.*

This lower bound can also be extended to the M-FASS setting, and additionally holds under imperfect M-anonymity and imperfect secrecy; see Section 6.2.1 for details.

Let us give two useful corollaries of the above lower bound. First, letting $d = 2$ and $m = (n - 1)/2$, the access structure requires U-FASS share size $2^{\Omega(n)}$, while having a small monotone CNF representation with just $O(n)$ clauses. This establishes an exponential separation between standard secret sharing and U-FASS (or M-FASS). Second, for every constant $m \geq 1$, there is an access structure with $O(n^m)$ minterms, for which our upper bound is $\tilde{O}(n^m)$ and the lower bound is $\Omega(n^m)$, showing that our upper bound is nearly tight.

The access structures in the above theorem correspond to *complete* partite graphs. A complete (k, n) partite graph is defined via k disjoint sets, each consisting of n parties. An authorized set consists of at least one party from each of the k sets. We show that any M-FASS scheme for sharing a single bit with perfect share anonymity must incur a per-party share size $\Omega(n^k)$. This stands in sharp contrast to standard secret sharing where it is easy to obtain a solution with *constant-size* shares.⁵

⁵To share a secret bit s , we additively share it into k parts $s_1, \dots, s_k$ and distribute s_i to each party in the i -th set. Correctness and secrecy follow from inspection.

Computational FASS: Generic Compiler. In light of the above lower bounds, we turn to computational assumptions to build FASS with smaller share size. Our first result is a generic compiler from any secret sharing scheme with information-theoretic or computational secrecy to M-FASS, assuming the existence of compute-and-compare obfuscation [WZ17, GW17] and injective pseudorandom generators (PRGs) with polynomial stretch. Our compiler preserves the share size of the underlying scheme (up to a polynomial overhead in the security parameter).

Theorem 1.3 (Informal, see Theorem 7.16.). *For any access structure $\mathcal{A} = \{\mathcal{A}_\lambda\}$ involving $n = n(\lambda)$ parties, if there exists a (information-theoretically or computationally secure) secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ with per-party share size bounded by S bits, then assuming the existence of the following:*

1. *Multi-bit compute-and-compare obfuscation for any distribution $\mathcal{P}_{\text{MBCC}}^{\text{TM}}$ that satisfies α -pseudo-entropy for any polynomial $\alpha = \alpha(\lambda)$.*
2. *Any injective PRG in NC^1 with polynomial stretch $m = m(\lambda) \geq \alpha(\lambda)$.*

there exists a computationally secure M-FASS scheme for $\mathcal{A}$ with per-party share size $S \cdot \text{poly}(\lambda, \log T)$ bits, where T is the maximum runtime of Recon.

Compute-and-compare obfuscation can be instantiated under the learning with errors (LWE) assumption [Reg05], while injective PRGs can be instantiated from several standard assumptions. Altogether, the above theorem can be instantiated under the LWE assumption.

Computational FASS: Graph Access Structures. We next present FASS schemes for specific graph access structures using computational assumptions other than LWE. Specifically, we consider: (i) the star graph access structure, where a minimal authorized set consists of the star node and any leaf node; and (ii) the complete bipartite graph access structure, where the nodes are divided into “left” and “right” sets and a minimal authorized set consists of one node from each of the two sets. For both of these access structures, our lower bound implies that information-theoretic M-FASS requires per-party share size that grows linearly with the number of parties. We show that it is possible to achieve per-party share size *independent* of the number of parties in both cases.

We first consider the complete bipartite graph access structure. Under standard assumptions on bilinear maps over prime-order groups, we construct an U-FASS scheme for complete bipartite graphs with per-party share size independent of the number of parties.

Theorem 1.4 (Informal, see Theorem 7.4.). *Let λ be a security parameter. Assuming the hardness of the Decisional Linear Assumption, there exists a computational U-FASS scheme for 1-bit secrets for the complete bipartite graph access structure over $n = n(\lambda)$ parties with per-party share size $O(\lambda)$.*

Our construction implicitly builds and uses a form of “anonymous” secret-key encryption scheme where any ciphertext can be decrypted using multiple, distinct but indistinguishable keys corresponding to different users. While such a scheme is easy to construct with parameters growing linearly in the number of users, achieving *sublinear-size* parameters is non-trivial.

Next, we observe that assuming only one-way functions, there is a simple U-FASS scheme for the star-graph access structure with share size independent of the number of parties.

Theorem 1.5 (Informal, see Theorem 7.1.). *Let λ be a security parameter. Assuming the existence of one-way functions, there exists a computational U-FASS scheme for 1-bit secrets for the star graph access structure with $n = n(\lambda)$ leaf nodes where the per-party share size is $O(\lambda)$.*

Robust FASS. Motivated by [EBG⁺24], we define a notion of robust reconstruction that captures the ability to recover *all* secrets given any mixture of their authorized shares in the multiple dealer setting. We show that all of our computational FASS schemes admit efficient, robust reconstruction for any polynomially *unbounded* number of secrets in the presence of polynomially unbounded random noise.

Theorem 1.6 (Informal, see Theorem 8.3, Theorem 8.4, Theorem 8.5). *All computational schemes in this paper have efficient robust reconstruction.*

Previously, robust FASS was only known for threshold access structures in the ramp setting and for only a *constant* number of dealers [EBG⁺24].

Application: Stalker Detection in Tracking Tags Ecosystem. We recall the application of FASS to stalker detection in the tracking tags ecosystem [EBG⁺24]. Tracking tags such as Apple AirTags are bluetooth-enabled devices that emit periodic messages that can be used to physically locate the tags with the help of large-scale volunteer networks consisting of smartphones and other devices. Unfortunately, stalkers have abused this tool, placing hidden tags, which we refer to as “stalker tags”, on the property of unsuspecting individuals.

To protect individuals from stalkers and to obtain legal evidence of abuse, [EBG⁺24] proposed a mechanism that allows a victim to deanonymize the identity of the owner of a stalker tag if the victim is in the proximity of the tag for some time window, which we refer to as the *detection window*. The key challenge in the design of such a system is the need to balance safety with privacy: on the one hand, we want to detect stalker tags, and on the other hand, we want to ensure privacy for honest users of tracking tags. For example, it should not be possible for a passive privacy adversary to deanonymize the owner of a tracking tag who is sitting in a coffee shop for an hour. At first glance, it seems that these two requirements are contradictory since relative to the detection window, there is no difference between a stalker and an honest user. However, as observed in [EBG⁺24], a key difference between stalker tags and honest tags is that stalker tags are in *continuous proximity to the victim for long periods*, while honest tags are likely to move out of proximity of a passive privacy adversary after short periods.

Eldridge et al. crucially leverage this asymmetry to devise a solution using FASS. The main idea is simple: during every detection window, a tracking tag generates a fresh secret sharing of its owner’s identity using *threshold* FASS and emits a new share every few seconds. Given a threshold number of shares, a victim can reconstruct the tag owner’s identity at the end of the detection window. However, due to the anonymity property of FASS, a privacy adversary who views fewer than a threshold number of broadcasts from an honest user tag cannot link the broadcasts together. Furthermore, their scheme also achieves *robustness*, i.e., stalker detection is possible even in the presence of noise which may consist of auxiliary share values emitted by other devices.

The solution of [EBG⁺24] suffers from two shortcomings: First, they achieve robust reconstruction only when there are a small constant number of stalker tags in the vicinity of the victim. Secondly, the threshold access structure only allows for a coarse delineation between what constitutes a stalker tag and a honest tag, i.e. stalkers are those that appear frequently enough in the detection window, and honest tags are those that do not. For example, consider a 12-hour detection window. An individual who works in a coffee shop in the morning, leaves for a few hours, and returns in the evening may be able to deanonymize a barista’s identity from her honest tag by combining the shares received from two separate sessions at the coffee shop. Attempted fixes such as shortening the detection window so that the shares emitted by the honest tag in the morning and evening are from independent, fresh sharings undesirably redefines stalkers as those who spend an entire morning in the coffee shop. Another failed fix is to increase the threshold, negatively impacting stalker detection by requiring a higher number of shares to be recovered among noisy shares emitted by other devices. A more reasonable fix is to use the *tribes* (threshold-of-threshold) access

structure where we divide the detection window into smaller sub-windows and require reconstruction to only succeed if one can observe sufficiently many broadcasts within each sub-window. In this case, tag detection can only succeed if one is in proximity to the tag for the entire detection window. The tribes access structure, however, is not supported by the prior work.

Our LWE-based solution for general access structures addresses both concerns by (1) enabling the detection of an arbitrary number of stalker tags, and (2) achieving FASS for any monotone access structure including the tribes access structure.

Application: Anonymous Conditional Disclosure of Secrets. In a conditional disclosure of secrets (CDS) scheme [GIKM98], n servers holding boolean inputs $x_1, \dots, x_n$, respectively, share a private random string as well as a secret value s . For any public function f , each server i can compute a message and send it to a referee. The main requirement is that the referee learns s if and only if $f(x_1, \dots, x_n) = 1$. CDS can be viewed as a “one-time” version of (secret-key) *multi-user* attribute-based encryption where a decryption key jointly decrypts multiple ciphertexts.

A robust version of n -server CDS [ABNP20] is implied by a secret sharing for partite graphs consisting of n sets, where each set consists of two nodes. Each authorized set contains one node x_i from each set such that $f(x_1, \dots, x_n) = 1$. The security requirement of CDS maps to the secrecy requirement of secret sharing. CDS, however, has no secrecy requirement on the inputs $x_1, \dots, x_n$ of the servers: the server messages may reveal their inputs to the referee. Furthermore, the referee might, in fact, require the knowledge of the inputs in order to compute the function output.

We observe that S-FASS for partite graphs yields a new notion of *anonymous CDS* where the referee does not learn any information about the server inputs $x_1, \dots, x_n$ when $f(x_1, \dots, x_n) = 0$. This notion can be viewed as a one-time version of (secret-key) multi-user *predicate* encryption [GVW15].

1.2 Related Work

Secret Sharing with Anonymous Reconstruction. Stinson and Vanstone were the first to study anonymous reconstruction [SV88] in the context of the threshold access structure and in the model where schemes have perfect correctness, perfect secrecy, and distinct share values. The works of Phillips and Phillips [PP92] studied the feasibility of anonymous reconstruction ideal secret sharing—secret sharing in which the size of each party’s share is equal to the size of the secret—showing that such a scheme is only possible in the case of (t, n) -threshold for $t = 1$ or $t = n$, or the closure of a complete bipartite graph. The work of Blundo and Stinson [BS97] study anonymous reconstruction in the context of the (t, n) -threshold access structure and an infinite class of non-threshold access structures. In both these cases, the strongest lower bound they can show—for certain structures—is that the bit length of the shares must be an $\Omega(\log n)$ additive factor larger than the number of bits needed to represent the secret. This lower bound is slightly improved by [KOKO02] so that it is tight for case of $t = 2$.

Lower Bounds on Share Size. The best known lower bound for per-party share sizes for general (non-anonymous) secret sharing is due to Csirmaz [Csi97], who showed the existence of an access structure for which *any* secret sharing scheme has per-party share size $\Omega(n/\log n)$, where n is the number of parties. Better lower bounds exist for secret sharing schemes whose reconstruction is structured or bounded. One example is linear secret sharing schemes, in which the shares are constructed in a way such that the secret can be reconstructed by computing a linear combination of the shares. A line of work [KW93, BGW99, PR17] shows an exponential lower bound on total share size for secret sharing schemes that follow this restricted construction template. Another example is the work of Larsen and Simkin [LS20], which gives an exponential lower bound for secret sharing schemes with bounded reconstruction—either computationally bounded or with bounded description size. The lower bounds we show in this paper have no such syntactic

constraints, and apply to very simple and explicit access structures, giving an unexpected exponential separation between DNF and CNF structures of the same description size.

In the positive direction, there is a large body of work aimed towards constructing secret sharing schemes with small share size. We refer the reader to the recent work of [ABI⁺23] for a survey, both in the information-theoretic and computational regimes.

Cryptography with Traceability. Our secret-sharing scheme for complete bipartite graphs from bilinear maps implicitly builds and uses a form of anonymous secret-key encryption scheme where multiple, distinct but indistinguishable keys can decrypt the same ciphertext. Such schemes are also implicitly constructed in the design of private linear broadcast encryption (PLBE) [BSW06], which in turn, yields traitor tracing schemes. As observed in recent works (see, e.g., [KMUW18]), such schemes can be obtained from function-hiding secret-key functional encryption schemes for comparison functions over a polynomial-size domain. In particular, [KMUW18, Zha24] constructed such schemes from one-way functions with security for *two* ciphertexts. In contrast, in our application, we require security for *multiple* ciphertexts.

We also mention a recent work on traceable secret sharing [BPR24] that uses the large-field variant of Shamir’s secret sharing scheme, which is fully anonymous, to construct traceable secret sharing for threshold access structures [GSS21].

Secret Sharing with Uniform Shares. Finally, we also mention the recent work of [BR24] that uses the uniform share property for constructing secret sharing schemes with certifiable deletion.

2 Technical Overview

We begin by providing an overview of our notions of anonymity through the perspective of a particular graph access structure to demonstrate an inherent tradeoff between share anonymity and share size. In a graph access structure, parties are represented by nodes in a simple undirected graph, and every edge constitutes a minimal authorized set of size two. Consider the $(1, n)$ star graph access structure (see Def. 6.4) that consists of a single center node with an edge to each of the n remaining parties, which we term the leaf nodes.

Single-dealer Anonymity. A simple idea to share a bit $s \in \{0, 1\}$ is to additively secret share $s = s_0 \oplus s_1$ and give the share value s_0 to the center node the same share value s_1 to every leaf node. We immediately observe that any adversary who knows the secret value s and one of the two share values s_b for some $b \in \{0, 1\}$ does not have any advantage over randomly guessing which party this share belongs to. More generally, for any two unauthorized sets of shares of the same size (possibly identical sets), there cannot be any guessing advantage in deciding between these two sets even when given the secret value s . This is exactly the notion of S-anonymity⁶ (see Def. 4.4). However, S-anonymity provides extremely weak anonymity guarantees and is insufficient for real world settings [EBG⁺24].

Multi-dealer Anonymity. In the setting of multiple share dealers (or equivalently sharings), we desire an additional anonymity guarantee that an unauthorized set of shares from a single dealer cannot be correlated. This is captured by the notion of M-anonymity⁷ (see Def. 4.2 and Def. A.1). M-anonymity states that an adversary, who knows two secret values s_0 and s_1 (possibly equal to s_0) and is given as input

⁶S-anonymity was previously studied as a similar definition called Share Indistinguishability in [PO20]. Our definition restricts the guarantee to unauthorized sets.

⁷M-anonymity was previously studied as a similar definition called Decisional Share-Unpredictability in [PO20]. Our definition restricts the guarantee to unauthorized sets.

a set of share values corresponding to an unauthorized set of parties, does not have any guessing advantage for deciding whether these shares are from a single dealing of s_0 , or if they are a mixture of shares from a dealing of s_0 and an independent dealing of s_1 .

Observe that the S-anonymous secret sharing scheme for the $(1, n)$ -star graph does not satisfy M-anonymity: An unauthorized set consisting of all the leaf node shares from a single dealer would contain only identical share values, a clear violation of M-anonymity. While the above S-anonymous secret sharing scheme involves 1-bit per-party share sizes, we will later prove a lower bound of $\Omega(n)$ bit per-party shares for any (im)perfect M-anonymous secret sharing scheme for the $(1, n)$ star graph access structure (see Section 6.3). These two observations formally separate M-anonymity from S-anonymity.

Uniformity. Finally, uniformity (or U-anonymity, see Def. 4.1) is the simplest and strongest possible notion of anonymity which states that every unauthorized set of shares is distributed uniform randomly over the set of share values. Our technical focus is on achieving U-anonymity and M-anonymity due to their strong anonymity guarantees.

Departure from Standard Secret Sharing Techniques. Standard secret sharing techniques for general access structures completely fail to achieve even weak anonymity guarantees. In particular, a common decomposition-based technique is to represent the access structure by a monotone circuit over simpler gates, such as ANDs and ORs, where ANDs are implemented via additive secret sharing and ORs via repetition of share values. Such decomposition techniques, used in the works of [ISN87, BL90, BI01, LV18, ABF⁺19], and similar techniques that result in giving several parties the same repeated values [BI93], are commonly used for realizing general access structures. The first issue of share repetition has already been concretely described in the $(1, n)$ -star graph access structure.

The second issue with decomposition of an access structure is that unauthorized sets can be partially authorized for sub-access structures, so partial reconstruction gives rise to detectable correlated intermediate values in the reconstruction process. To make this concrete, consider the access structure on $2n$ many parties partitioned into two sets P_1, P_2 each containing n parties that consists of the AND of two (t, n) -threshold access structures. That is, a minimal authorized set of shares must contain shares from at least t parties from P_1 and at least t parties from P_2 . A standard secret sharing scheme for this access structure is to first additively secret share $s = s_1 + s_2$, then apply a (t, n) -threshold secret sharing scheme for each of s_1 and s_2 , and give every party in P_1 a (t, n) -share of s_1 and every party in P_2 a (t, n) -share of s_2 . Even if we use a U-FASS scheme for each threshold access structure, the composed scheme fails to satisfy even weak anonymity due to correlation attacks: Given an unauthorized set of $t + 1$ parties, an adversary can identify whether all these $t + 1$ parties belong to a single partition element P_i . In the case they all belong to the same partition element P_i , any t -subset of these $t + 1$ parties will reconstruct to the same value s_i .

2.1 Near-Optimal Information-Theoretic FASS

We will now describe the main ideas underlying our near-optimal construction (see Construction 5.7). For expository purposes, keep in mind our $\Omega(n)$ lower bound per-party share size for the $(1, n)$ -star graph access structure as a target for our upper bound whose proof we delay to Section 6.3.

We begin by reviewing the key idea in Construction 1 in [PO20], a FASS construction for arbitrary access structures represented by a monotone DNF that achieves perfect correctness, perfect M-anonymity, and statistical U-anonymity. The key technical idea, also used in the related context of distribution designs [BGIK16], produces shares for a single minterm of parties $\{1, \dots, t\} \subseteq [n]$ by sampling the following set of vectors:

1. Choose $n - 1$ linearly independent vectors over $\mathbb{F}_q^m$, say $\mathbf{v}_2, \dots, \mathbf{v}_n$ for a dimension m sufficiently large.

2. Set $\mathbf{v}_1 = \sum_{i=2}^t \mathbf{v}_i$.

Observe that there is a unique minimal set of linearly dependent vectors, namely $\{\mathbf{v}_1, \dots, \mathbf{v}_t\}$. Any other subset of vectors not containing this exact set is distributed identically to a set of random linearly independent vectors. Moreover, we can easily plant a recoverable secret in this linear dependence by choosing the coefficients in this linear dependence based on a secret s . Therefore, we can use the vector $\mathbf{v}_i$ as the i -th party's share s_i . When there are several minterms, we can repeat this process independently for each such minterm, resulting in the prior construction.

The prior construction sets the field size to be $q = 2$ and vector space dimension $m = O(n\ell)$ where ℓ is the number of minterms. For the $(1, n)$ -star graph access structure, this translates into a per-party share size of $O(n^3)$, which has a quadratic gap to the aforementioned lower bound of $\Omega(n)$.

Towards Optimal Share Size. Our objective is to obtain a matching upper bound for our lower bound of $\Omega(n)$ per-party share size for the $(1, n)$ -star graph access structure. To reduce this gap, we could consider reducing the dimension of the vector space m to be as small as $n - 1$. This would result in per-party share size of $O(n^2)$, which still has a linear gap to the aforementioned lower bound of $\Omega(n)$. While the gap to the lower bound has shrunk, this parameter setting only achieves perfect M-anonymity and the share values are only a constant statistical distance from being uniform random. This is because the probability that a set of at most $n - 1$ random $\mathbb{F}_2$ -vectors of dimension $n - 1$ is linearly independent is bounded by $\frac{1}{2}$. Our first observation, therefore, is that modifying the above construction with sufficiently large field size q allows us to use a smaller vector space dimension $m = n - 1$ while still obtaining statistical U-anonymity. This variant construction (see Theorem 5.1) achieves perfect correctness, perfect secrecy, and statistical U-anonymity. When we apply this construction to the $(1, n)$ -star graph access structure, we obtain a per-party share size of $\tilde{O}(n^2)$ bits while maintaining statistical U-anonymity.

An Upper Bound Matching the Lower Bound. Our main positive contribution to the information-theoretic setting of FASS is a generic construction for monotone DNF access structures with near perfect correctness, perfect U-anonymity, and per-party share sizes that are optimal up to polylogarithmic factors (see Theorem 5.6). That is, for the $(1, n)$ -star graph access structure, this main construction obtains a per-party share size of $\tilde{O}(n)$ bits and satisfies perfect U-anonymity.

To motivate our construction, we observe that in the previous construction each party receives a vector of $O(n)$ field elements for every minterm. Can we reduce the vector length? If we reduce the vector length, we unfortunately introduce new linear dependencies other than the planted linear dependence. However, observe that for correctness we are only concerned about having a unique linear dependency on sets of size at most t for a minterm of degree t . Since we obviously cannot sample $n - 1$ random linearly independent vectors of length $t \log n$, let us try sampling $n - 1$ uniform random vectors of length $t \log n$. This length was chosen to ensure that the probability of a linear dependency of size at most t existing remains small and to allow for a union bound argument showing that no such linear dependency exists among any of the ℓ -many minterms. As a result of this modification, we trade perfect correctness for perfect U-anonymity and reduce the per-party share size to be ℓ -many vectors over $\mathbb{F}_q$ of length $t \log n$.

Curiously, our new construction no longer has efficient reconstruction. Nevertheless, we can show that for reasonably long secrets s of length $\Omega(n/t)$ where t is the maximum size of any minterm in the DNF, we can obtain both efficient reconstruction and near optimal information rate⁸ (see Cor. 5.11).

⁸In the secret sharing setting, information rate refers to the per-party share size divided by the bit length of the secret.

2.2 Lower Bounds on Share Size

We now finally come to the lower bound on the per-party share size for the $(1, n)$ -star graph access structure (see Theorem 6.13). Consider any U-FASS scheme that achieves perfect correctness, perfect secrecy, and perfect U-anonymity for the $(1, n)$ -star graph for a secret in $\{0, 1\}$. Perfect secrecy implies that the uncertainty about the secret is not diminished by any single party's share. In particular, this applies to every leaf node's share. Perfect U-anonymity implies that two unauthorized shares corresponding to two leaf nodes are independently, identically distributed uniformly over some share space. In particular, these two unauthorized shares cannot be correlated. Finally, perfect correctness tell us that the center node's share must contain enough information to reconstruct the secret from any of the n -many leaf nodes' shares.

Since the leaf nodes' shares cannot be correlated, the center node's share needs to contain enough information for n *different* ways to reconstruct the secret s . This suggests that the center node should contain at least n bits of information. Moreover, if the center node has at least n bits of information, every leaf node must also have at least n bits of information because perfect U-anonymity implies that these unauthorized shares are independently and identically distributed. In other words, every party's share must be at least $\Omega(n)$ bits long! In fact, this exact reasoning applies not only to U-anonymity, but also to the weaker notion of M-anonymity. M-anonymity, like U-anonymity, implies that the distributions of two unauthorized shares are independent of one another even when conditioned on a fixed value of the secret s . This independence forces the center node to account for multiple different reconstruction paths. Moreover, we show that this same reasoning is robust to imperfect correctness, imperfect secrecy, and imperfect M-anonymity.

Exponential Sized Lower Bounds on Per-party Share Size. The general observation made in the above lower bound is that in an access structure, if there are multiple reconstruction paths that involve a common unauthorized party critical for reconstruction, then this particular critical party's share must contain enough information for all these reconstruction paths. M-anonymity (and U-anonymity) implies that these reconstruction paths are independently distributed, forcing the critical party's share to be large. We can therefore extend this lower bound to any access structure with this property.

For example, we can generalize the same argument to (k, n) -star graph access structures where k satisfies any relation with n (see Theorem 6.18). The generalization proceeds by an analogous iterative argument on the k sets of n leaf parties. Enumerating these k sets, we observe that the entropy of every share value corresponding to a party in the first set must be at least the entropy of the secret $H(S)$ due to the secrecy property. As we did for the $(1, n)$ -star graph access structures, we can then use the fact every party in the second set must have at least n many independent reconstruction paths, where each path involves a different single party from the first set. Since the share values of the parties in the first set have large entropy, by employing the correctness property and the share independence property implied by the M-anonymity, we can show that the entropy of the share values of the parties in the second set have appropriately larger entropy to account for those values, namely $\Omega(n \cdot H(S))$ bits of entropy. This argument continues until we reach the center node, arriving at $\Omega(n^k \cdot H(S))$ bits of entropy for the center node. However, every single share value is by itself unauthorized, and therefore must be identically distributed. Thus, every party's share must have $\Omega(n^k \cdot H(S))$ bits of entropy.

As an immediate consequence, our U-FASS construction for DNFs has optimal per-party share size up to polylogarithmic factors for a family of DNFs, namely those corresponding to the (k, n) -star graph (see Corollary 6.19). Observe that if λ is a security parameter, if $k = \lambda$ and $n = 2$, then the (k, n) -star graph is represented by a DNF with $O(\lambda)$ variables, yet the per-party share size is subject to a lower bound of $2^{\Omega(\lambda)}$ bits for 1-bit secrets.

Lower Bounds on Anonymous Reconstruction. We also give a lower bound showing that anonymous reconstruction cannot be implemented by a linear reconstruction algorithm for all non-trivial cases of the (t, n) -threshold access structure (see Theorem 6.3). In particular, the intuition is that anonymous reconstruction implies that the linear reconstruction is permutation invariant. That is, if the linear reconstruction algorithm is given by taking a dot product with some vector $\mathbf{a}$, permutating the order should not change the reconstruction output. This implies that anonymous linear reconstruction is effectively restricted to the case of additive secret sharing, which provably cannot be used for non-trivial (t, n) -threshold access structures. This result formally shows the difficulty of attaining anonymous reconstruction even for the threshold case.

2.3 Computational FASS

We now provide an overview of our results in the computational setting, where the information-theoretic security requirement is relaxed to computational indistinguishability. Our computational FASS schemes circumvent the information-theoretic lower bounds, and can additionally allow robust reconstruction.

Computational U-FASS for Star Graph. We first give a one-way function-based solution for the $(1, n)$ -star graph access structure that beats the lower bound on per-party share size of $\Omega(n)$ bits for information-theoretic U-FASS schemes for sharing 1-bit secrets (see Theorem 7.1).

We use a secret-key encryption scheme with pseudorandom ciphertexts based on pseudorandom functions (PRFs). Our design template is as follows: To share a secret $s \in \{0, 1\}$, we first sample a key k of a secret-key encryption scheme. The share of a leaf party consists of a fresh encryption of the secret s padded with a string of 0's under the key k . The share of the central party, in contrast, consists of the key k .

Given any authorized set containing the key and one or more ciphertexts, the reconstruction procedure can iterate over all pairs of shares, and attempt to use one share as a key and the other as a ciphertext to recover a string ending in 0's. This procedure recovers the secret s without knowing which share corresponds to the central party. Secrecy follows from the semantic security of the encryption scheme. Finally, U-anonymity can be achieved if the ciphertext and key are both pseudorandom. Note that the resulting scheme has per-party share size independent of the number of parties, beating the information theoretic lower bound of $\Omega(n)$ bits assuming only one-way functions.

2.3.1 The Complete Bipartite Graph Access Structure

We next construct a computational U-FASS scheme for the complete bipartite graph access structure (see Theorem 7.4). This graph access structure consists of a complete bipartite graph, in which every left node is connected to every right node. A minimal authorized set is therefore a pair of a left subset party and right subset party. This access structure is subject to the $\Omega(n)$ lower bound on the per-party share size for information-theoretic U-FASS schemes.

Following our approach to construct a computational U-FASS scheme for the $(1, n)$ -star graph access structure, we first consider the black-box use of a secret-key encryption scheme with “suitable” properties to achieve U-anonymity. More specifically, consider the template where every party in the left partition of the graph receives a fresh key, and every party in the right partition receives a fresh encryption of the secret. For correctness, we require that every key can decrypt every ciphertext. For share anonymity, we require that the joint distribution over a tuple of ciphertexts or a tuple of keys is pseudorandom.

It is easy to extend the construction for the $(1, n)$ -star graph as follows: Sample a fresh key for each party in the left partition, resulting in n keys. Now, for every party in the right partition, create a fresh ciphertext that consists of n ciphertexts where the i -th ciphertext contains an encryption of the secret under the key of the i -th party in the left partition. While it is easy to argue correctness and share anonymity,

the main issue is that this scheme achieves per-party share size linear in n . Indeed, we do not know how to achieve sublinear share size using only symmetric key assumptions.

Our Approach. We now present our solution using bilinear maps. We first consider the use of composite order groups to explain the key idea. Generate any bilinear group $G = \langle g \rangle$ of order $N = p_1 p_2 p_3$ for primes p_1, p_2, p_3 , where g_i denotes a generator for the subgroup of order p_i , with pairing $e(\cdot, \cdot) : G \times G \rightarrow G_T$ for some target group G_T of order N . To share a secret s , we first multiplicatively secret share it as $s = s_1 s_2$. Next, we produce keys that are a function of s_1 and ciphertexts which are a function of s_2 . More specifically, a key will be of the form $g_1^{s_1} g_2^r$ for a freshly chosen random r , and a ciphertext will be of the form $g_1^{s_2} g_3^{r'}$ for a freshly chosen r' .

First, note that the per-party share size is independent of the number of parties. We now describe the (anonymous) reconstruction procedure and argue correctness. Given a minimally authorized set consisting of a key and a ciphertext, we can use the pairing operation:

$$e(g_1^{s_1} g_2^r, g_1^{s_2} g_3^{r'}) = e(g_1, g_1)^{s_1 s_2} = e(g_1, g_1)^s.$$

We can then iterate over the exponent space to recover the secret which lies in a polynomially bounded subset. The anonymous reconstruction procedure for an overauthorized set consisting of multiple keys and ciphertexts is more involved and iterates over all pairs of shares (see the technical section for more details). Finally, by the standard subgroup hiding assumption, it follows that keys and ciphertexts are indistinguishable to random group elements.

The Problem with Composite Order Bilinear Groups. This encryption scheme seemingly yields a computational U-FASS scheme for the bipartite graph access structure. However, the subtle impediment is that with a composite order subgroup, the sharing algorithm must possess the necessary subgroup generators to produce the shares. These subgroup generators cannot sit in a public common random string because the subgroup hiding assumption is false if the subgroup generators are public. What if the sharing algorithm generates this pairing group itself? In a world with multiple share dealers, to ensure correctness, this sharing algorithm must also output the bilinear group description along with its shares, ruining even the M-anonymity property as every share would have to be tagged with the same bilinear group description!

Using Prime Order Bilinear Groups. It is known that many constructions based on composite order bilinear groups can be done instead over prime order bilinear groups [Fre10]. These assumptions are captured by the dual vector space pairings framework and reduce to the widely assumed hardness of the Decisional-Linear (DLIN) assumption [OT09, Lew12, OT15]. With prime order bilinear groups, there is a single fixed prime order bilinear group that every share dealer uses. Generating the shares no longer requires knowledge of a subgroup generator, and instead the sharing algorithm can sample uniform random vectors to generate shares from the correct distribution.

In the technical section, we present a U-FASS scheme for complete bipartite graphs using prime order bilinear groups. The key intuition behind the scheme is the same as in the composite order bilinear group setting.

2.3.2 A General Compiler from LWE

Finally, we show a generic compiler that transforms any efficient information-theoretic or computational secret sharing scheme for an access structure $\mathcal{A}$ into a computational M-FASS scheme for the same access structure $\mathcal{A}$.

All-or-nothing. As discussed in the beginning of Section 2, a key challenge in the design of FASS schemes is that the standard composition approach used in secret sharing does not work. This is because an adversary can use partial reconstruction attempts to detect correlations between shares to distinguish between different unauthorized sets. To prevent such attacks, we desire an *all-or-nothing* property: On any unauthorized set, the reconstruction algorithm should output $\perp$ or a uniform random value. The best-known cryptographic tool that is associated with such an all-or-nothing property is program obfuscation.

We now describe the ideas behind our general compiler. For intuition, we first consider the ideal obfuscation model, i.e. a model in which every party has black-box oracle access to a function f of our choice. We will later describe how we can instantiate our construction idea from standard assumptions.

Warmup in the Ideal Obfuscation Model. In the ideal obfuscation model, we can construct a computational U-FASS scheme $\widetilde{\Pi} = (\widetilde{\text{Share}}, \widetilde{\text{Recon}})$ from any standard secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ in the following way.

1. $\widetilde{\text{Share}}(s)$ secret shares s with the standard secret sharing scheme, $((x_1, sh_1), \dots, (x_n, sh_n)) \leftarrow \text{Share}(s)$ where x_i denotes the i th party's identity. The algorithm will pad every share to be of the same length with any reasonable padding convention. It uses any pseudorandom secret-key encryption scheme, e.g. any PRF based secret-key scheme, to encrypt these padded shares, obtaining ciphertexts $(ct_1, \dots, ct_n)$ under some key sk . Finally, it defines the function $f_{sk,s}$, with sk and s hardcoded.
 - This function takes as input some set of ciphertexts. It decrypts the ciphertexts, strips the padding, and runs Recon on the unpadded plaintexts.
 - If Recon outputs s , then $f_{sk,s}$ outputs s . Otherwise, it outputs $\perp$.

$\widetilde{\text{Share}}(s)$ outputs the shares $\{\widetilde{sh}_i\}_{i \in [n]}$ and gives oracle access to $f_{sk,s}$ for every party.

2. To reconstruct, any set of parties queries the ideal obfuscation on the tuple of their shares.

In the ideal obfuscation model, every unauthorized set of shares is pseudorandom and the only query output that is likely to be obtained is $\perp$. Thus the scheme achieves U-anonymity in the ideal obfuscation model.

Compute-and-Compare Obfuscation. Can we instantiate this ideal obfuscation with an actual primitive? While it may be natural to attempt an indistinguishability obfuscation (*iO*) based construction, this function could belong to a function class for which it suffices to use a weaker primitive in the family of obfuscation primitives.

Indeed, we observe that $f_{sk,s}$ looks like a multi-bit compute-and-compare function! That is, $f_{sk,s}(x)$ looks to be of the form $\text{MBCC}[g_{sk}, y, z](x)$: “If $g_{sk}(x) = y$, then output z . Else output $\perp$ ”, where $y = z = s$ and where g_{sk} decrypts the input, strips off the padding, runs Recon on the unpadded plaintexts, and outputs the result. Prior work on program obfuscation for this class of functions obtains virtual black-box security guarantees under stringent conditions on either the entropy or the computational unpredictability of the target point y . These constructions are based on the polynomial hardness of Learning with Errors (LWE) [WZ17, GKW17].

With regards to the stringent conditions on the target point y , we will require that y has high computational entropy (HILL entropy [HILL99]) even conditioned on the function description g_{sk} (which may reveal sk), z , and some set of unauthorized shares consisting of ciphertexts. Clearly the entropy requirement cannot be satisfied if $y = z$. To address this issue, we modify $\widetilde{\text{Share}}(s)$ to first sample a uniform

random secret r , and instead secret share r with Share. Now, $y = r$ and $z = s$, where r is chosen independently of s , so z leaks no information about y . It remains to check that y still has high computational entropy given sk and the some set of unauthorized shares consisting of ciphertexts whose plaintexts are shares of y . We can assume that the secret key sk is leaked by g_{sk} , so it suffices to check that y has high computational entropy conditioned on a set of unauthorized set of shares of y . This condition is true by the secrecy of the standard secret sharing scheme. In summary, when we consider a single unauthorized party, a share consisting of a compute-and-compare obfuscation and encrypted padded share value looks indistinguishable from a virtual black-box simulated program and uniform random string.

Achieving M -Anonymity. Taking one step back, to instantiate the ideal obfuscation model, we have to we give each party some program. What program do we give? Can we give every party the same compute-and-compare obfuscation, $\text{Obf}(\text{MBCC}[g_{sk}, y, z])$? No! As noted in the beginning of Section 2, repetition destroys M -anonymity if we give each party the same copy of a compute-and-compare obfuscation. What if we instead gave each party a freshly obfuscated compute-and-compare program of the same function $\text{MBCC}[g_{sk}, y, z]$? Unfortunately, we can no longer appeal to the virtual black-box security of the compute-and-compare obfuscation for unauthorized sets of size at least two. In particular, we would have to argue that the computational entropy of the target point y remains large even when conditioned on a compute-and-compare obfuscation $\text{Obf}(\text{MBCC}[g_{sk}, y, z])$, a circular argument.

To address this issue, we make use of a pseudorandom generator (PRG). For every party $i \in [n]$, we sample a fresh PRG g_i , and party i will now receive a compute-and-compare obfuscation of the program $\text{MBCC}[g_i \circ g_{sk}, g_i(y), z]$. Observe now that we can argue that the computational entropy of the target point $g_i(y)$ remains large even when conditioned on a compute-and-compare obfuscation $\text{Obf}(\text{MBCC}[g_j \circ g_{sk}, g_j(y), z])$ for $i \neq j$, conditioned on z , and conditioned on some unauthorized set of shares for the secret y . Here, z is completely independent of y , and the secrecy of the secret sharing scheme tells us that this unauthorized set of shares is indistinguishable from an unauthorized set of shares for some independent secret y' . Finally, we appeal to the pseudorandomness of the PRG g_i to argue that $g_i(y)$ is indistinguishable from uniform, allowing us to argue that the computational entropy is maximal.

To summarize, every party i 's share consists of an obfuscated program $\text{Obf}(\text{MBCC}[g_i \circ g_{sk}, g_i(y), z])$ and a ciphertext ct_i of a padded share of y (including the party identity) produced by the original standard secret sharing scheme. Observe, however, that while we can argue that the ciphertext ct_i is pseudorandom for an appropriate choice of secret-key encryption, we cannot argue that the compute-and-compare obfuscated program is pseudorandom. In fact, in the work of [WZ17], the description of these obfuscated compute-and-compare programs contain low-norm matrices that make them far from random. Therefore, we obtain only M -anonymity, rather than U -anonymity.

2.3.3 Robust Reconstruction

In the standard (non-anonymous) secret sharing setting, robust reconstruction is easily handled by having each dealer use a common tag on every share. To combat against random insertion errors, a dealer can use a digital signature scheme to sign each share and add the verification key as a common tag. However, in the anonymous setting, such tagging procedures conflict with anonymity requirements and we therefore require new approaches for robust reconstruction. Indeed, prior work [EBG⁺24] obtains efficient robust reconstruction for ramp secret sharing for the threshold access structure by using list-decoding, allowing robust reconstruction in the presence of a constant number of dealers. Their work leaves open efficient robust reconstruction for non-ramp threshold schemes and for an unbounded polynomial number of dealers.

In our work, we show that all three of our computational solutions additionally satisfy efficient robust reconstruction even against random insertion errors (see Section 8). Our LWE-based compiler, in

particular, positively answers the above two open questions.

In the case of our OWF-based computational U-FASS scheme for the $(1, n)$ -star graph access structure and our DLIN-based computational U-FASS scheme for the complete bipartite graph access structure, when reconstruction is attempted on unauthorized sets or on two shares from two separate dealers, the result is a random value. How can we distinguish between a random value and a well-formed message? In these schemes, the secret space is specified to be an exponentially small fraction of all possible reconstruction values. For example, in the OWF-based construction, the binary secret space is the set of $\{0^\lambda, 10^{\lambda-1}\} \subseteq \{0, 1\}^\lambda$ for a security parameter $\lambda \in \mathbb{N}$. Similarly, in the DLIN-based scheme, the binary secret space is $\{0, 1\} \subseteq \mathbb{F}_q$ for an exponential sized q . Abstracting this property into a cryptographic primitive, we effectively construct an encryption scheme that outputs $\perp$ when we attempt to decrypt a ciphertext with a uniform random key. Moreover, this robust reconstruction is efficient for these two access structures: For these two access structures, the brute force method of reconstructing all pairs of shares is polynomial-time because these are graph access structures so all minimal authorized sets are of size 2. See Theorem 8.3 and Theorem 8.4 for a more formal statement using this intuition.

In the case of the LWE-based computational U-FASS compiler for arbitrary access structures, recall that every share is parsed into a ciphertext and an obfuscated function. To find all the secrets corresponding to authorized sets from a common dealer, we run each obfuscated function on the set of all ciphertexts. Hardcoded in each obfuscated function is a secret key for an SKE scheme, and the obfuscated function decrypts every ciphertext. If we use the above abstracted variant SKE scheme, instead of the standard SKE, decrypting a ciphertext with the wrong secret key will result in a $\perp$ value since these secret keys are uniformly distributed. Then, we can modify the function being obfuscated to ignore all $\perp$ values and attempt to run the underlying standard secret sharing reconstruction only on the non- $\perp$ values. For any obfuscated circuit generated by a dealer D , the output of the circuit will be exactly either the secret s if the ciphertexts contain an authorized set of shares generated by same dealer D . We can then output a list of all such secrets s . See Theorem 8.5 for the formal details.

2.4 Open Questions and Future Directions

Our work leaves open several interesting questions for future research. We outline a few of them here.

Information-Theoretic FASS. The first open question is the possibility of *perfect* U-FASS for general access structures, namely U-FASS with both perfect uniformity and perfect correctness. The question is open even for threshold structures with reconstruction threshold $3 \leq t \leq n - 2$ and even for the weaker notion of perfect M-FASS. In particular, it is open in the 3-out-of-5 threshold case. Recall that the known U-FASS for threshold structures, based on a large-field variant of Shamir’s scheme with random interpolation points, does *not* achieve the perfect notion of U-FASS, compromising either perfect correctness (if the interpolation points are independent)⁹ or perfect uniformity (if they are guaranteed to be distinct). This, in turn, requires the share size to grow with the level of security or correctness. In contrast, a perfect U-FASS scheme must provide anonymity with “constant-size” shares, depending only on the access structure. Interesting progress on this question has been recently made by Con [Con24], who shows that Shamir’s original scheme can be instantiated to give a perfect U-FASS scheme for *gap threshold* structures, where the reconstruction threshold is bigger than the secrecy threshold by roughly a factor of 2.

A second set of questions considers the feasibility of an improved information-theoretic M- or U-FASS schemes for the single-minterm access structure. Consider a monotone DNF given by a single minterm ϕ of size t . Our lower bound given in Theorem 6.18 does not extend to this access structure because this access structure has only one reconstruction path. On the other hand, our best upper bound for the per-party

⁹To achieve perfect uniformity, the polynomial values on repeated points are replaced by uniform and independent field elements.

share size for this access structure is $\tilde{O}(t\lambda)$. Can we give a non-general information-theoretic construction for ϕ with a smaller upper bound? Secondly, for $t = O(n)$, it may take an exponential time to identify the minimal authorized set of size t from a super authorized set, say all n shares. In the setting of sharing a single-bit secret, can we obtain efficient reconstruction for this access structure?

Another open question concerns share size of FASS for *weighted threshold* access structures. The standard virtualized-version of Shamir’s secret sharing for weighted threshold access structures does not achieve S-FASS due to share-size asymmetry. Namely, Shamir’s secret sharing scheme readily yields, via virtualization, a secret sharing scheme for this access structure: Each party with weight w_i receives w_i shares of Shamir’s secret sharing scheme. Note that the share size of party i in this scheme grows linearly with its weight w_i . It is easy to see that this scheme does *not* achieve S-Anonymity. This is because the share size of each party leaks information about its identity (since the weights are public). Note that naive attempts to pad the shares of the party to achieve share-size symmetry may break correctness. Our generic compiler based on compute-and-compare obfuscation resolves this challenge since the padding can be hidden inside the ciphertexts and easily detected by the obfuscated program but not “outside” the obfuscation. This, however, requires the use of computational assumptions. We leave open the problem of constructing *information-theoretic* S-FASS schemes for weighted threshold access structures – with arbitrary variance between the party weights¹⁰ – that achieve similar efficiency as the one obtained from Shamir’s secret sharing.

Computational FASS. Finally, we leave open the question of diversifying the assumptions for computational FASS. We show that even for simple representation classes such as CNF, efficient information-theoretic U-FASS schemes do not exist, which implies the necessity of one-way functions. Are one-way functions sufficient? Alternatively, does efficient computational U-FASS for CNF imply public-key encryption or some other standard primitive not known to follow from one-way functions?

3 Preliminaries

Notation Throughout this paper, λ denotes the security parameter. We use $\mathbf{x} \parallel \mathbf{y}$ to denote vector concatenation. For $\mathbf{x} \in \{0, 1\}^n$, we use $\text{wt}(\mathbf{x})$ to denote the Hamming weight of $\mathbf{x}$. We denote the set of participating parties by $[n] = \{1, 2, 3, \dots, n\}$. The power set of S is denoted 2^S . Script capital letters $\mathcal{K} = \{\mathcal{K}_\lambda\}_{\lambda \in \mathbb{N}}$ denote a collection of sets $\mathcal{K}_\lambda$, which we sometimes refer to as an ensemble.

3.1 Secret Sharing

We recall several classic definitions for secret sharing and introduce the related notation that we will use throughout the paper.

Definition 3.1 (Access Structure). *An n -party access structure is a collection of subsets of parties $\Gamma \subseteq 2^{[n]}$. We refer to a set of parties $A \in \Gamma$ as authorized and to other sets as unauthorized. We assume access structures to be monotone in the sense that if $A \in \Gamma$ and $A \subseteq B$, then $B \in \Gamma$.*

We now define a standard notion of secret sharing, which can be easily modified to fit both the computational and the information-theoretic variants. It will be implicitly understood that the number of parties $n = n(\lambda)$ is a function of the security parameter λ , and for every value of λ , one can consider a monotone access structure Γ_λ and a secret domain $\mathcal{S}_\lambda$. To simplify the presentation, without loss of generality, a secret sharing scheme will sometimes be presented in this paper with respect to a single access structure instead

¹⁰For “balanced” weights, one can appeal to list-decoding to achieve correctness for some parameter choices by using random evaluation points for padding.

of an ensemble of access structures $\Gamma = \{\Gamma_\lambda\}_{\lambda \in \mathbb{N}}$ and a secret space $\mathcal{S} = \{\mathcal{S}_\lambda\}_{\lambda \in \mathbb{N}}$. To avoid burdening the reader with notation, we avoid explicitly defining the ensemble in each definition.

Definition 3.2 (Standard Secret Sharing Syntax). *A secret sharing scheme for an access structure Γ and secret space $\mathcal{S}$, consists of two algorithms (Share, Recon) with the following syntax:*

1. $\text{Share}(1^\lambda, s)$: a PPT algorithm that takes as input a secret $s \in \mathcal{S}_\lambda$ and outputs shares ss .
2. $\text{Recon}(1^\lambda, \mathbf{x}, ss)$: is a deterministic algorithm that takes as input the party identities $\mathbf{x} \in \{0, 1\}^\lambda$, and a tuple of shares ss and outputs a value $s' \in \mathcal{S}_\lambda$. The string $\mathbf{x} \in \{0, 1\}^n$ is an indicator vector for a subset $A \subseteq [n]$ so that $\mathbf{x}_A \in \{0, 1\}^n$ is a vector where $\mathbf{x}_{A,i} = 1$ for all $i \in A$ and $\mathbf{x}_{A,i} = 0$ otherwise.

For any set $A \in 2^{[n]}$, we will use the notation sh_A to denote the tuple $(sh_i)_{i \in A}$ where sh_i is the i -th party's share.

To be able to refer to the randomness used in the PPT algorithm Share, we sometimes write $\text{Share}(1^\lambda, k; r)$ where r is the randomness to an otherwise deterministic process.

Definition 3.3 (Correctness). *For a function $\varepsilon : \mathbb{N} \rightarrow [0, 1]$, a secret sharing scheme (Share, Recon) with access structure Γ and secret space $\mathcal{S}$ is $(1 - \varepsilon)$ -correct if for all $\lambda \in \mathbb{N}$, for all $s \in \mathcal{S}_\lambda$, for all sets $A \in \Gamma_\lambda$,*

$$\Pr_{sh \leftarrow \text{Share}(1^\lambda, s)} \left[\text{Recon}(1^\lambda, \mathbf{x}_A, sh_A) = s \right] \geq 1 - \varepsilon(\lambda)$$

where the probability statement is over the coins of Share. When $\varepsilon(\lambda) = 0$, then we say that the scheme satisfies perfect correctness.

Definition 3.4 (Secrecy). *A statistically (resp. computationally) secret sharing scheme (Share, Recon) with access structure space $\Gamma = \{\Gamma_\lambda\}$ is secret if for every computationally unbounded (in the computational setting, polynomial-time non-uniform) algorithm $\mathcal{A}$, there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$, such that for all sufficiently large $\lambda \in \mathbb{N}$, for all unauthorized sets $B \in 2^{[n]} \setminus \Gamma_n$, for all $s_0, s_1 \in \mathcal{S}_\lambda$,*

$$\left| \Pr_{sh^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, B, s_0, s_1, sh_B^{(0)}) = 1 \right] - \Pr_{sh^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)} \left[\mathcal{A}(1^\lambda, B, s_0, s_1, sh_B^{(1)}) = 1 \right] \right| \leq \mu(\lambda)$$

Definition 3.5 (Anonymous Reconstruction Secret Sharing Syntax [BS97]). *A secret sharing scheme with anonymous reconstruction for an access structure Γ and secret space $\mathcal{S}$, consists of two algorithms (Share, Recon) with the following syntax:*

1. $\text{Share}(1^\lambda, s)$: a PPT algorithm that takes as input a secret $s \in \mathcal{S}_\lambda$ and outputs shares ss .
2. $\text{Recon}(1^\lambda, ss)$: is a deterministic algorithm that takes as input a tuple of shares ss and outputs a value $s' \in \mathcal{S}_\lambda$.

For any set $A \in 2^{[n]}$, we will use the notation sh_A to denote the tuple $(sh_i)_{i \in A}$ where sh_i is the i -th party's share.

Remark 3.1. The correctness and secrecy properties for anonymous secret sharing schemes is defined identically except syntactically the reconstruction algorithm Recon does not take the party identities as an input.

3.2 Information Theory

Definition 3.6 (Statistical Distance). *Let X and Y be two random variables with probability distributions p_X and p_Y . The statistical distance (also called the total variation distance, the L^1 -distance, or the Kolmogorov distance) of X and Y is*

$$\Delta(X, Y) := \frac{1}{2} \sum_{z \in \text{supp}(X) \cup \text{supp}(Y)} |p_X(z) - p_Y(z)|.$$

Definition 3.7 (Shannon Entropy). *The Shannon entropy of any random variable X is*

$$\mathbf{H}(X) := -\mathbb{E}_X[\log p_X(x)] = - \sum_{x \in \text{supp } X} p_X(x) \log p_X(x).$$

For $\alpha \in [0, 1]$, the binary Shannon entropy is

$$\mathbf{H}(\alpha) := -\alpha \log \alpha - (1 - \alpha) \log(1 - \alpha) \in [0, 1].$$

The conditional Shannon entropy between random variables X and Y is

$$\mathbf{H}(X | Y) = \mathbf{H}(X, Y) - \mathbf{H}(Y).$$

Definition 3.8 (Mutual Information). *If X and Y are two finite discrete random variables, then their mutual information is*

$$I(X; Y) := \mathbf{H}(X) + \mathbf{H}(Y) - \mathbf{H}(X, Y).$$

The conditional mutual information with respect to a finite discrete random variable Z is defined as

$$I(X; Y | Z) := \sum_{z \in \text{supp}(Z)} p_Z(z) \cdot I(X; Y | Z = z) = \sum_{z \in \text{supp}(Z)} p_Z(z) \cdot I(X | Z = z; Y | Z = z).$$

Definition 3.9 (Fano's Inequality [CT⁺91]). *For any estimator $\hat{X}$ such that $X - Y - \hat{X}$ is a Markov chain, with $P_e = \Pr[X \neq \hat{X}]$, we have*

$$\mathbf{H}(P_e) + P_e \log |X| \geq \mathbf{H}(X | \hat{X}) \geq \mathbf{H}(X | Y).$$

Definition 3.10 (Average Conditional Min-entropy [DORS08]). *For random variables A, B , the average conditional min-entropy is defined to be*

$$\tilde{\mathbf{H}}_\infty(A | B) := -\log \left(\sum_b \Pr[B = b] \cdot \max_a \Pr[A = a | B = b] \right).$$

Remark 3.2 (Interpretation of Average Conditional Min-entropy). *If $A - B$ is a Markov chain, then the average conditional min-entropy is the negative logarithm of the probability that a maximum likelihood decoder is able to recover correctly the value of the random variable A from the value of B . If this value is small, then this means the algorithm is able to correctly guess the value of random variable A with large probability. In particular, if $\tilde{\mathbf{H}}_\infty(A | B) \leq \varepsilon$, then there exists a maximum likelihood decoder that correctly recovers A with probability at least $2^{-\varepsilon}$ over the choice of B .*

Lemma 3.11 (Average Conditional Min-entropy Bounded Above by Conditional Shannon Entropy). *For random variables A, B , $\tilde{\mathbf{H}}_\infty(A | B) \leq \mathbf{H}(A | B)$ where $\tilde{\mathbf{H}}_\infty$ is the average conditional min-entropy (Def. 3.10) and $\mathbf{H}$ refers to Shannon entropy.*

Proof. Recall the definition of conditional min-entropy for any fixed value of b ,

$$\mathbf{H}_\infty(A \mid B = b) := -\log \max_a \Pr[A = a \mid B = b]$$

and recall that $\mathbf{H}(A \mid B = b) \geq \mathbf{H}_\infty(A \mid B = b)$. Then,

$$\begin{aligned} \mathbf{H}(A \mid B) &= \sum_b \Pr[B = b] \cdot \mathbf{H}(A \mid B = b) \\ &\geq \sum_b \Pr[B = b] \cdot \mathbf{H}_\infty(A \mid B = b) \\ &= \sum_b \Pr[B = b] \cdot \left(-\log \max_a \Pr[A = a \mid B = b] \right) \\ &\geq -\log \left(\sum_b \Pr[B = b] \cdot \max_a \Pr[A = a \mid B = b] \right) \\ &= \mathbf{H}_\infty(A \mid B) \end{aligned}$$

where the last inequality is due to Jensen's inequality applied on the convex function $(-\log)$. $\square$

Lemma 3.12 (Mutual Information Upper Bounded by Statistical Distance [Zha07]). *Let X and Y be two finite discrete random variables with probability distributions p_X and p_Y , and with support size m and n respectively, and let p_{XY} be their joint distribution. Let $\delta = \Delta(p_{XY}, p_X p_Y)$ be their statistical distance. Then*

$$I(X; Y) \leq \delta \cdot \log(mn - 1) + \mathbf{H}(\delta).$$

We now prove three simple general facts about entropy that will facilitate our lower bound proofs in Section 6.

Lemma 3.13 (Discarding Lemma). *Let A, B, C be three discrete random variables. Then $I(A; B, C) \geq I(A; B)$.*

Proof of Lemma 3.13. Observe that $\mathbf{H}(A \mid B, C) \leq \mathbf{H}(A \mid B)$ because conditioning cannot increase entropy. Therefore,

$$\begin{aligned} I(A; B, C) &= \mathbf{H}(A) - \mathbf{H}(A \mid B, C) \\ &\geq \mathbf{H}(A) - \mathbf{H}(A \mid B) \\ &= I(A; B). \end{aligned}$$

$\square$

Lemma 3.14. *Let A, B, C be three discrete random variables. If $\mathbf{H}(B) - \mathbf{H}(B \mid C) \leq \rho$, then $I(A; B \mid C) \geq I(A, C; B) - \rho$. When $\rho = 0$, equality holds.*

Proof of Lemma 3.14. By definition of mutual information and conditional mutual information:

$$\begin{aligned} I(A; B \mid C) &= I(B; A \mid C) = \mathbf{H}(B \mid C) - \mathbf{H}(B \mid A, C) \\ &\geq \mathbf{H}(B) - \mathbf{H}(B \mid A, C) - \rho \\ &= I(B; A, C) - \rho \\ &= I(A, C; B) - \rho. \end{aligned}$$

$\square$

Lemma 3.15. Let A, B, C be three discrete random variables. $I(A; B, C) \geq I(A; B \mid C)$.

Proof of Lemma 3.15. Since conditioning cannot increase entropy, $H(A \mid C) \leq H(A)$. Therefore,

$$\begin{aligned} I(A; B, C) &= H(A) - H(A \mid B, C) \\ &\geq H(A \mid C) - H(A \mid B, C) \\ &= I(A; B \mid C). \end{aligned}$$

□

3.3 Compute-and-Compare Obfuscation

Definition 3.16 (Distributional Indistinguishability [WZ17]). An obfuscation Obf for the distribution class $\mathcal{D}$ over a family of programs $\mathcal{P}$ satisfies distributional indistinguishability if there exists a (non-uniform) PPT simulator Sim , such that for all distribution ensembles $\mathcal{D} = \{\mathcal{D}_\lambda\} \in \mathcal{D}$, we have

$$\left(\text{Obf}(1^\lambda, P), \text{aux} \right) \approx_c \left(\text{Sim}(1^\lambda, P.\text{params}), \text{aux} \right)$$

where $(P, \text{aux}) \leftarrow \mathcal{D}_\lambda$.

Definition 3.17 (Multi-bit Compute-and-compare [WZ17]). Given a function $f : \{0, 1\}^{\ell_{\text{in}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}$ along with a target point $y \in \{0, 1\}^{\ell_{\text{out}}}$, and a message $z \in \{0, 1\}^{\ell_{\text{msg}}}$, we define the multi-bit compute-and-compare program:

$$\text{MBCC}[f, y, z](x) = \begin{cases} z & \text{if } f(x) = y \\ \perp & \text{otherwise} \end{cases}.$$

We define the class $\mathcal{P}_{\text{MBCC}}^{\text{TM}}$ to consist of canonical descriptions of programs $\text{MBCC}[f, y, z]$, where f is represented as a Turing Machine.

Definition 3.18 (Conditional (HILL) Pseudo-Entropy [HILL99, HLR07]). Let $X = \{X_\lambda\}$, $Y = \{Y_\lambda\}$ be ensembles of jointly distributed random variables. We define the conditional pseudo-entropy of X conditioned on Y to be at least $\ell(\lambda)$, denoted by $\mathbf{H}_{\text{HILL}}(X \mid Y) \geq \ell(\lambda)$ if there exists some $X' = \{X'_\lambda\}$ jointly distributed with Y such that $(X, Y) \approx_c (X', Y)$ and $\tilde{\mathbf{H}}_\infty(X'_\lambda \mid Y_\lambda) \geq \ell(\lambda)$.

Definition 3.19 (α -Pseudo-Entropy [WZ17]). For a function $\alpha(\lambda)$, the class of α -pseudo-entropy distributions $\mathcal{D}_{\alpha\text{-PE}}$ consists of ensembles $\mathcal{D} = \{D_\lambda\}$ such that $(\text{MBCC}[f, y, z], \text{aux}) \leftarrow D_\lambda$ satisfies $\mathbf{H}_{\text{HILL}}(y \mid (f, z, \text{aux})) \geq \alpha(\lambda)$.

Definition 3.20 (Class of Pseudo-entropic Distributions [WZ17]). For any polynomial $\alpha : \mathbb{N} \rightarrow \mathbb{N}$, we define the class of α -pseudo-entropic distributions $\mathcal{D}_{\alpha\text{-PE}}$ to consist of ensembles $\mathcal{D} = \{D_\lambda\}$ over $(\text{MBCC}[f, y, z], \text{aux})$ such that y is $\alpha(\lambda)$ -pseudo-entropic given (f, z, aux) .

Definition 3.21 (PRG for Pseudo-entropic Seeds [WZ17]). Let $\mathcal{D}$ be a class of distributions $\mathcal{D} = \{D_\lambda\}$ where $(s, \text{aux}) \leftarrow D_\lambda$ outputs a seed $s \in \{0, 1\}^{n(\lambda)}$ and some auxiliary information aux . A PRG $(\text{Gen}, \mathcal{G})$ for the class $\mathcal{D}$ consists of a family of polynomial-time functions $\mathcal{G} = \{G : \{0, 1\}^n \rightarrow \{0, 1\}^m\}$ sampled via $G \leftarrow \text{Gen}(1^n, 1^m)$. We require that for every $D \in \mathcal{D}$ and for every polynomial $m(\lambda)$ we have

$$(G, G(s), \text{aux}) \approx_c (G, u, \text{aux})$$

where $G \leftarrow \text{Gen}(1^{n(\lambda)}, 1^{m(\lambda)})$, $(s, \text{aux}) \leftarrow D_\lambda$, and $u \xleftarrow{\$} \{0, 1\}^{m(\lambda)}$.

- A PRG for α -**pseudo-entropy seeds** is a PRG for the class $\mathcal{D}$ of distributions where $H_{\text{HILL}}(s \mid \text{aux}) \geq \alpha(\lambda)$.
- A PRG is in NC^1 if for all $G \leftarrow \text{Gen}(1^n, 1^m)$, we can compute G by a circuit of depth $d(n) = O(\log n)$. In particular, the depth is independent of m .
- A PRG is **injective** if there exists some polynomial $m^*(n)$ such that for all $m \geq m^*(n)$ the function $G \leftarrow \text{Gen}(1^n, 1^m)$ is injective with overwhelming probability. A PRG is **strongly injective** if G is injective even when restricted to only the first $m^*(n)$ bits of output.

Remark 3.3 (Injective PRGs for Pseudo-Entropic Seeds in NC^1 from Lattices [GKPV10, BPR12, AKPW13, WZ17]). Assuming the polynomial hardness of LWE with secret dimension $n = n(\lambda)$, sample count $m = m(\lambda)$, modulus $q = q(\lambda)$ and Gaussian width $\sigma = \alpha q$ where $\alpha = 1/\text{poly}(n)$, there exists injective PRGs in NC^1 for λ^ε -Pseudo-Entropic Seeds for any $\varepsilon > 0$.

Theorem 3.22 (Succinct TM MBCC Obfuscator [WZ17]). Under the LWE assumption, and assuming the existence of fully homomorphic encryption¹¹, there exists an obfuscator for $\mathcal{P}_{\text{MBCC}}^{\text{TM}}$ that satisfies distributional indistinguishability (Def. 3.16) for all α -pseudo-entropic distributions $D_{\alpha\text{-PE}}$ for polynomials $\alpha : \mathbb{N} \rightarrow \mathbb{N}$.

Moreover, this obfuscator is succinct: the runtime of the obfuscation procedure and the size of obfuscation is

$$\text{poly}(\lambda, |g|, |y|, \log t)$$

where $|g|$ is the TM description size, and t is the fixed runtime bound on g .

4 Fully Anonymous Secret Sharing

In this section, we introduce several notions of anonymity in secret sharing. Informally, there are two types of anonymity that we consider:

1. **[Reconstruction Anonymity]** Firstly, that the reconstruction algorithm does not make use of party identities (See Def. 3.5).
2. **[Unauthorized Set Anonymity]** Secondly, any unauthorized set of shares does not reveal the identity of the parties involved.

Any secret sharing scheme that satisfies both notions of anonymity is a *fully anonymous* secret sharing (FASS) scheme. The notion of unauthorized set anonymity is presented in varying strengths.

1. **[Uniformity (U-Anonymity)]** The strongest notion of unauthorized set anonymity is that unauthorized sets of shares are uniformly distributed or pseudorandom.
2. **[Multiple Dealer Anonymity (M-Anonymity)]** A strictly weaker notion is that a set of unauthorized shares containing a mixture of shares from independent fresh sharings of different secrets is indistinguishable from a set of the same cardinality of unauthorized shares from a single sharing of a secret.
3. **[Single Dealer Anonymity (S-Anonymity)]** Finally, the weakest notion is that an unauthorized set of shares from a single sharing of a secret is indistinguishable from any other unauthorized set of shares of the same cardinality.

¹¹This requires additionally assuming circular security.

A secret sharing scheme with anonymous reconstruction (Def. 3.5) that satisfies U-anonymity is a U-fully anonymous secret sharing scheme (U-FASS), and the terminology M-FASS and S-FASS are similarly defined. We now formally present these notions of unauthorized set anonymity in the order of the strongest anonymity guarantees to the weakest guarantees.

4.1 Uniformity

First, we introduce a natural and strong form of secrecy: Any unauthorized set of shares is uniform randomly distributed or pseudorandom.

Definition 4.1 (Uniformity or U-Anonymity). *A secret sharing scheme (Share, Recon) for an access structure Γ is statistically (resp. computationally) μ -uniform or μ -U-Anonymous if for all computationally unbounded algorithms $\mathcal{A}$ (in the computational setting, non-uniform polynomial-time), for all sufficiently large $\lambda \in \mathbb{N}$, for all secrets $s \in \mathcal{S}_\lambda$, for any unauthorized set $B \in 2^{[n]} \setminus \Gamma_\lambda$, sh_B for $\text{sh} \leftarrow \text{Share}(1^\lambda, s)$, we have*

$$\left| \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, \text{sh}_B) = 1 \right] - \Pr_{U \sim \text{Unif}(\Omega_\lambda)^{|B|}} \left[\mathcal{A}(1^\lambda, s, B, U) = 1 \right] \right| \leq \mu(\lambda)$$

where the set Ω_λ is the efficiently samplable and publicly known support of a single share produced by the sharing algorithm over all choices of secrets. The notation $\text{Unif}(\Omega_\lambda)^{|B|} = \underbrace{\text{Unif}(\Omega_\lambda) \times \text{Unif}(\Omega_\lambda) \times \cdots \times \text{Unif}(\Omega_\lambda)}_{|B| \text{ times}}$

is used above.

Remark 4.1 (Uniformity Implies Secrecy). Perfect (respectively, statistical or computational) uniformity implies perfect (resp. statistical or computational) secrecy.

Remark 4.2 (Default Conventions). As usual, statistical (resp. computational) U/M/S-Anonymity without a specified parameter refers to a definition in which for every unbounded adversary (resp. non-uniform polynomial-time) $\mathcal{A}$, there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$ for which the rest of the statement holds.

4.2 Multi-dealer Anonymity

We now introduce a notion strictly weaker than uniformity, termed multi-dealer anonymity (M-Anonymity). The definition of M-Anonymity captures the inability of an adversary to detect whether an unauthorized set of shares consists of shares produced by a single sharing of a secret s , or a mixture of shares from multiple sharings of possibly different secrets. Our M-Anonymity definition considers the hardest distinguishing task, that is, the task of distinguishing between shares coming from a single sharing and all-but-one shares coming from a sharing of some secret s_0 and one share produced by the sharing of some secret s_1 . This statement is justified because M-Anonymity implies a stronger notion of anonymity, Strong M-Anonymity, whose definition we relegate to Appendix A in Definition A.1.

Definition 4.2 (M-Anonymity). *A secret sharing scheme (Share, Recon) for access structure Γ , and secret space $\mathcal{S}$ is μ -M-Anonymous if for every computationally unbounded (in the computational setting, non-uniform polynomial-time) algorithm $\mathcal{A}$, there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$ such that for all sufficiently large $\lambda \in \mathbb{N}$, for all $B \in 2^{[n]} \setminus \Gamma_\lambda$ and for any $B' \subset B$, $|B'| = |B| - 1$, and for any singleton $\{a\} \in 2^{[n]} \setminus \Gamma_\lambda$, for all secrets $s_0, s_1 \in \mathcal{S}_\lambda$,*

$$\left| \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B', a, \text{sh}_B^{(0)}) = 1 \right] - \Pr_{\substack{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0) \\ \text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)}} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B', a, \text{sh}_{B'}^{(0)} \parallel \text{sh}_{\{a\}}^{(1)}) = 1 \right] \right| \leq \mu(\lambda).$$

Lemma 4.3 (Uniform Unauthorized Shares Implies Multi-secret Anonymous Secret Sharing). *If a secret sharing scheme satisfies μ -U-Anonymity against computationally unbounded adversaries (in the computational setting, non-uniform polynomial time adversaries), then it satisfies (3μ) -M-Anonymity against computationally unbounded adversaries (resp. non-uniform polynomial time adversaries).*

Proof. The μ -U-Anonymity property directly implies that for any computationally unbounded adversaries (resp. non-uniform polynomial time adversaries) $\mathcal{A}$, for all $B \in 2^{[n]} \setminus \Gamma_\lambda$, for any $B' \subset B$ such that $|B'| = |B| - 1$, for any unauthorized $a \in [n]$, for all secrets $s_0, s_1 \in \mathcal{S}_\lambda$,

$$\left| \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B' \cup \{a\}, \text{sh}_B^{(0)}) = 1 \right] - \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B' \cup \{a\}, \text{Unif}(\Omega_\lambda)^{|B|}) = 1 \right] \right| \leq \mu(\lambda). \quad (1)$$

where Ω_λ is the support of the share space domain, which is publicly known and efficiently uniformly samplable. The μ -U-Anonymity property also directly implies that

$$\left| \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B' \cup \{a\}, \text{Unif}(\Omega_\lambda)^{|B|}) = 1 \right] - \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B' \cup \{a\}, \text{sh}_{B'}^{(0)} \parallel \text{Unif}(\Omega_\lambda)) = 1 \right] \right| \leq \mu(\lambda). \quad (2)$$

since the uniform values in Ω_λ can be sampled by $\mathcal{A}$. Finally, the μ -U-Anonymity property also implies that

$$\left| \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B' \cup \{a\}, \text{sh}_{B'}^{(0)} \parallel \text{Unif}(\Omega_\lambda)) = 1 \right] - \Pr_{\substack{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0) \\ \text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)}} \left[\mathcal{A}(1^\lambda, s_0, s_1, B, B' \cup \{a\}, \text{sh}_{B'}^{(0)} \parallel \text{sh}_{\{a\}}^{(1)}) = 1 \right] \right| \leq \mu(\lambda) \quad (3)$$

where Equation 3 follows from a straightforward formal reduction from the μ -U-Anonymity property. If there exists a computationally unbounded algorithm (resp. non-uniform polynomial time algorithm) $\mathcal{A}$ that violates Equation 3 for some family of secrets $\{s_{0,\lambda}, s_{1,\lambda}\}_{\lambda \in \mathbb{N}}$ and some family of unauthorized sets $\{B_\lambda, B'_\lambda \cup \{a_\lambda\}\}_{\lambda \in \mathbb{N}}$ satisfying the above property that $B_\lambda, B'_\lambda \cup \{a_\lambda\}$ differ by at most one element, then we can construct a computationally unbounded algorithm (resp. non-uniform polynomial time algorithm) $\mathcal{D}$ that breaks μ -U-Anonymity.

The reduction $\mathcal{D}$ is given as input $(1^\lambda, s_{1,\lambda}, \{a_\lambda\}, X)$ where X is either sh_{a_λ} or $\text{Unif}(\Omega_\lambda)$ where sh_{a_λ} is the a_λ -th share for some fresh sharing of $s_{1,\lambda}$ performed by the challenger from the definition of μ -U-Anonymity. This algorithm $\mathcal{D}$ on input 1^λ has the advice string $s_{0,\lambda}, B_\lambda, B'_\lambda \cup \{a_\lambda\}$ all hard-coded in its description. It produces shares $\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_{0,\lambda})$, and outputs the value output by $\mathcal{A}(1^\lambda, s_{0,\lambda}, s_{1,\lambda}, B_\lambda, B'_\lambda \cup \{a_\lambda\}, \text{sh}_{B'}^{(0)} \parallel X)$. If Equation 3 were greater than $\mu(\lambda)$, then $\mathcal{D}$ has greater than $\mu(\lambda)$ advantage in distinguishing between whether X was sh_{a_λ} or a random value from Ω_λ . Then, the existence of $\mathcal{D}$ contradicts the μ -U-Anonymity assumption. Therefore, no such $\mathcal{A}$ can exist and Equation 3 holds.

Observe that Equation 1, Equation 2, and Equation 3 continue to hold even if $\mathcal{A}$ is given additionally as input a by the same reductions. By applying the triangle inequality applied on Equation 1, Equation 2, and Equation 3, we obtain the desired Lemma statement. $\square$

4.3 Single-dealer Anonymity

The strictly weaker notion of anonymity than M-Anonymity is single-dealer anonymity (S-Anonymity), which intuitively states that the share values of an unauthorized set for a single sharing of a secret s does not reveal the party identities. We define S-Anonymity to be the hardest such task—distinguishing between shares of two unauthorized sets who differ only by at most one party.

Definition 4.4 (S-Anonymity). For $\mu : \mathbb{N} \rightarrow [0, 1]$, a secret sharing scheme (Share, Recon) for access structure Γ , secret space $\mathcal{S}$, is μ -S-anonymous if for all computationally unbounded (in the computational setting, non-uniform polynomial-sized) adversaries $\mathcal{A}$, for all sufficiently large $\lambda \in \mathbb{N}$, for all $B \in 2^{[n]} \setminus \Gamma_\lambda$, for all $B' \subseteq B$ such that $|B'| = |B| - 1$, for all $a \in [n]$ such that $B' \cup \{a\} \in 2^{[n]} \setminus \Gamma_\lambda$,

$$\left| \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_B) = 1 \right] - \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_{B' \cup \{a\}}) = 1 \right] \right| \leq \mu(\lambda).$$

Remark 4.3 (Non-triviality of S-anonymity). The above definition of S-anonymity requires the existence of at least two unauthorized sets of the same size. But what if an access structure contains a single unauthorized set? We note that this is only possible when this unauthorized set is a singleton (a single party), which is an uninteresting access structure. Indeed, any unauthorized set of size two or more implies (due to monotonicity) at least two unauthorized sets of smaller size, where the above definition is applicable.

Lemma 4.5 (M-Anonymity Implies S-Anonymity). A secret sharing scheme (Share, Recon) for $n = n(\lambda)$ parties that satisfies statistical (resp. computational) μ -M-Anonymity also satisfies statistical (resp. computational) $(2n\mu)$ -S-Anonymity.

Proof. Suppose that a secret sharing scheme (Share, Recon) satisfies statistical (resp. computational) μ -M-Anonymity. We proceed by a hybrid argument and we will show that for any adversary $\mathcal{A}$, for any secret s and for unauthorized sets B and $B' \subseteq B$, $|B'| = |B| - 1$, and for any $a \in [n]$ such that $B' \cup \{a\}$ is unauthorized,

$$\left| \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_B) = 1 \right] - \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_{B' \cup \{a\}}) = 1 \right] \right| \leq \mu(\lambda).$$

As a quick overview of this hybrid argument, we begin by considering the the case when the adversary $\mathcal{A}$ takes as input $(1^\lambda, s, B, B', a, \text{sh}_B)$. Let $b_1, \dots, b_{|B|}$ be an enumeration of all the parties of B . In each subsequent hybrid, we will replace one of the shares of $\text{sh}_B = (\text{sh}_{b_1}, \dots, \text{sh}_{b_{|B|}})$, say share sh_{b_i} , by a share $\text{sh}_{b_i}^{(i)}$ for party b_i where $\text{sh}^{(i)} \leftarrow \text{Share}(1^\lambda, s^{(i)})$ is a sharing of a random secret value $s^{(i)}$. At the end of $|B|$ -many hybrids, we will reach the setting where the adversary's view consists of $|B|$ -many shares where each share comes from a secret sharing of a random value. The statistical (resp. computational) indistinguishability between consecutive hybrids follows from a reduction from the statistical (resp. computational) M-Anonymity property. A symmetrical and otherwise identical argument shows that $\text{sh}_{B' \cup \{a\}}$ can be switched to shares where each shares come from a secret sharing of a random value. Therefore, we show only the first direction.

To prove our desired claim, it suffices to show the indistinguishability of the follow hybrids:

- Hybrid $H_0^{\mathcal{A}}(1^\lambda, s, B, B', a)$: Let $\text{sh} \leftarrow \text{Share}(1^\lambda, s)$. The adversary $\mathcal{A}$ takes as input $(1^\lambda, s, B, B', a, \text{sh}_B)$. The output of the hybrid $H_0(1^\lambda, \mathcal{A}, s, B, B', a)$ is the output of the adversary $\mathcal{A}$.

- For $i \in [|B|]$, define hybrid $H_i^{\mathcal{A}}(1^\lambda, s, B, B', a)$: Let $\text{sh} \leftarrow \text{Share}(1^\lambda, s)$. For $1 \leq j \leq i$, sample a uniform random secret $s^{(j)}$ and let $\text{sh}^{(j)} \leftarrow \text{Share}(1^\lambda, s^{(j)})$. The adversary $\mathcal{A}$ takes as input

$$\left(1^\lambda, s, B, B', a, \text{sh}_{b_1}^{(1)} \parallel \text{sh}_{b_2}^{(2)} \parallel \cdots \parallel \text{sh}_{b_i}^{(i)} \parallel \text{sh}_{\{b_i, b_{i+1}, \dots, b_{|B|}\}}\right).$$

The output of the hybrid $H_i^{\mathcal{A}}(1^\lambda, \mathcal{A}, s, B, B', a)$ is the output of the adversary $\mathcal{A}$.

Observe that in hybrid $H_{|B|}^{\mathcal{A}}(1^\lambda, s, B, B')$, the adversary $\mathcal{A}$'s view consists of the input

$$\left(1^\lambda, s, B, B', a, \text{sh}_{b_1}^{(1)} \parallel \text{sh}_{b_2}^{(2)} \parallel \cdots \parallel \text{sh}_{b_{|B|}}^{(|B|)}\right).$$

We claim that if there exists that any adversary $\mathcal{D}$ that satisfies

$$\left| \Pr \left[H_i^{\mathcal{D}}(1^\lambda, s, B, B', a) = 1 \right] - \Pr \left[H_{i+1}^{\mathcal{D}}(1^\lambda, s, B, B', a) = 1 \right] \right| > \delta(\lambda), \quad (4)$$

then we can construct an adversary $\mathcal{A}$ such that for infinitely many λ , there exists at least one secret s_1 such that for participant sets $C = \{b_{i+1}, \dots, b_{|B|}\}$, $b = b_{i+1}$, and $C' = C \setminus \{b\}$,

$$\left| \Pr_{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)} \left[\mathcal{A} \left(1^\lambda, s_0, s_1, C, C', b, \text{sh}_C^{(0)} \right) = 1 \right] - \Pr_{\substack{\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0) \\ \text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)}} \left[\mathcal{A} \left(1^\lambda, s_0, s_1, C, C', b, \text{sh}_{C'}^{(0)} \parallel \text{sh}_{\{b\}}^{(1)} \right) = 1 \right] \right| > \delta(\lambda)$$

The existence of this s_1 follows from a straightforward averaging argument. Since the difference between the view of $\mathcal{D}$ in the hybrids $H_i^{\mathcal{D}}$ and $H_{i+1}^{\mathcal{D}}$ consists of only a single share for the party b_{i+1} , where this share is generated by the same sharing of the secret s in the former hybrid $H_i^{\mathcal{D}}$ and where this share is generated by a fresh sharing of a uniform random secret chosen over the randomness of the hybrid/experiment $H_{i+1}^{\mathcal{D}}$, for these two hybrids to be distinguishable with probability at least $\delta(\lambda)$, there must exist at least one secret value s_1 for the uniformly sampled secret $s^{(i+1)}$, for which the distinguishing probability of $\mathcal{D}$ is at least $\delta(\lambda)$. If not, then for all values of $s^{(i+1)}$, $\mathcal{D}$ has strictly less than $\delta(\lambda)$ distinguishing advantage, a contradiction to the assumption in Equation 4. We have therefore established that for these particular values $s_1 = s_{1,\lambda}$, where we abuse notation by omitting λ when it is clear that a specific value of λ is being considered, $\mathcal{D}$ succeeds with probability at least $\delta(\lambda)$. Then, with probability over the remaining randomly chosen secrets $s^{(1)}, \dots, s^{(i)}$ and the random coins of the sharing algorithm Share of all the secrets $s^{(1)}, \dots, s^{(i)}, s_0, s_1$, we have that

$$\left| \Pr \left[\mathcal{D} \left(1^\lambda, s_0, C, C', b, \text{sh}_{b_1}^{(1)} \parallel \text{sh}_{b_2}^{(2)} \parallel \cdots \parallel \text{sh}_{b_i}^{(i)} \parallel \text{sh}_{b_{i+1}} \parallel \text{sh}_{b_{i+2}} \parallel \cdots \parallel \text{sh}_{|B|} \right) \right] - \Pr \left[\mathcal{D} \left(1^\lambda, s_0, C, C', b, \text{sh}_{b_1}^{(1)} \parallel \text{sh}_{b_2}^{(2)} \parallel \cdots \parallel \text{sh}_{b_i}^{(i)} \parallel \widehat{\text{sh}^{(i+1)}}_{b_{i+1}} \parallel \text{sh}_{b_{i+2}} \parallel \cdots \parallel \text{sh}_{|B|} \right) \right] \right| > \delta(\lambda) \quad (5)$$

where $\widehat{\text{sh}^{(i+1)}}_{b_{i+1}}$ is the b_{i+1} th share of a sharing of this particular choice of s_1 . This concludes the averaging argument.

Therefore, we construct an algorithm $\mathcal{A}$ that on input $(1^\lambda, s_0, s_1, C, C', b, X)$ generates exactly the view for $\mathcal{D}$ corresponding to either $H_i \left(1^\lambda, \mathcal{D}, s_0, C, C', b, \text{sh}_C^{(0)} \right)$ or $H_{i+1} \left(1^\lambda, \mathcal{D}, s_0, C, C', \text{sh}_C^{(0)} \right)$ by sampling random secrets $s^{(j)}$ for $1 \leq j \leq i$ and producing secret shares $\text{sh}^{(j)} \leftarrow \text{Share}(1^\lambda, s^{(j)})$. The adversary $\mathcal{A}$ will output

$$\mathcal{D} \left(1^\lambda, s_0, C, C', b, \text{sh}_{b_1}^{(1)} \parallel \text{sh}_{b_2}^{(2)} \parallel \cdots \parallel \text{sh}_{b_i}^{(i)} \parallel X \right).$$

By Equation 5, the distinguishing advantage of $\mathcal{A}$ is therefore at least $\delta(\lambda)$. Finally observe, that if $\mathcal{D}$ is computationally unbounded or computationally bounded, then $\mathcal{A}$ is respectively computationally unbounded or computationally bounded. This concludes the proof that for all $i \in \{0, \dots, |B| - 1\}$, H_i and H_{i+1} are δ -indistinguishable.

A symmetric sequence of $|B|$ many hybrids allows us to start from a hybrid that outputs $\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_{B' \cup \{a\}})$ and end up at $H_{|B|}^{\mathcal{A}}$. By the triangle inequality,

$$\left| \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_B) = 1 \right] - \Pr_{\text{sh} \leftarrow \text{Share}(1^\lambda, s)} \left[\mathcal{A}(1^\lambda, s, B, B', a, \text{sh}_{B' \cup \{a\}}) = 1 \right] \right| \leq 2 \cdot |B| \cdot \delta(\lambda).$$

□

5 Information-Theoretic U-FASS for DNFs

We now present two information-theoretic U-FASS schemes for access structures defined by an arbitrary monotone DNF formula. We assume without loss of generality that there are no minterms of size 1¹². The first scheme we present is essentially a large-field variant of the construction in [PO20]. While this scheme is conceptually identical to that in [PO20], the exact implementation details differ since we focus on U-anonymity, which is not studied in the previous work. Moreover, our notions of share anonymity syntactically differ from those in [PO20]. Since we prove strong lower bounds on share size in this work, we additionally analyze the per-party share size, which is not explicitly studied in the previous work. In a nutshell, our use of a large-field variant allows us to reduce the share size towards optimality while maintaining the statistical U-anonymity property in comparison to the prior work.

5.1 Statistical U-FASS for DNFs

We first present a U-FASS scheme for DNFs with perfect correctness, perfect secrecy, and statistical U-anonymity by taking a large field version of [PO20].

Theorem 5.1 (Statistical U-FASS for DNFs, Implicit in [PO20]). *For any monotone DNF formula $\Phi(x_1, \dots, x_n) = m_1 \vee m_2 \vee m_3 \vee \dots \vee m_\ell$ and statistical security parameter λ , there exists a U-FASS scheme for the access structure defined by Φ that satisfies*

1. *Perfect correctness and secrecy.*
2. *$2^{-\lambda}$ -U-anonymity.*
3. *A per-party share size of $O(\ell n(\log \ell + \lambda))$ bits for 1-bit secrets.*
4. *Both Share and Recon can be implemented in time $\text{poly}(n, \ell, \lambda)$.*

We prove Theorem 5.1 using the following construction, Construction 5.2.

Construction 5.2. *Let $\Phi(x_1, \dots, x_n) = m_1 \vee m_2 \vee m_3 \vee \dots \vee m_\ell$ be a monotone DNF formula and λ a statistical security parameter. The construction of the secret sharing scheme for the access structure Φ begins by first setting $q \geq \ell \cdot (2^\lambda + 1)$ such that q is not a power of two and not a power of three.*

- $\text{Share}(s \in \{0, 1\})$:

¹²Any minterm of size one is removed from the formula Φ , and the corresponding party is given the secret s . This does not affect correctness, nor secrecy, nor anonymity.

1. For $i \in [\ell]$, let $m_i = x_{i_1} \wedge x_{i_2} \wedge \dots \wedge x_{i_{t_i}}$.

(a) Sample $n - 1$ random linearly independent vectors $\{\mathbf{v}_j^{(i)} \in \mathbb{F}^{n-1}\}_{j \in [n], j \neq i_1}$.

(b) Set

$$\mathbf{v}_{i_1}^{(i)} \leftarrow (s + 1) \cdot \mathbf{v}_{i_2}^{(i)} + \sum_{j=3}^{t_i} \mathbf{v}_{i_j}^{(i)} \in \mathbb{F}^{n-1}.$$

2. Set the share of party $k \in [n]$ to be $\text{sh}_k \leftarrow (\mathbf{v}_k^{(1)}, \dots, \mathbf{v}_k^{(\ell)})$.

• Recon($\text{sh}_1, \dots, \text{sh}_t$):

1. For $k \in [t]$, parse sh_k as a tuple of vectors $(\mathbf{v}_k^{(i)})_{i \in [\ell]}$.

2. Let $i \in [\ell]$ be such that $\mathbf{v}_1^{(i)}, \dots, \mathbf{v}_t^{(i)}$ are linearly dependent. If no such i exists, then output $\perp$. Otherwise, linear algebraically find the minimal linearly dependent set S of size $t' \leq t$. For all t' ($t' - 1$) choices of $\mathbf{w}_1, \mathbf{w}_2$ from S , arbitrarily set the remaining vectors $\mathbf{w}_3, \dots, \mathbf{w}_{t'}$ to the remaining distinct vectors in S , and output 0 if

$$-\mathbf{w}_1 + \sum_{j=2}^{|S|} \mathbf{w}_j = 0. \quad (6)$$

Otherwise, output 1 if

$$-\mathbf{w}_1 + 2\mathbf{w}_2 + \sum_{j=3}^{|S|} \mathbf{w}_j = 0. \quad (7)$$

Remark 5.1 (Reconstruction Time). Checking the linear independence of k -many vectors can be done with $O(k^\omega)$ field operations for linear algebra constant $\omega < 3$. For each minterm index $i \in [\ell]$, Step 2 takes at most t linear-independence checks to identify the minimal sized linearly dependent set. Then for each of the $O(t^2)$ pairs $(\mathbf{w}_1, \mathbf{w}_2)$, the algorithm must perform at most $O(t)$ field operations to verify which of the two linear dependencies holds. Therefore, Recon requires in the worst-case $O(\ell \cdot (t \cdot t^\omega + t^2 \cdot t)) = O(\ell \cdot n^4)$ field operations since $t \leq n$.

Remark 5.2 (Share Size). The share size in Construction 5.2 of each party is $\ell(n - 1)$ many field elements or $\ell(n - 1) \log q$ many bits. We have the requirement that the field size $q \geq \ell \cdot (2^\lambda + 1)$, and a standard fact is that there exists at least one prime $q \in [n, 2n]$ for any positive integer n , so this gives a per-party share size of $O(\ell n(\log \ell + \lambda))$.

Proof of Theorem 5.1. The statement follows from Lemma 5.3, Lemma 5.3, and Lemma 5.5. □

Lemma 5.3 (Perfect Correctness). *Construction 5.2 satisfies perfect correctness.*

Proof. For any unauthorized set of shares B , there does not exist any minterm whose corresponding set of vectors from B has a linear dependence. Therefore the reconstruction algorithm in Construction 5.2 will always output $\perp$.

For any authorized set A of t many shares, there exists at least one minterm m_i of degree t_i whose corresponding set of vectors $S = \{\mathbf{w}_1, \dots, \mathbf{w}_t\}$ in the shares of A are linearly dependent because $\{\mathbf{v}_{i_j}^{(i)}\}_{j \in [t_i]} \subseteq S$. Moreover, $\{\mathbf{v}_{i_j}^{(i)}\}_{j \in [t_i]}$ is the minimal linearly dependent subset of every linearly dependent subset of S by

construction, so it can be efficiently found linear algebraically. The ordering $(\mathbf{w}_j = \mathbf{v}_{i_j}^{(i)})_{j \in [t_i]}$ will satisfy exactly one of the two equations, Equation 6 and Equation 7, so for that ordering the reconstruction algorithm outputs the correct secret.

We now argue that the algorithm can only output the correct secret value. That is, no other ordering results in the reconstruction algorithm outputting $1 - s$. It suffices to consider the case when $s = 0$.

1. First consider any minterm of degree two, say $m_i = x_{i_1} \wedge x_{i_2}$. By construction we have that $-\mathbf{v}_{i_1}^{(i)} + \mathbf{v}_{i_2}^{(i)} = 0$, which implies that $\mathbf{v}_{i_1}^{(i)} - \mathbf{v}_{i_2}^{(i)} = 0$. Observe that $-\mathbf{v}_{i_1}^{(i)} + 2\mathbf{v}_{i_2}^{(i)} = 0$ is impossible because together with the previous constraint, this would imply that $\mathbf{v}_{i_2}^{(i)} = 0$, contradicting the fact that $\mathbf{v}_{i_2}^{(i)}$ was sampled as part of linearly independent set of vectors. For the same reason, the equation $2\mathbf{v}_{i_1}^{(i)} - \mathbf{v}_{i_2}^{(i)} = 0$ cannot simultaneously hold. Therefore, when $s = 0$, regardless of the choice of ordering of vectors $(\mathbf{w}_1, \mathbf{w}_2)$, only Equation 6 can be satisfied.
2. This argument extends to any minterm $m_i = x_{i_1} \wedge \dots \wedge x_{i_{t_i}}$ of degree $t_i > 2$. By construction we have that

$$-\mathbf{v}_{i_1}^{(i)} + \sum_{j=2}^{t_i} \mathbf{v}_{i_j}^{(i)} = 0.$$

Assume for sake of contradiction that there exists a vector $\mathbf{c} = (c_1, \dots, c_{t_i})$ that is a permutation of the length t_i vector $(-1, 2, 1, \dots, 1)$ such that

$$\sum_{j=1}^{t_i} c_j \cdot \mathbf{v}_{i_j}^{(i)} = 0.$$

Then there exists a non-zero vector $\mathbf{d} = (d_1, \dots, d_{t_i}) \in \{0, \pm 1, \pm 2, \pm 3\}^{t_i}$ that is in the linear span $\text{Span}(\mathbf{c}, (-1, 1, \dots, 1))$ such that $d_1 = 0$. Namely, take $\mathbf{d} = \mathbf{c} + c_1 \cdot (-1, 1, \dots, 1)$. By construction of $\mathbf{d}$, we have $d_1 = 0$ and there exists at least one index $i \neq 1$ such that $d_i \in \{\pm 1, \pm 2, \pm 3\}$. More specifically, if $c_1 = \pm 1$, then for the index i such that $c_i = 2$, we have $d_i \neq 0$. If $c_1 = 2$, then $d_i \in \{\pm 1, \pm 2, \pm 3\}$ for all $i \neq 1$. Then observe that

$$\sum_{j=1}^{t_i} d_j \cdot \mathbf{v}_{i_j}^{(i)} = \sum_{j=2}^{t_i} d_j \cdot \mathbf{v}_{i_j}^{(i)} = 0$$

because $\mathbf{d}$ is in the linear span of $\mathbf{c}$ and $(-1, 1, \dots, 1)$. Since $(d_2, \dots, d_{t_i})$ is a non-zero vector (here, we use the fact that the field of not of characteristic two nor three), this non-zero linear combination implies the linear dependence of $\{\mathbf{v}_{i_j}^{(i)}\}_{j \in \{2, \dots, t_i\}}$ a contradiction. Therefore, no such $\mathbf{c}$ can exist, so Equation 7 cannot be satisfied regardless of the ordering $(\mathbf{w}_1, \dots, \mathbf{w}_{|S|})$ and the reconstruction algorithm cannot incorrectly output 1.

The case of $s = 1$ follows by symmetry. □

Lemma 5.4 (Perfect Secrecy). *Construction 5.2 satisfies perfect secrecy.*

Proof. The unauthorized sets in the case that $s = 0$ are distributed identically to the case of $s = 1$. In both cases, for each minterm, say with index $i \in [\ell]$, the corresponding set of vectors $\{\mathbf{v}_j^{(i)}\}_{j \in B}$, where B is the unauthorized set of parties, are distributed as uniformly chosen linearly independent vectors of $\mathbb{F}_q^{n-1}$. The vectors corresponding to each minterm i are independently sampled from minterm i' for $i \neq i'$. □

Lemma 5.5 (Statistical U-anonymity). *Construction 5.2 satisfies $2^{-\lambda}$ -statistical U-anonymity.*

Proof. As noted in the proof of secrecy, for any unauthorized set, regardless of the value of s , the set of vectors corresponding to a minterm in the unauthorized set are distributed as uniformly chosen linearly independent vectors of $\mathbb{F}_q^{n-1}$, and each minterm's corresponding set of vectors are independently sampled from another minterm's corresponding set of vectors.

The probability that a uniformly drawn set of k vectors from $\mathbb{F}_q^{n-1}$ are linearly independent is

$$\frac{(q^{n-1} - 1)(q^{n-1} - q) \cdots (q^{n-1} - q^{k-1})}{q^{k(n-1)}} = \prod_{i=0}^{k-1} \left(1 - q^{i-(n-1)}\right).$$

A non-trivial unauthorized set can be as large as $n-1$, so $k \leq n-1$. Of these possible values of k , the lowest probability of linear independence is when $k = n-1$, in which case the above probability is bounded below by

$$1 - \sum_{i=1}^{n-1} q^{-i} \geq 1 - \frac{1}{q-1}.$$

If W is the number of $(n-1)$ linearly independent vectors in $\mathbb{F}_q^{n-1}$, then we conclude

$$W \geq q^{(n-1)^2} \cdot \left(1 - \frac{1}{q-1}\right).$$

Let P denote the uniform distribution over all $(n-1)$ linearly independent vectors over $\mathbb{F}_q^{n-1}$ and let Q denote the uniform distribution over $(\mathbb{F}_q^{n-1})^{n-1}$. A direct statistical distance computation shows that the statistical distance is

$$\Delta(P, Q) = 1 - \frac{W}{q^{(n-1)^2}} \leq \frac{1}{q-1}.$$

There are ℓ -many minterms, whose corresponding vectors are sampled independently of each other. Let $P_1, P_2, \dots, P_\ell$ be independent and identical distributions to P and similarly $Q_1, Q_2, \dots, Q_\ell$ be independent and identical distributions to Q . Since statistical distance of independent samples is sub-additive, the total statistical distance from

$$\Delta((P_1, \dots, P_\ell), (Q_1, \dots, Q_\ell)) \leq \frac{\ell}{q-1} \leq 2^{-\lambda}.$$

□

Remark 5.3 (Perfect U-FASS and Statistical Correctness). One can modify the above construction to sample uniform random vectors $\{\mathbf{v}_{i_j} \in \mathbb{F}_q^{n-1}\}_{j \in \{2, \dots, n\}}$ instead of linearly independent ones. The resulting scheme has perfect secrecy and perfect U-FASS, where the latter property immediately implies the former, but only statistical correctness with correctness parameter $1 - \frac{\ell}{q} \geq 1 - \frac{1}{2^\lambda}$.

5.2 Main Construction with Near-Optimal Share Sizes

We now present our main positive information-theoretic contribution that obtains near-optimal share sizes. We modify Construction 5.2 so that for each minterm $m_i = x_{i_1} \wedge x_{i_2} \wedge \cdots \wedge x_{i_{t_i}}$, instead of sampling a uniform random linearly independent set of $n-1$ vectors $\{\mathbf{v}_j^{(i)} \in \mathbb{F}_q^{n-1}\}_{j \in [n], j \neq i_1}$, we will sample $\{\mathbf{v}_j^{(i)} \in \mathbb{F}_q^{t_i \log n}\}_{j \in [n], j \neq i_1}$ as $n-1$ many independently sampled uniform random vectors. Each minterm of degree t_i instead contributes to each party's share size on the order of $O(t_i(\log n)(\log q))$ bits instead of $O(n \log q)$ bits. This modification will satisfy statistical correctness, perfect secrecy, and perfect U-anonymity, however, we will lose efficient reconstruction because there will no longer be a common minimal linearly dependent set that is a subset of every linearly dependent set.

Theorem 5.6 (Reduced Share Size for U-FASS for DNFs). *For any monotone DNF formula $\Phi(x_1, \dots, x_n) = m_1 \vee m_2 \vee \dots \vee m_\ell$ where $t := \max_i (\text{degree}(m_i))$ and statistical correctness parameter λ , there exists a U-FASS scheme for the access structure defined by Φ that satisfies*

1. $(1 - 2^{-\lambda})$ -correctness.
2. Perfect secrecy and perfect U-anonymity.
3. A per-party share size of $O(\ell t(\log n)(\lambda + \log n + \log \ell))$ bits for 1-bit secrets.
4. Share can be implemented in time $\text{poly}(n, \ell, \lambda)$.

We prove Theorem 5.6 using the following construction, Construction 5.7.

Construction 5.7 (Near-optimal Share-size Construction). *Let $\Phi(x_1, \dots, x_n) = m_1 \vee m_2 \vee m_3 \vee \dots \vee m_\ell$ be a monotone DNF formula where $t := \max_i (\text{degree}(m_i))$ and λ a statistical correctness parameter. The alternative share-size construction of a perfect U-FASS scheme for the access structure Φ begins by setting $\mathbb{F} = \mathbb{F}_q$ for $q \geq \ell \cdot n \cdot 2^\lambda$.*

• **Share**($s \in \{0, 1\}$):

1. For $i \in [\ell]$, let $m_i = x_{i_1} \wedge x_{i_2} \wedge \dots \wedge x_{i_{t_i}}$.

(a) Sample independently $n - 1$ uniform random vectors $\left\{ \mathbf{v}_j^{(i)} \in \mathbb{F}^{t \log n} \right\}_{j \in [n], j \neq i_1}$.

(b) Set

$$\mathbf{v}_{i_1}^{(i)} \leftarrow (s + 1) \cdot \mathbf{v}_{i_2}^{(i)} + \sum_{j=3}^{t_i} \mathbf{v}_{i_j}^{(i)} \in \mathbb{F}^{t \log n}.$$

2. Set the share of party $k \in [n]$ to be $\text{sh}_k \leftarrow (\mathbf{v}_k^{(1)}, \dots, \mathbf{v}_k^{(\ell)})$.

• **Recon**($\text{sh}_1, \dots, \text{sh}_t$):

1. For $k \in [t]$, parse sh_k as a tuple of vectors $(\mathbf{v}_k^{(i)})_{i \in [\ell]}$.
2. Let $i \in [\ell]$ be such that $\mathbf{v}_1^{(i)}, \dots, \mathbf{v}_t^{(i)}$ are linearly dependent. If no such i exists, then output $\perp$. For all t_i -size subsets S of $\mathbf{v}_1^{(i)}, \dots, \mathbf{v}_t^{(i)}$, for all orderings of S , $(\mathbf{w}_1, \dots, \mathbf{w}_{t_i})$, output 0 if

$$-\mathbf{w}_1 + \sum_{j=2}^{|S|} \mathbf{w}_j = 0. \tag{8}$$

Otherwise, output 1 if

$$-\mathbf{w}_1 + 2\mathbf{w}_2 + \sum_{j=3}^{|S|} \mathbf{w}_j = 0. \tag{9}$$

Proof of Theorem 5.6. The statement follows from Lemma 5.8, Lemma 5.9, Lemma 5.10. □

Lemma 5.8 (Correctness of Near-optimal Construction). *Construction 5.7 satisfies $(1 - 2^{-\lambda})$ correctness.*

Proof. The probability that an uniform random set of t_i -many vectors sampled from $\mathbb{F}^{t \log n}$ is linearly dependent is at most $q^{t-t \log n}$ since $t_i \leq t$. Any t_i -subset of $\left\{ \mathbf{v}_j^{(i)} \right\}_{j \in [n]}$ corresponding to a minterm $m_i = x_{i_1} \wedge \dots \wedge x_{i_{t_i}}$ is distributed as a uniform random set of t_i -many vectors sampled from $\mathbb{F}^{t \log n}$ if this subset is not the planted linearly dependent set of vectors $\{\mathbf{v}_{i_1}^{(i)}, \dots, \mathbf{v}_{i_{t_i}}^{(i)}\}$. There are at most n^t many such non-planted t_i -subsets. By a union bound, the probability that there exists a minterm m_i for which there exists a non-planted t_i -sized linearly dependent subset is $\ell \cdot n^t \cdot q^{t-t \log n}$. For the choice of $q \geq \ell \cdot n \cdot 2^\lambda$, we have $\ell \cdot n^t \cdot q^{t-t \log n} \leq 2^{-\lambda-1} \leq 2^{-\lambda}$.

In other words, for any unauthorized set of shares B , with probability at least $1 - 2^{-\lambda}$, the reconstruction algorithm in Construction 5.7 will output $\perp$.

For any authorized set of shares A , with probability at least $1 - 2^{-\lambda-1}$, for each minterm m_i that is satisfied, the only linear dependent sets of size t_i are the planted linear dependent sets. For the planted linear dependent set, only one of Equation 8 and Equation 9 can hold regardless of the ordering on the vectors. To argue this, if both Equation 8 holds for some ordering $(\mathbf{w}_1, \dots, \mathbf{w}_{t_i})$ and Equation 9 holds for some ordering $(\mathbf{w}'_1, \dots, \mathbf{w}'_{t_i})$, then some proper subset of these vectors are linearly dependent (see the proof of Lemma 5.3). This strict subset of vectors is distributed identically as independently sampled uniform random vectors over $\mathbb{F}^{t \log n}$. With the same choice of q , one can show by a nearly identical union bound that there does not exist any minterm m_i for which any *at most* $t_i - 1$ -sized subset of the corresponding vectors to m_i are linearly dependent with probability at least $1 - 2^{-\lambda-1}$ for any q of our choice. Therefore, with probability at least $1 - 2^{-\lambda}$, the reconstruction algorithm will output the correct value of the secret. $\square$

Lemma 5.9 (Perfect Secrecy of Near-optimal Construction). *Construction 5.7 satisfies perfect secrecy.*

Proof. Regardless of the choice of s , the unauthorized set of shares is such that for every minterm m_i , the corresponding vectors in the unauthorized set are distributed as independent random vectors from $\mathbb{F}^{t \log n}$. $\square$

Lemma 5.10 (Perfect U-anonymity of Near-optimal Construction). *Construction 5.7 satisfies perfect U-anonymity.*

Proof. Regardless of the choice of s , the unauthorized set of shares is such that for every minterm m_i , the corresponding vectors in the unauthorized set are distributed as independent random vectors from $\mathbb{F}^{t \log n}$. $\square$

Remark 5.4 (Tightness of the Upper Bound for Certain Families of DNFs). The $(1, n)$ -star graph access structure (Def. 6.4) can be formulated as a DNF with $n + 1$ variables and n many minterms each of degree 2. Construction 5.7 with correctness parameter $\lambda = \log^2 n$ has $(1 - n^{\log n})$ -correctness, perfect secrecy and perfect U-anonymity, and a per-party share size of $\tilde{O}(n)$ for 1-bit secrets, matching up to polylogarithmic factors the lower bound on the per-party share size of $\Omega(n)$ for 1-bit secrets.

The same tightness up to polylogarithmic factors between the upper and lower bound on the total share size holds for the DNF corresponding to the (k, n) -star graph access structure (Def. 6.4). See Theorem 1.1.

Remark 5.5 (Simultaneous Efficient Reconstruction and Near-Optimal Share Size for Long Secrets). A straightforward combination of the two schemes gives a U-FASS scheme that for all monotone DNFs that has both efficient reconstruction and near-optimal per-party information rate¹³. We give the formal theorem statement in Corollary 5.11.

The observation motivating this construction of the hybrid scheme is that the non-optimal share size construction for DNFs, Construction 5.2, has an efficient reconstruction algorithm that identifies a minimal

¹³Again, information rate refers to share size in bits per bit of the secret.

authorized set as an intermediate step. The hybrid construction for T -bit secrets $\mathbf{s} = (s_1, \dots, s_T) \in \{0, 1\}^T$ gives each party the corresponding share for a sharing of 0 and gives each party T -many shares produced by using the near-optimal share size construction for DNFs, Construction 5.7, for each bit of the secret $\mathbf{s}$. To reconstruct, the parties can first run the efficient reconstruction algorithm on the shares corresponding to a sharing of 0 to identify which parties form a minimal authorized set of some size t' . Then, to recover each bit s_i of the secret $\mathbf{s}$, the reconstruction process can brute force a bit of the secret s_i using a $O(t')$ -time brute forcing of the secret s_i described exactly in the Recon algorithm of Construction 5.7.

For sufficiently large polynomially bounded T , say $T = \omega(n/t)$ where t is the maximum size of any minterm, the per-party share size is asymptotically dominated by the T -many shares produced using the near-optimal share size construction, Construction 5.7. Therefore, the share is asymptotically dominated by $O(T \cdot \ell t (\log n)(\lambda + \log n + \log \ell))$, which results in per-party information rate $\tilde{O}(\ell t \lambda)$. This is a near-optimal information rate due to Corollary 6.19.

Corollary 5.11 (Hybrid Construction for Long Secrets). *For any monotone DNF formula $\Phi(x_1, \dots, x_n) = m_1 \vee m_2 \vee \dots \vee m_\ell$ where $t := \max_i (\text{degree}(m_i))$ and statistical correctness parameter λ , and statistical security parameter η , for secrets $\mathbf{s} \in \{0, 1\}^T$ such that $T = \omega(n/t)$, there exists a U-FASS scheme for the access structure defined by Φ that satisfies*

1. $(1 - T \cdot 2^{-\lambda})$ -correctness.
2. Perfect secrecy.
3. $(1 - 2^\eta)$ U-anonymity.
4. A per-party information rate of $\tilde{O}(\ell t \lambda)$.
5. Both Recon and Share can be implemented in time $\text{poly}(n, \ell, T, \eta, \lambda)$.

Proof. Use Construction 5.2 to share 0 to obtain shares $z_1, \dots, z_n$. For each $i \in [T]$, share the i -th bit of the secret $\mathbf{s} = (s_1, \dots, s_T) \in \{0, 1\}^T$ with Construction 5.7 to share s_i to obtain shares $x_1^{(i)}, \dots, x_n^{(i)}$. For $j \in [n]$ set the j th party's share to be $\text{sh}_j \leftarrow (z_j, x_j^{(1)}, x_j^{(2)}, \dots, x_j^{(T)})$. To reconstruct on a set of shares $(\text{sh}_i)_{i \in S \subseteq [n]}$, run the algorithm Recon from Construction 5.2 on the set of shares. This algorithm as an intermediate computation identifies the indices $\mathcal{J}$ of a minimal authorized set of size t . For $i \in [T]$, run the reconstruction algorithm Recon from Construction 5.7 on these share $(x_j^{(i)})_{j \in \mathcal{J}}$; this step is efficient at it requires at most $O(t^2)$ many attempts of selecting the correct choice of $\mathbf{w}_1$ and $\mathbf{w}_2$ in Equation 8 and Equation 9. The correctness follows by applying a union bound with the $(1 - 2^{-\lambda})$ correctness of Construction 5.7 for each $i \in [T]$. The perfect secrecy follows immediately, We have perfect U-anonymity for the share values $\{x_j^{(i)}\}_{j \in [n]}$ for $i \in [T]$, and we have $(1 - 2^\eta)$ U-anonymity for the shares $\{z_j\}_{j \in [n]}$, resulting in an overall $(1 - 2^\eta)$ -U-anonymity. Finally the per-party information rate is given by:

$$O\left(\frac{\ell n (\log \ell + \eta)}{T} + \ell t (\log n)(\lambda + \log n + \log \ell)\right)$$

which is dominated by the second term when $T = \omega(n/t)$. □

6 Information-theoretic Lower Bounds

6.1 Linear Anonymous Reconstruction is Impossible for Threshold Schemes

We now briefly show that any fully anonymous secret sharing scheme cannot have linear anonymous reconstruction. First we recall the definition of linear secret sharing schemes and then we precisely state

what we mean by linear anonymous reconstruction.

Definition 6.1 (Linear Secret Sharing Scheme [Bei96]). *Let $\mathbb{F}$ be any finite field and let $\mathcal{S} \subseteq \mathbb{F}$ be the secret domain and let the set of parties be $[n]$. A secret sharing scheme Π for access structure $\Gamma \subseteq [n]$ is a linear secret sharing scheme if the two following properties hold:*

1. Every party's share $\mathbf{v}_i$ is a vector over $\mathbb{F}$.
2. **[Linear Reconstruction]** *For any authorized set $A = \{a_1, \dots, a_\ell\} \in \Gamma$, there exists a set of vectors $\{\mathbf{b}_{A,i}\}_{i \in [\ell]}$ such that for all secrets $s \in \mathcal{S}$, and over all possible choices of randomness used to produce the shares $\{\mathbf{v}_i\}_{i \in [n]}$ from s , the shares satisfy $s = \sum_{i \in [\ell]} \mathbf{b}_{A,i} \cdot \mathbf{v}_{a_i}$.*

Observe above that the notion of linear reconstruction implicitly assumes that the reconstruction algorithm can identify the party identity of each share, that is, it is able to match $\mathbf{b}_{A,i}$ to $\mathbf{v}_{a_i}$. In the anonymous reconstruction setting, the party identities are hidden, which implies two things:

1. The reconstruction algorithm must use the same set of vectors $\{\mathbf{b}_i\}_{i \in [\ell]}$ for all authorized sets of size ℓ .
2. For any authorized set of size ℓ , the reconstruction algorithm must successfully recover the secret s for any permutation π on the position of these ℓ shares.

Definition 6.2 (Linear Anonymous Reconstruction). *We use the same syntax as in Definition 6.1. For every $\ell \in [n]$, there exists a set of vectors $\{\mathbf{b}_{\ell,i}\}_{i \in [\ell]}$ such that the following holds. For any authorized sequence $A = (a_1, \dots, a_\ell)$, for all secrets $s \in \mathcal{S}$, and for all choices of randomness used to produce the shares $\{\mathbf{v}_i\}_{i \in [n]}$ from s , the shares satisfy $s = \sum_{i \in [\ell]} \mathbf{b}_{\ell,i} \cdot \mathbf{v}_{a_i}$.*

We now show that this property is too much to ask for in a (t, n) -threshold linear secret sharing scheme for any $n > t > 1$.

Theorem 6.3. *For any finite field $\mathbb{F}$, for any non-trivial secret space $\mathcal{S} \subseteq \mathbb{F}$, for n parties, there does not exist a linear secret sharing scheme with linear anonymous reconstruction for the (t, n) -threshold access structure for $n > t > 1$.*

Proof. It suffices to consider the $(2, 3)$ -threshold setting. Suppose there existed two vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ such that for any sharing $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ of any secret s , for all $i \neq j \in [3]$,

$$\mathbf{b}_1 \mathbf{v}_i + \mathbf{b}_2 \mathbf{v}_j = s$$

Then observe that for all $i \neq j \neq k$

$$\begin{aligned} (\mathbf{b}_1 \mathbf{v}_i + \mathbf{b}_2 \mathbf{v}_k) - (\mathbf{b}_1 \mathbf{v}_j + \mathbf{b}_2 \mathbf{v}_k) &= \mathbf{b}_1 (\mathbf{v}_i - \mathbf{v}_j) = 0 \\ (\mathbf{b}_1 \mathbf{v}_k + \mathbf{b}_2 \mathbf{v}_i) - (\mathbf{b}_1 \mathbf{v}_k + \mathbf{b}_2 \mathbf{v}_j) &= \mathbf{b}_2 (\mathbf{v}_i - \mathbf{v}_j) = 0. \end{aligned}$$

In other words,

$$\begin{aligned} \mathbf{b}_1 \mathbf{v}_1 &= \mathbf{b}_1 \mathbf{v}_2 = \mathbf{b}_1 \mathbf{v}_3 \\ \mathbf{b}_2 \mathbf{v}_1 &= \mathbf{b}_2 \mathbf{v}_2 = \mathbf{b}_2 \mathbf{v}_3. \end{aligned}$$

But this is a contradiction since this implies that $\mathbf{v}_1$ alone is authorized since

$$(\mathbf{b}_1 + \mathbf{b}_2) \mathbf{v}_1 = \mathbf{b}_1 \mathbf{v}_1 + \mathbf{b}_2 \mathbf{v}_2 = s.$$

The exact same argument can be made for any (t, n) where $1 < t < n$. Namely, for any (t, n) reconstruction, we will instead have the expression that for any distinct t indices $\{i_1, \dots, i_t\} \subseteq [n]$,

$$\sum_{j=1}^t \mathbf{b}_j \mathbf{v}_{i_j} = s.$$

Then observe that by taking the appropriate differences we again obtain that for all $i \in [t]$

$$\mathbf{b}_i \mathbf{v}_1 = \dots = \mathbf{b}_i \mathbf{v}_n.$$

Therefore, any single share can reconstruct the secret. □

6.2 Share Size Lower Bounds

In this section, we will concentrate on the (k, n) -star graph access structure and show lower bounds on the per-party share size of any M-FASS scheme for this access structure. Note that every U-FASS scheme is an M-FASS scheme, so these lower bounds apply to every U-FASS scheme. These lower bounds are tight and match the upper bound given in Construction 5.7 up to polylogarithmic factors.

Definition 6.4 ((k, n) -star Access Structure). *A (k, n) -star access structure consists of $kn + 1$ parties. One party is the center party. The rest of the parties are partitioned into k equal-sized sets of cardinality n , $P_1, \dots, P_k$. A minimal authorized set consists of the center party, and one party from each of $P_1, \dots, P_k$.*

Remark 6.1 (DNF Formulation). The DNF formulation of a (k, n) -star access structure consists of a DNF with $kn + 1$ variables and n^k many minterms corresponding to the minimal authorized sets of parties from the (k, n) -star access structure.

We provide an informal overview of the results given in this section:

1. In Section 6.2.1, we introduce the entropic versions of the definition of correctness, secrecy, and U-Anonymity/M-Anonymity. We show that the perfect non-entropic versions of correctness, secrecy, and anonymity are equivalent to the perfect entropic versions, and that the imperfect version of correctness implies the imperfect version of entropic correctness, establishing the applicability of the lower bounds to any information-theoretic scheme achieving imperfect correctness, perfect secrecy, and perfect U-Anonymity/M-Anonymity.
We additionally show that imperfect U-Anonymity/M-Anonymity implies imperfect entropic anonymity. For a detailed discussion on this extension, see Remark 6.4.
2. In Section 6.3, we prove a lower bound of $\Omega(n)$ on the per-party share size of any M-FASS scheme for sharing a 1-bit secret with (im)perfect correctness, (im)perfect statistical secrecy, and (im)perfect statistical U-Anonymity/M-Anonymity for the $(1, n)$ -star access structure. The proof technique presented here is the template for all subsequent lower bounds.
3. In Section 6.4, we prove a lower bound of $\Omega(n^2)$ on the per-party share size of any M-FASS scheme for sharing 1-bit secrets with (im)perfect correctness, (im)perfect statistical secrecy, and (im)perfect statistical U-Anonymity/M-Anonymity for the $(2, n)$ -star access structure by introducing a way to bootstrap the previously introduced proof technique.

4. In Section 6.5, we prove a lower bound of $\Omega(n^k)$ on the per-party share size of any M-FASS scheme for sharing 1-bit secrets with (im)perfect correctness, (im)perfect¹⁴ statistical secrecy, (im)perfect¹⁵ statistical U-Anonymity/M-Anonymity for the (k, n) -star access structure by inductively bootstrapping the base technique introduced.

6.2.1 Entropic Correctness, Secrecy, and Anonymity

Perfect entropic correctness and perfect entropic secrecy are standard notions used to prove lower bounds in the study of standard perfect secret sharing [KGH83, CSGV91], e.g. see the survey of Beimel [Bei11]. In this section, we review and extend these entropic notions beyond the perfect setting to allow the lower bounds to apply to secret sharing schemes with imperfect entropic correctness and imperfect entropic secrecy. Moreover, we introduce entropic anonymity, which captures the conditional independence of random variables.

Entropic Correctness First, we generalize perfect entropic correctness to $(1 - \epsilon)$ -entropic correctness.

Definition 6.5 ($(1 - \epsilon)$ -Entropic Correctness). *Let S be random variable that is uniformly distributed over some secret space. A secret sharing scheme satisfies $(1 - \epsilon)$ -entropic correctness if for all authorized set of shares represented by random variable A , $H(S | A) \leq \epsilon$. Equivalently, $I(A; S) \geq H(S) - \epsilon$.*

Lemma 6.6 (Definitional Equivalence of Correctness). *If a secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ satisfies $(1 - \epsilon)$ -correctness (Def. 3.3), then Π satisfies $(1 - (H(\epsilon) + \epsilon \cdot \log(|S| - 1)))$ -entropic correctness (Def. 6.5) in which the secret value is sampled from the uniform distribution over the set S .*

Conversely, if Π satisfies $(1 - \eta)$ -entropic correctness (Def. 6.5) in which the secret value is sampled from the uniform distribution over the set S , then Π satisfies $2^{-\eta}$ -correctness (Def. 3.3).

Remark 6.2. A scheme satisfies perfect correctness if and only if it satisfies perfect entropic correctness. Moreover, whenever the imperfect correctness holds for a correctness parameter tending to 1 as $n \rightarrow \infty$, the scheme also satisfies imperfect entropic correctness for a parameter tending to 1 as $n \rightarrow \infty$.

Proof of Lemma 6.6. Assume Π satisfies $(1 - \epsilon)$ -correctness (Def. 3.3) for all authorized sets B . By Fano's inequality 3.9,

$$H(S | A) \leq H(S | \text{Recon}(A)) \leq H_2(\epsilon) + \epsilon \cdot \log(|S| - 1).$$

If n is the number of parties, when $\lim_{n \rightarrow \infty} \epsilon(n) = 0$ observe that the right hand side tends to 0 in the limit as well.

In the other direction, if Π satisfies $(1 - \eta)$ -entropic correctness meaning that $H(S | A) \leq \eta$, then by Lemma 3.11,

$$\tilde{H}_\infty(S | A) \leq H(S | A) \leq \eta,$$

which implies that maximum likelihood decoding succeeds with probability $2^{-\eta}$ over the probability of B . If n is the number of parties, when $\lim_{n \rightarrow \infty} \eta(n) = 0$ observe that the success probability tends to 1 in the limit as well. $\square$

Entropic Secrecy. We generalize the notion of entropic secrecy (previously defined in [KGH83, CSGV91, Bei11]) to ρ -entropic secrecy.

¹⁴This is technically nuanced. See Remark 6.4.

¹⁵See Footnote 14.

Definition 6.7 (ρ -Entropic Secrecy). *Let S be a random variable that is uniformly distributed over some secret space. A secret sharing scheme satisfies perfect secrecy if for all unauthorized set of shares represented by random variable B , $H(S | B) \geq H(S) - \rho$. Equivalently $I(S; B) \leq \rho$.*

We relate the notion of ρ -entropic secrecy back to secrecy (Def. 3.4) in two separate lemmas.

1. The first lemma, Lemma 6.8, is imported from Beimel [Bei11], who shows that perfect entropic secrecy is equivalent to perfect secrecy.
2. The second lemma, Lemma 6.9, implies that $(\mu(n))$ -imperfect secrecy (Def. 3.4) implies $\mu(n) \cdot t(n)$ -imperfect entropic secrecy if the sharing algorithm runs in time $t(n)$. This $t(n)$ appears due to a technical complication in upper bounding the mutual information by statistical distance.

First, the survey of Beimel gives us the following lemma.

Lemma 6.8 (Perfect Secrecy is Equivalent to Perfect Entropic Secrecy (Claim 1 in [Bei11])). *A secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ satisfies perfect secrecy (Def. 3.4) if and only if Π satisfies perfect entropic secrecy (Def. 6.7).*

Lemma 6.9 (Imperfect Secrecy Implies Imperfect Entropic Secrecy). *If a secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ has Share running in time $t(n)$ and satisfies statistical $\rho(n)$ -secrecy (Def. 3.4), then Π satisfies $(t \cdot \rho + H(\rho))$ -entropic secrecy.*

Proof. Let $S - B$ be a Markov chain where S is a uniform random secret and B is the shares of any unauthorized set from a sharing of S . Let p_B , p_S , p_{BS} , and $p_B \otimes p_S$ be respectively the probability distribution on B , on S , on the joint (B, S) , and on the product of the marginals B and S . Then ρ -secrecy implies that $\Delta(p_{BS}, p_B \otimes p_S) \leq \rho$. By Lemma 3.12, $I(S; B) \leq t \cdot \rho + H(\rho)$. $\square$

Entropic Anonymity. Our notion of entropic anonymity is chosen to be an entropic notion of conditional independence. The reason for this can be seen in the definitions of U-Anonymity (Def. 4.1) and M-Anonymity (Def. 4.2), which implicitly state that shares from an unauthorized set of shares are conditionally independent from other shares in the unauthorized set.

Definition 6.10 (η -Entropic Anonymity). *Let S be a random variable that is uniformly distributed over some secret space. A secret sharing scheme has η -entropic anonymity if all unauthorized sets of shares, represented without loss of generality by a pair of random variables (A, B) , where A, B represent an arbitrary partition of the set, satisfy $H(A | B, S) \geq H(A | S) - \eta$.*

Example 1. When $\eta = 0$, the above property is exactly the conditional independence of A and B namely that $H(A | B, S) = H(A | S)$. In the proofs of the lower bounds below, we typically consider A as a single party's share and B as the other parties' shares in the unauthorized set. In general, however, A and B can form any partition of the unauthorized shares. For singleton unauthorized sets of shares, the condition above is trivially satisfied since B is absent.

Remark 6.3 (S-Anonymity does not Imply Entropic Anonymity). S-Anonymity does not imply that the marginal distribution of the shares of two unauthorized sets are independent. A concrete example of this can be found in Remark 6.5.

Lemma 6.11 (Perfect M-Anonymity Equivalent to Perfect Entropic Anonymity). *A secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ satisfies perfect M-Anonymity (Def. 4.2) if and only if Π satisfies perfect entropic anonymity (Def. 6.10).*

Proof. Consider a Markov chain $S - (A, B)$ where A and B form a partition of an arbitrary unauthorized set of shares produced by a sharing of secret given by a uniform random variable S , and consider a random variable B' in the Markov chain $S - B'$ representing a fresh independent sharing of the same random variable S for the same parties as B . That is, the marginal distributions of B and B' conditioned on S are identical.

Perfect M-Anonymity implies perfect strong M-Anonymity (Def. A.1), which implies that for , for all $s^* \in \text{supp}(S)$, we have

$$\Delta(p_{AB|S=s^*}, p_{A|S=s^*} \otimes p_{B'|S=s^*}) = \Delta(p_{AB|S=s^*}, p_{A|S=s^*} \otimes p_{B|S=s^*}) = 0.$$

Therefore, A and B are independent conditioned on S , and perfect entropic anonymity holds. The other direction is immediate and is obtained by reversing the above argument. $\square$

Analogous to our exploration of imperfect secrecy, we give a separate lemma statement relating imperfect statistical M-Anonymity to imperfect entropic anonymity. Recall that U-Anonymity implies M-Anonymity, so the lemma statement applies to all schemes that satisfy U-Anonymity.

Lemma 6.12 (Imperfect M-Anonymity Implies Imperfect Entropic Anonymity). *Any secret sharing scheme where Share runs in time $t(n)$ that satisfies statistical $\mu(n)$ -M-Anonymity (Def. 4.2), also satisfies $(2n \cdot \mu \cdot t + H(2n\mu))$ -entropic anonymity.*

Proof. To simplify the proof, we will use the Lemma A.2 to note that statistical μ -M-Anonymity implies statistical $(2n \cdot \mu)$ -Strong M-Anonymity. Let (A, B) be random variables where A and B form a partition of an arbitrary unauthorized set of shares produced by a sharing of secret given by a uniform random variable S . For B' be a random variable produced by a fresh independent sharing of S that represents the share values for the same parties as B . Then observe that $(2n \cdot \mu)$ -Strong M-Anonymity implies that for all values s^* of S , the distribution of A and B conditions on $S = s^*$, which we denote by $P_{s^*} = (A, B \mid S = s^*)$ and the distribution of A and B' conditioned on $S = s^*$, which we denote by $Q_{s^*} = (A, B' \mid S = s^*)$ have small statistical distance $d := \Delta(P_{s^*}, Q_{s^*}) \leq 2n\mu$. By Lemma 3.12,

$$\begin{aligned} H(A, B' \mid S) - H(A, B \mid S) &= (H(A \mid S) + H(B' \mid S)) - (H(B \mid S) + H(A \mid B, S)) \\ &= H(A \mid S) - H(A \mid B, S) \\ &\leq d \cdot \log(|\text{supp } A| \cdot |\text{supp } B| - 1) + H(d) \\ &\leq d \cdot t + H(d). \end{aligned}$$

$\square$

Remark 6.4 (Lower Bounds for Imperfect Schemes). The perfect secrecy and perfect M-Anonymity conditions imply respectively perfect entropic secrecy and perfect entropic anonymity, and schemes with perfect secrecy and perfect M-Anonymity are subject unconditionally to our lower bounds. In this work, we extend our lower bounds to schemes that satisfy statistical secrecy and statistical M-Anonymity by introducing Lemma 6.9 and Lemma 6.12. Due to the use of Lemma 3.12 in the proofs of these two lemmas, applying these two lemmas requires considering a fixed time bound $t(n)$ on the sharing algorithm runtime where n is the number of parties. Here are two examples of natural settings for which we obtain lower bounds for schemes with imperfect secrecy and imperfect M-Anonymity for the star graph access structures we consider in this paper:

1. For any scheme with an efficient polynomial-time Share algorithm, i.e. $t(n) = \text{poly}(n)$ where n is the number of parties, and with negligible statistical secrecy and negligible statistical U/M-Anonymity.

2. For any scheme with an exponential-time $t(n)$ Share algorithm, the lower bounds apply to the scheme if $(\rho \cdot t)(n)$ and $(\eta \cdot t)(n)$ are subconstant in the number of parties n where ρ is the entropic secrecy parameter and where η is entropic anonymity parameter.

6.3 Lower Bound for the Star Graph

We now show an information-theoretic lower bound on the share size of any uniform secret sharing scheme for the secret space $\{0, 1\}$ for the $(1, n)$ -star undirected graph access structure (Def. 6.4).

Theorem 6.13. *Any M-FASS scheme for the $(1, n)$ -star graph access structure with $(1 - \varepsilon)$ -correctness, statistical ρ -secrecy, and statistical η -M-Anonymity for $\varepsilon \in [0, 1)$ and for negligible functions $\rho(n), \eta(n)$ has per-party share size $\Omega(n \cdot \ell)$ bits for sharing ℓ -bit secrets.*

Remark 6.5 (No Extension to S-FASS). One cannot hope to extend this lower bound to S-FASS schemes as there exists a simple S-FASS scheme for the $(1, n)$ -star graph with per-party share size $O(1)$ bits for sharing a 1-bit secret. The scheme is as follows: For a secret s , additively secret share $s = s_1 + s_2$ and give the center node s_1 and give all leaf nodes s_2 . The technical barrier as to why the lower bound does not extend to S-FASS schemes is that S-Anonymity does not imply entropic anonymity.

To give some intuition about this lower bound, the set of vertices incident to every edge in the star graph is a minimal authorized set so that there are n -many possible minimal authorized sets whose shares all reconstruct to the same secret. Since the center party's share is part of every minimal authorized set, the center party's share must allow for reconstruction when used with any of the other n -many parties' shares. These other parties' shares are uniformly sampled and each individual share, when considered alone or in any subset without the center party's share, must hide the secret that has at least one bit of entropy. This implies that the center party's share size must be large enough to accommodate enough information for n possible reconstruction paths that are sampled from a high entropy space.

Let S be a random variable for a uniform random T -bit secret so that $H(S) = T$. Let $C, L_1, \dots, L_n$ denote random variables for shares produced by a fully anonymous sharing scheme. These random variables, which are functions of the share randomness and S , correspond to the share value of the center node and the share values of the respective leaf nodes.

Lemma 6.14 (Lower Bound for the Star Graph). *Let S be a random variable uniformly distributed over support $\{0, 1\}^T$. For any information-theoretic fully anonymous secret sharing scheme satisfying ε -entropic correctness, ρ -entropic secrecy, and η -entropic anonymity, $H(C | S) \geq (T - \varepsilon - 2\rho - \eta) \cdot n$.*

Proof. To demonstrate the proof, let us first prove it for the case of perfect entropic correctness, perfect entropic secrecy and perfect entropic anonymity. We begin by using the chain rule of Shannon entropy in two different ways,

$$\begin{aligned} H(C, L_1, \dots, L_n | S) &= H(C | S) + H(L_1 | C, S) + H(L_2 | L_1, C, S) + \dots + H(L_n | L_{n-1}, \dots, L_1, C, S) \\ &= H(L_1 | S) + H(L_2 | L_1, S) + \dots + H(L_n | L_{n-1}, \dots, L_1, S) + H(C | L_n, \dots, L_1, S). \end{aligned}$$

By isolating the term $H(C | S)$ we have that,

$$\begin{aligned} H(C | S) &= H(C | L_n, \dots, L_1, S) + \sum_{i=1}^n (H(L_i | L_{i-1}, \dots, L_1, S) - H(L_i | L_{i-1}, \dots, L_1, C, S)) \\ &\geq \sum_{i=1}^n (H(L_i | L_{i-1}, \dots, L_1, S) - H(L_i | L_{i-1}, \dots, L_1, C, S)) \end{aligned} \tag{10}$$

where the inequality follows by the non-negativity of Shannon entropy. Recall that for any secret $s \in \{0, 1\}$, any unauthorized set of shares satisfies that each of the shares in this set is uniformly distributed over some fixed support Ω , and hence are independent. Then observe that for all $i \in [n]$, $\{L_i, L_{i-1}, \dots, L_1\}$ are random variables for an unauthorized set of shares by the definition of the star graph access structure. Therefore, for all $i \in [n]$, we have that

$$\begin{aligned} \mathbf{H}(L_i \mid L_{i-1}, \dots, L_1, S) &= \sum_{s \in \{0,1\}^T} \Pr[S = s] \cdot \mathbf{H}(L_i \mid L_{i-1}, \dots, L_1, S = s) \\ &= \sum_{s \in \{0,1\}^T} \Pr[S = s] \cdot \mathbf{H}(L_i \mid S = s) \\ &= \mathbf{H}(L_i \mid S). \end{aligned} \quad (11)$$

Then recall that conditioning on a random variable cannot increase the entropy, therefore for all $i \in [n]$,

$$\mathbf{H}(L_i \mid L_{i-1}, \dots, L_1, C, S) \leq \mathbf{H}(L_i \mid C, S). \quad (12)$$

Combining Equation 10, Equation 11, and Equation 12, we see that

$$\mathbf{H}(C \mid S) \geq \sum_{i=1}^n (\mathbf{H}(L_i \mid S) - \mathbf{H}(L_i \mid C, S)) = \sum_{i=1}^n I(L_i; C \mid S). \quad (13)$$

We now lower bound each term in the sum by 1. Perfect secrecy implies that

$$\begin{aligned} \forall i \in [n], I(L_i; S) &= \mathbf{H}(S) - \mathbf{H}(S \mid L_i) = 0 \\ \wedge I(C; S) &= \mathbf{H}(S) - \mathbf{H}(S \mid C) = 0. \end{aligned}$$

Perfect correctness implies that for all $i \in [n]$, $\mathbf{H}(S \mid L_i, C) = 0$, so that

$$I(L_i, C; S) = \mathbf{H}(S) - \mathbf{H}(S \mid L_i, C) = T.$$

Then, for all $i \in [n]$,

$$\begin{aligned} I(L_i; C \mid S) &= I(L_i, S; C). && \text{(Lemma 3.14 with } \mathbf{H}(C) = \mathbf{H}(C \mid S).) \\ &= I(C; L_i, S) && \text{(Symmetry of mutual information.)} \\ &\geq I(C; S \mid L_i) && \text{(Lemma 3.15.)} \\ &= I(L_i, C; S) && \text{(Lemma 3.14 with } \mathbf{H}(L_i) = \mathbf{H}(L_i \mid S).) \\ &\geq T && \text{(Perfect correctness.)} \end{aligned}$$

Applying this inequality to each term in Equation 13 gives $\mathbf{H}(C \mid S) \geq T \cdot n$.

For the imperfect case when ε, ρ, η are possibly not 0, the first differing inequality is that the use of η -entropic anonymity instead of perfect entropic anonymity changes Equation 11 to be

$$\mathbf{H}(L_i \mid L_{i-1}, \dots, L_1, S) \geq \mathbf{H}(L_i \mid S) - \eta.$$

This change implies that Equation 13 is instead

$$\mathbf{H}(C \mid S) \geq \sum_{i=1}^n (\mathbf{H}(L_i \mid S) - \mathbf{H}(L_i \mid C, S) - \eta) = \left(\sum_{i=1}^n I(L_i; C \mid S) \right) - n \cdot \eta. \quad (14)$$

The next change is that Lemma 3.14 is applied with ρ -entropic secrecy instead of perfect secrecy and $(1 - \varepsilon)$ -entropic correctness, so that for all $i \in [n]$,

$$\begin{aligned} I(L_i; C \mid S) &\geq I(L_i, S; C) - \rho. \\ &= I(C; L_i, S) - \rho \\ &\geq I(C; S \mid L_i) - \rho \\ &\geq I(L_i, C; S) - 2\rho \\ &\geq T - \varepsilon - 2\rho. \end{aligned}$$

Then, applying this inequality to each term in the summation of the RHS of Equation 14, we have

$$H(C \mid S) \geq n \cdot (T - \varepsilon - 2\rho - \eta).$$

□

Proof of Theorem 6.13. Since every party alone is unauthorized, the statistical M-Anonymity property implies the statistical S-Anonymity property which implies that the statistical distance between the marginal distribution on the share C and share L_i for any $i \in [n]$ is negligible. Lemma 6.14 implies that the central node's share has at least $\Omega(T \cdot n)$ bits of entropy. Without loss of generality, the share domain has polynomial bit description, so by Lemma 3.12, the difference in entropy between $H(C \mid S)$ and $H(L_i \mid S)$ is negligible in n . This implies that $H(L_i \mid S)$ is $\Omega(T \cdot n)$ bits of entropy. □

6.4 A Quadratic Lower Bound for Per-Party Share Size

We now show how to extend the lower bound for the $(1, n)$ -star graph access structure to the $(2, n)$ access structure. Namely, we prove the following theorem.

Theorem 6.15. *Any M-FASS scheme for the $(2, n)$ -star graph access structure with $(1 - \varepsilon)$ -correctness, statistical ρ -secrecy, and statistical η -M-Anonymity for $\varepsilon \in [0, 1)$ and for negligible functions $\rho(n), \eta(n)$ has per-party share size $\Omega(T \cdot n^2)$ bits for sharing a T -bit secret.*

We can view the $2n + 1$ parties in this access structure as forming a center party, and a disjoint left and right set of size n . Our quadratic lower bound for the per-party share size of any M-FASS scheme for the $(2, n)$ -star graph access structure (Def. 6.4) proceeds in two steps. First, we prove that the entropy of any share in one of the two disjoint sets is $\Omega(T \cdot n)$. Second, we prove that the center party's share has entropy $\Omega(T \cdot n^2)$. Statistical M-Anonymity implies statistical S-Anonymity, which then implies that every party's share has entropy $\Omega(T \cdot n^2)$.

Let S be a random variable for a uniform random T -bit secret so that $H(S) = T$. Let $C, X_1, \dots, X_n, Y_1, \dots, Y_n$ denote random variables for shares produced by a M-FASS scheme. These random variables, which are functions of the share randomness and S , are such that C corresponds to the share value of the center node, $\{X_i\}_{i \in [n]}$ corresponds to the share values of the “left” partition, and $\{Y_i\}_{i \in [n]}$ corresponds to the share values of the “right” partition. We first prove the following technical lemma.

Lemma 6.16. *Let S be a random variable uniformly distributed over $\{0, 1\}^T$. For any information-theoretic secret sharing scheme for the $(2, n)$ -star access structure that satisfies $(1 - \varepsilon)$ -entropic correctness, ρ -entropic secrecy, η -entropic anonymity, for all $i \in [n]$,*

$$H(X_i \mid S) - H(X_i \mid Y_n, \dots, Y_1, C, S) \geq (T - \varepsilon - 2\rho - \eta) \cdot n.$$

The proof of Lemma proceeds analogously to that of the lower bound for the star graph.

Proof of Lemma 6.16 in the Perfect Setting. We will prove the case where $\varepsilon = \rho = \eta = 0$, i.e. perfect entropic correctness, perfect entropic secrecy, and perfect entropic anonymity. The proof where these values are non-zero follows by a simple modification as demonstrated in proof of Lemma 6.14 and is omitted because it imparts no additional intuition and requires us to carry these values throughout the proof.

It suffices to prove the case of $i = 1$. We begin by using the chain rule of Shannon entropy in two ways,

$$\begin{aligned} \mathbf{H}(X_1, Y_n, \dots, Y_1, C \mid S) &= \mathbf{H}(C \mid S) + \mathbf{H}(Y_1 \mid C, S) + \sum_{j=2}^n \mathbf{H}(Y_j \mid Y_1, \dots, Y_{j-1}, C, S) + \mathbf{H}(X_1 \mid Y_1, \dots, Y_n, C, S) \\ &= \mathbf{H}(X_1 \mid S) + \mathbf{H}(C \mid X_1, S) + \mathbf{H}(Y_1 \mid X_1, C, S) + \sum_{j=2}^n \mathbf{H}(Y_j \mid Y_1, \dots, Y_{j-1}, X_1, C, S). \end{aligned}$$

Rewrite the above to obtain,

$$\begin{aligned} \mathbf{H}(X_1 \mid S) - \mathbf{H}(X_1 \mid Y_1, \dots, Y_n, C, S) &= \mathbf{H}(C \mid S) - \mathbf{H}(C \mid X_1, S) \\ &\quad + \sum_{j=1}^n \mathbf{H}(Y_j \mid Y_1, \dots, Y_{j-1}, C, S) - \mathbf{H}(Y_j \mid Y_1, \dots, Y_{j-1}, X_1, C, S) \end{aligned} \quad (15)$$

Since (C, X_1) is an unauthorized set, the entropic anonymity property implies $\mathbf{H}(C \mid S) = \mathbf{H}(C \mid X_1, S)$, therefore,

$$\mathbf{H}(C \mid S) - \mathbf{H}(C \mid X_1, S) = 0. \quad (16)$$

Let $Z_0 := C$ and for all $j \in [n]$, let $Z_{j-1} = (Y_1, \dots, Y_{j-1}, C)$. For $k \in \{1, \dots, n-1\}$, let $\vec{Y}_k = (Y_1, \dots, Y_k)$. We briefly preface the calculations. For $j \geq 2$, (X_1, Z_{j-1}) corresponds to an authorized set while the application of Lemma 3.14 can only be used when the involved random variables form an unauthorized set. Fortunately, (Y_j, Z_{j-1}) corresponds to an unauthorized set, so to avoid the involvement of X_1 , we first appeal to the symmetry of mutual information. To formally invoke Lemma 3.14, we observe that

$$\mathbf{H}(Y_j \mid Z_{j-1}, S) = \mathbf{H}(Y_j \mid S) = \mathbf{H}(Y_j) \quad (17)$$

where the first equality is due to entropic anonymity, and the second equality is due to entropic secrecy. Subsequently, we need the observation that

$$\mathbf{H}(S \mid X_1, C) = \mathbf{H}(S), \quad (18)$$

which follows directly from entropic secrecy since (X_1, C) is unauthorized. Note that the second invocation of Lemma 3.14 uses the Discarding Lemma (Lemma 3.13) to again avoid the superficial problem posed by conditioning on a random variable corresponding to an authorized set. Thus, for all $j \in [n]$,

$$\begin{aligned} \mathbf{H}(Y_j \mid \vec{Y}_{j-1}, C, S) - \mathbf{H}(Y_j \mid \vec{Y}_{j-1}, X_1, C, S) &= I(Y_j; X_1 \mid Z_{j-1}, S) \\ &= I(X_1; Y_j \mid Z_{j-1}, S) && \text{(Symmetry of mutual information.)} \\ &= I(X_1, Z_{j-1}, S; Y_j) && \text{(Lemma 3.14 with Equation 17.)} \\ &= I(Y_j; X_1, Z_{j-1}, S) && \text{(Symmetry of mutual information.)} \\ &\geq I(Y_j; X_1, C, S) && \text{(Discarding Lemma (Lemma 3.13).)} \\ &\geq I(Y_j; S \mid X_1, C) && \text{(Lemma 3.15.)} \\ &= I(Y_j, X_1, C; S) && \text{(Lemma 3.14 with Equation 18.)} \\ &\geq T. && \text{(Entropic Correctness.)} \end{aligned}$$

Combining Equation 15, Equation 16, and the inequality¹⁶ immediately above, we conclude that

$$\mathbf{H}(X_1 | S) - \mathbf{H}(X_1 | Y_1, \dots, Y_n, C, S) \geq T \cdot n.$$

□

Lemma 6.16 implies that every non-center party's share has entropy $\Omega(T \cdot n)$. These non-center party shares are split into a “left” set and a “right” set corresponding to the $(2, n)$ -star access structure. As can be seen in the above proof, we argued this lower bound for the entropy of non-center party shares by matching a non-center party share with each of the shares in the opposite side set.

To lower bound the entropy of the center party's share, we will match the center party's share with each of the shares in, without loss of generality, the left set. Each of these left set parties has already been matched with each of the right set parties in the proof of Lemma 6.16. Therefore, our lower bound makes use of all n^2 many pairs, where each pair corresponds to a possible reconstruction path for the central party. We now use Lemma 6.16 to produce a lower bound for the entropy of the center party's share.

Lemma 6.17. *Let S be a random variable uniformly distributed over $\{0, 1\}$. For any information-theoretic secret sharing scheme for the $(2, n)$ -star access structure that satisfies $(1 - \varepsilon)$ -entropic correctness, ρ -entropic secrecy, and η -entropic anonymity, $\mathbf{H}(C | S) \geq (T - \varepsilon - 2\rho - \eta) \cdot n^2 - \rho \cdot n = \Omega(T \cdot n^2)$.*

Proof. We will use the shorthand $\vec{X}_i := (X_i, \dots, X_1)$ and $\vec{Y}_i := (Y_i, \dots, Y_1)$ for $i \in [n]$. We begin by using the chain rule of Shannon entropy in two ways,

$$\begin{aligned} \mathbf{H}(C, \vec{X}_n, \vec{Y}_n | S) &= \mathbf{H}(C | S) + \mathbf{H}(\vec{Y}_n | C, S) + \sum_{i=1}^n \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, C, S) \\ &= \mathbf{H}(\vec{Y}_n | S) + \sum_{i=1}^n \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, S) + \mathbf{H}(C | \vec{X}_n, \vec{Y}_n, S). \end{aligned}$$

Rearranging the above gives,

$$\begin{aligned} \mathbf{H}(C | S) - \mathbf{H}(C | \vec{X}_n, \vec{Y}_n, S) &= \mathbf{H}(\vec{Y}_n | S) - \mathbf{H}(\vec{Y}_n | C, S) \\ &\quad + \sum_{i=1}^n \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, S) - \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, C, S) \\ &\geq \sum_{i=1}^n \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, S) - \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, C, S) \end{aligned}$$

where the last inequality is since mutual information is always non-negative, i.e. $I(\vec{Y}_n; C | S) = \mathbf{H}(\vec{Y}_n | S) - \mathbf{H}(\vec{Y}_n | C, S) \geq 0$. Then, for all $i \in [n]$,

$$\begin{aligned} \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, S) - \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, C, S) &\geq \mathbf{H}(X_i | S) - \mathbf{H}(X_i | \vec{X}_{i-1}, \vec{Y}_n, C, S) - \rho \quad (\text{Entropic Secrecy.}) \\ &\geq \mathbf{H}(X_i | S) - \mathbf{H}(X_i | \vec{Y}_n, C, S) - \rho \\ &\quad (\text{Conditioning only decreases entropy.}) \\ &\geq (T - \varepsilon - 2\rho - 2\eta)n - \rho. \quad (\text{Lemma 6.16.}) \end{aligned}$$

Therefore,

$$\mathbf{H}(C | S) - \mathbf{H}(C, \vec{X}_n, \vec{Y}_n | S) \geq (T - \varepsilon - 2\rho - \eta) \cdot n^2 - \rho \cdot n.$$

¹⁶In the setting of $(1 - \varepsilon)$ -entropic correctness, ρ -entropic secrecy, η -entropic anonymity, we will have $I(Y_j; X_1 | Z_{j-1}, S) \geq 1 - \varepsilon - 2\rho - \eta$ instead. So the below inequality will have $(1 - \varepsilon - 2\rho - \eta) \cdot n$ on the right hand side.

By the non-negativity of Shannon entropy,

$$H(C \mid S) \geq (T - \varepsilon - 2\rho - \eta) \cdot n^2 - \rho \cdot n.$$

□

Proof of Theorem 6.15. This proof is identical to the proof of Theorem 6.13 except that we use Lemma 6.17 instead of Lemma 6.14. □

6.5 An Exponential Lower Bound

As mentioned before, the above strategy extends to the (k, n) -star access structure, for any $k, n \in \mathbb{N}$, where k and n satisfy any arbitrary relation. We will show that this access structure for $O(nk)$ parties has a lower bound on the per-party share size of $\Omega(n^k)$ for any M-FASS scheme that satisfies $(1 - \varepsilon)$ -entropic correctness, perfect entropic secrecy, and perfect entropic anonymity. In particular, in the setting of a $t = 1$ bit secret, $n = 2$ and $k = O(\lambda)$ for some security parameter λ , we obtain an exponential sized lower bound of $\Omega(2^\lambda)$ for the per-party share size per 1-bit secret in the number of parties $O(\lambda)$.

Theorem 6.18. *Any M-FASS scheme for the (k, n) -star graph access structure with $(1 - \varepsilon)$ -correctness, perfect secrecy, and perfect M-Anonymity for $\varepsilon \in [0, 1)$ has per-party share size $\Omega(T \cdot n^k)$ bits for sharing T -bit secrets, where k can have any arbitrary relation with n .*

Corollary 6.19. *For every $T \in \mathbb{N}$, for every $m, d \in \mathbb{N}$, there is an n -party access structure $\mathcal{A}_{m,d}$ with $n = md + 1$, represented by a monotone DNF with $\ell = d^m$ minterms of size $t = O(m)$, such that:*

- *Every M-FASS scheme for sharing T -bit secrets for $\mathcal{A}_{m,d}$ has per-party share size $\Omega(T \cdot \ell)$.*
- *$\mathcal{A}_{m,d}$ has a monotone CNF with $O(m)$ clauses of size $O(d)$.*

Proof. Consider the (m, d) -star graph access structure. This is an access structure with a monotone DNF representation $n = md + 1$ variables, with $\ell := d^m$ minterms of size $t = O(m)$, and it has a monotone CNF representation of with $O(m)$ clauses of size $O(d)$. Now Theorem 6.18 to obtain a per-party share size of $\Omega(T \cdot \ell)$ bits. □

Remark 6.6 (Tightness of Theorem 6.18). Theorem 6.18 implies that Construction 5.7 is optimal for the (k, n) -star graph up to polylogarithmic factors. In particular, for every constant $k \geq 1$, there is an access structure with $O(n^k)$ minterms, for which our upper bound is $\tilde{O}(n^k)$ and the lower bound is $\Omega(n^k)$, showing that our upper bound is nearly tight.

Theorem 6.20 (Polynomial Lower Bound for Imperfect Schemes). *Any M-FASS scheme for the (k, n) -star graph access structure, where $k(n) = O(1)$, with $(1 - \varepsilon)$ -correctness, statistical ρ -secrecy, and statistical η -M-Anonymity, for $\varepsilon \in [0, 1)$ and negligible functions $\rho(n), \eta(n)$, has per-party share size $\Omega(T \cdot n^k)$ bits for sharing T -bit secrets.*

Remark 6.7 (Imperfect Secrecy and Imperfect M-Anonymity). As mentioned in Remark 6.4, while we can extend Theorem 6.18 to the imperfect Secrecy and imperfect M-Anonymity for the case of $k(n) = \text{poly}(n)$, for technical reasons we need a stronger assumption on $\rho(n), \eta(n)$. We suspect, however, that this is not the true state of affairs.

The proofs of Theorem 6.18 and Theorem 6.20 follow the exact same template as the proof of Theorem 6.15. Let $C, \{P_{i,j}\}_{i \in [k], j \in [n]}$ denote random variables for the shares produced by a fully anonymous secret sharing scheme for the (k, n) -star access structure where $P_{i,j}$ denotes the j th party in the set P_i . For shorthand, we use $\vec{V}_i := (P_{i,1}, \dots, P_{i,n})$ and $\vec{V}_{i,k} := (P_{i,1}, \dots, P_{i,k})$ for $k \leq n$. That is, $\vec{V}_i$ contains all the parties in set P_i and $\vec{V}_{i,k}$ contains the first k parties of in the set P_i .

Lemma 6.21. *Let S be a random variable uniformly distributed over $\{0, 1\}^T$. For any information-theoretic secret sharing scheme for the (k, n) -star access structure that satisfies $(1 - \varepsilon)$ -entropic correctness, ρ -entropic secrecy, and η -entropic anonymity, for $\varepsilon \in [0, 1)$, for any negligible $\rho(n)$ and negligible $\eta(n)$ satisfies $\mathbf{H}(C | S) \geq \Omega(T \cdot n^k)$.*

The high-level approach to prove Lemma 6.21 is identical to that of Lemma 6.17.

Proof of Lemma 6.21 in the Perfect Setting. We focus on the setting where $\varepsilon = \rho = \eta = 0$ to make the exposition more manageable for the reader. The proof proceeds inductively on the partitions from $i = 1$ to $i = k$. The usage of entropic correctness, entropic secrecy, and entropic anonymity are used in nearly the same way as previous proofs, but in a slightly different order as one will see. Namely, we argue inductively that the shares in P_i have Shannon entropy at least $\Omega(T \cdot n^{i-1})$.

For the base case where $i \in [n]$, for any $i_2, \dots, i_k \in [n]$, we have

$$\begin{aligned}
\mathbf{H}(P_{1,i} | S) - \mathbf{H}(P_{1,i} | P_{2,i_2}, \dots, P_{k,i_k}, C, S) &= I(P_{1,i}; P_{2,i_2}, \dots, P_{k,i_k}, C | S) && \text{(Definition of M.I.)} \\
&= I(P_{1,i}, S; P_{2,i_2}, \dots, P_{k,i_k}, C) && \text{(Lemma 3.14.)} \\
&= I(P_{2,i_2}, \dots, P_{k,i_k}, C; P_{1,i}, S) && \text{(Symmetry of M.I.)} \\
&\geq I(P_{2,i_2}, \dots, P_{k,i_k}, C; S | P_{1,i}) && \text{(Lemma 3.15.)} \\
&= I(P_{1,i}, P_{2,i_2}, \dots, P_{k,i_k}, C; S) && \text{(Lemma 3.14.)} \\
&\geq T. && \text{(Entropic correctness.)}
\end{aligned}$$

where the first usage of Lemma 3.14 requires the observation that $\mathbf{H}(P_{2,i_2}, \dots, P_{k,i_k}, C) = \mathbf{H}(P_{2,i_2}, \dots, P_{k,i_k}, C | S)$ due to entropic secrecy, the second usage of Lemma 3.14 requires the observation that $\mathbf{H}(S) = \mathbf{H}(S | P_{1,i})$ since $I(S; P_{1,i}) = 0$ by entropic secrecy, and the final inequality requires the observation that $(P_{1,i}, P_{2,i_2}, \dots, P_{k,i_k}, C)$ is an authorized set of shares. Observe that entropic anonymity was not used, as arguing that any single unauthorized share has at least one bit of entropy only requires correctness and secrecy. We will use entropic anonymity to grow this trivial base case in the inductive step.

The inductive hypothesis is that for any positive integer $\ell < k$, for all $i \in [n]$, for all $i_{\ell+1}, i_{\ell+2}, \dots, i_k \in [n]$,

$$\mathbf{H}(P_{\ell,i} | S) - \mathbf{H}(P_{\ell,i} | P_{\ell+1,i_{\ell+1}}, P_{\ell+2,i_{\ell+2}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, C, S) \geq T \cdot n^{\ell-1}. \quad (19)$$

Using the inductive hypothesis, we prove the following statement for $\ell + 1$, for all $i \in [n]$, for all $i_{\ell+2}, i_{\ell+3}, \dots, i_k \in [n]$,

$$\mathbf{H}(P_{\ell+1,i} | S) - \mathbf{H}(P_{\ell+1,i} | P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell}, C, S) \geq T \cdot n^{\ell}. \quad (20)$$

For any $i \in [n]$, for any $i_{\ell+2}, i_{\ell+3}, \dots, i_k \in [n]$, proving the inductive step begins by using the chain rule of Shannon entropy as follows,

$$\begin{aligned}
&\mathbf{H}(P_{\ell+1,i}, \vec{V}_1, \dots, \vec{V}_{\ell}, P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, C | S) \\
&= \mathbf{H}(C | S) \\
&\quad + \mathbf{H}(\vec{V}_{\ell} | C, S) \\
&\quad + \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} | \vec{V}_{\ell}, C, S) \\
&\quad + \mathbf{H}(P_{\ell+1,i} | P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell}, C, S).
\end{aligned}$$

Similarly, we can use the chain rule of Shannon entropy on the same left-hand side expression to obtain,

$$\begin{aligned}
& \mathbf{H}(P_{\ell+1,i}, \vec{V}_1, \dots, \vec{V}_\ell, P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, C \mid S) \\
&= \mathbf{H}(P_{\ell+1,i} \mid S) \\
&\quad + \mathbf{H}(C \mid P_{\ell+1,i}, S) \\
&\quad + \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid P_{\ell+1,i}, C, S) \\
&\quad + \mathbf{H}(\vec{V}_\ell \mid P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, P_{\ell+1,i}, C, S).
\end{aligned}$$

Therefore,

$$\begin{aligned}
& \mathbf{H}(P_{\ell+1,i} \mid S) - \mathbf{H}(P_{\ell+1,i}, \vec{V}_1, \dots, \vec{V}_\ell, P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, C \mid S) \\
&= [\mathbf{H}(C \mid S) - \mathbf{H}(C \mid P_{\ell+1,i}, S)] \\
&\quad + [\mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid \vec{V}_\ell, C, S) \\
&\quad \quad - \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid P_{\ell+1,i}, C, S)] \\
&\quad + [\mathbf{H}(\vec{V}_\ell \mid C, S) - \mathbf{H}(\vec{V}_\ell \mid P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, P_{\ell+1,i}, C, S)]. \tag{21}
\end{aligned}$$

First observe that $(C, P_{\ell+1,i})$ is an unauthorized set of shares, so the entropic anonymity implies, $\mathbf{H}(C \mid P_{\ell+1,i}, S) = \mathbf{H}(C \mid S)$. Then,

$$\mathbf{H}(C \mid S) - \mathbf{H}(C \mid P_{\ell+1,i}, S) = 0. \tag{22}$$

The observe that

$$(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_\ell, C)$$

is an unauthorized set as this set of shares is missing a share of a party from the partition $P_{\ell+1}$. Also,

$$(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, P_{\ell+1,i}, C)$$

is an unauthorized set because this set of shares is missing a share of a party from the partition P_ℓ . The entropic anonymity therefore implies

$$\begin{aligned}
\mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid \vec{V}_\ell, C, S) &= \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid S) \\
&= \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid P_{\ell+1,i}, C, S).
\end{aligned}$$

so that

$$\begin{aligned}
& \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid \vec{V}_\ell, C, S) \\
&\quad - \mathbf{H}(P_{\ell+2,i_{\ell+2}}, P_{\ell+3,i_{\ell+3}}, \dots, P_{k,i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1} \mid P_{\ell+1,i}, C, S) = 0. \tag{23}
\end{aligned}$$

Then, observe that $(\vec{V}_\ell, C)$ is an unauthorized set of shares, so

$$\begin{aligned}
\mathbf{H}(\vec{V}_\ell \mid C, S) &= \mathbf{H}(\vec{V}_\ell \mid S) && \text{(Entropic Anonymity.)} \\
&= \mathbf{H}(P_{\ell,1} \mid S) + \sum_{j=2}^n \mathbf{H}(P_{\ell,j} \mid \vec{V}_{\ell,j-1}, S) && \text{(Chain rule.)} \\
&= \sum_{j=1}^n \mathbf{H}(P_{\ell,j} \mid S). && \text{(Entropic Anonymity.)}
\end{aligned}$$

For notational brevity, let

$$\begin{aligned} Z &:= (P_{\ell+2, i_{\ell+2}}, P_{\ell+3, i_{\ell+3}}, \dots, P_{k, i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, P_{\ell+1, i}, C, S) \\ &= (P_{\ell+1, i}, P_{\ell+2, i_{\ell+2}}, P_{\ell+3, i_{\ell+3}}, \dots, P_{k, i_k}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, C, S). \end{aligned}$$

Then,

$$\begin{aligned} \mathbf{H}(\vec{V}_\ell \mid Z) &= \mathbf{H}(P_{\ell,1} \mid Z) + \sum_{j=2}^n \mathbf{H}(P_{\ell,j} \mid \vec{V}_{\ell,j-1}, Z) && \text{(Chain rule.)} \\ &\leq \sum_{j=1}^n \mathbf{H}(P_{\ell,j} \mid Z). && \text{(Conditioning cannot increase entropy.)} \end{aligned}$$

The induction hypothesis implies that for all $i_\ell \in [n]$, $\mathbf{H}(P_{\ell, i_\ell} \mid S) - \mathbf{H}(P_{\ell, i_\ell} \mid Z) \geq T \cdot n^{\ell-1}$. Therefore

$$\begin{aligned} \mathbf{H}(\vec{V}_\ell \mid C, S) - \mathbf{H}(\vec{V}_\ell \mid P_{\ell+2,1}, \dots, P_{k,1}, \vec{V}_1, \dots, \vec{V}_{\ell-1}, P_{\ell+1,i}, C, S) \\ &= \mathbf{H}(\vec{V}_\ell \mid C, S) - \mathbf{H}(\vec{V}_\ell \mid Z) \\ &\geq \sum_{j=1}^n [\mathbf{H}(P_{\ell,j} \mid S) - \mathbf{H}(P_{\ell,j} \mid Z)] \\ &\geq \sum_{j=1}^n T \cdot n^{\ell-1} = T \cdot n^\ell. \end{aligned} \tag{24}$$

By substituting Equation 22, Equation 23, and Equation 24 into Equation 21, we finish the proof of the induction step.

Finally, we observe by the same technique of taking the difference of two uses of the chain rule, we have

$$\begin{aligned} \mathbf{H}(C \mid S) - \mathbf{H}(C \mid \vec{V}_1, \dots, \vec{V}_k, S) \\ &= [\mathbf{H}(\vec{V}_1, \dots, \vec{V}_{k-1} \mid S) - \mathbf{H}(\vec{V}_1, \dots, \vec{V}_{k-1} \mid C, S)] \\ &\quad + [\mathbf{H}(\vec{V}_k \mid \vec{V}_1, \dots, \vec{V}_{k-1}, S) - \mathbf{H}(\vec{V}_k \mid \vec{V}_1, \dots, \vec{V}_{k-1}, C, S)] \\ &\geq T \cdot n^k. \end{aligned}$$

The inequality results from two observations. Firstly, $\mathbf{H}(\vec{V}_1, \dots, \vec{V}_k \mid S) = \mathbf{H}(\vec{V}_1, \dots, \vec{V}_{k-1} \mid C, S)$ by the entropic anonymity. Secondly,

$$\begin{aligned} \mathbf{H}(\vec{V}_k \mid \vec{V}_1, \dots, \vec{V}_{k-1}, S) - \mathbf{H}(\vec{V}_k \mid \vec{V}_1, \dots, \vec{V}_{k-1}, C, S) \\ &= \mathbf{H}(\vec{V}_k \mid S) - \mathbf{H}(\vec{V}_k \mid \vec{V}_1, \dots, \vec{V}_{k-1}, C, S) && \text{(Entropic anonymity.)} \\ &\geq T \cdot n^k, \end{aligned}$$

where the final inequality is attained by applying the chain rule, then applying the above lower bound that was proven inductively on all n terms of the post-chain-rule expression. Therefore, by the non-negativity of Shannon entropy,

$$\mathbf{H}(C \mid S) \geq T \cdot n^k.$$

□

Proofs of Theorem 6.18 and Theorem 6.20. The proof follows from Lemma 6.21 in the same exact manner as the proof of Theorem 6.13. $\square$

Remark 6.8 (An Exponential Lower Bound for the (s, t) -Connectivity Access Structure). We can show that the same technique extends to an exponential lower bound on the per-party share size for the (s, t) -connectivity access structure for any statistical M-FASS scheme with imperfect correctness, perfect secrecy, and perfect M-Anonymity. In this access structure, the edges of the undirected graph represent parties and a minimal authorized set of parties consists of a path from vertex s to vertex t .

The idea for the lower bound is to consider a graph that consists of a total of $n + 1$ “bottleneck” vertices ordered from $s = b_1, \dots, t = b_{n+1}$ and connect b_i to b_{i+1} with k edges. If we restrict ourselves to simple graphs, we can simply expand these edges by introducing k more “intermediate” vertices for every $(i, i + 1)$ pair so that instead of having k edges from b_i to b_{i+1} , the graph has each intermediate node connected to b_i and b_{i+1} . The idea is that from s to t , between each of the bottlenecks, there are k choices of edges to take, resulting in n^k many paths from s to t . To argue the lower bound, we use the same technique from before. We argue that every party that corresponds to an edge between b_n and b_{n+1} must have at least 1 bit entropy, then every party between b_{n-1} and b_n must have at least n bits of entropy, and so forth. The number of edges is $O(nk)$ while the minimum entropy of the shares will be $\Omega(T \cdot n^k)$ for T -bit secrets. Applying our lower bound technique to the case of (s, t) -connectivity demonstrates that this technique can apply to access structures other than the star graph.

7 Computational Upper Bounds

7.1 From OWF for Star graph

In this section we show a computational U-FASS scheme for the $(1, n)$ -star graph access structure with asymptotically better total share size than what is possible information-theoretically.

Theorem 7.1. *Let λ be a security parameter. Assuming the existence of one-way functions, there exists a computational U-FASS scheme for sharing a 1-bit secret with $1 - \mu(\lambda)$ correctness, computational secrecy, and computational U-Anonymity, for the $(1, n(\lambda))$ -star graph access structure where $\mu : \mathbb{N} \rightarrow [0, 1]$ is a negligible function with a per-party share size of $O(\lambda)$.*

We now introduce the following construction to prove Theorem 7.1.

Construction 7.2. *Let $F : \{0, 1\}^\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ be a PRF.*

- $\text{Share}(1^\lambda, s \in \{0, 1\})$:
 1. Sample a uniform random key $k \in \{0, 1\}^\lambda$. Sample another uniform random value $r_\star \in \{0, 1\}^\lambda$. Set $\text{sh}_\star \leftarrow (k \| r_\star)$.
 2. For $i \in [n(\lambda)]$, set $\text{sh}_i \leftarrow (r_i \| F(k, r_i) \oplus \bar{s})$ where $r_i \xleftarrow{\$} \{0, 1\}^\lambda$ and $\bar{s} := (s \| 0^{\lambda-1}) \in \{0, 1\}^\lambda$.
 3. Output $(\text{sh}_\star, \text{sh}_1, \dots, \text{sh}_n)$.
- $\text{Recon}(1^\lambda, \text{sh}'_1, \dots, \text{sh}'_t)$:
 1. For all $i \in [t]$, let k' be the first λ bits of sh'_i . Then for all $j \in [t] \setminus \{i\}$, parse sh'_j as $(r'_j \| c'_j)$. If the vector $F(k', r'_j) \oplus c'_j$ ends in $\lambda - 1$ many 0's output the first bit of the vector.
 2. Otherwise output $\perp$.

Proof of Theorem 7.1.

Correctness: For any authorized set, there will exist at least one choice out of $sh'_1, \dots, sh'_t$, such that choosing the key k' from the first λ bits of this share results in the reconstruction algorithm outputting s . When the wrong choice of “center” node is chosen, this key k' is uniformly distributed, and we have $F(k', r'_i) \oplus c'_j$ is computationally close to uniformly distributed. The probability that the last λ bits are 0 is therefore negligible in λ . Therefore, the algorithm will output s with overwhelming probability. For the same reason, for any unauthorized set of size at least two, the algorithm will output $\perp$ with overwhelming probability. On any unauthorized set of size one, the algorithm will trivially output $\perp$.

Computational Secrecy and Computational U-Anonymity For any unauthorized set of size 1, the share is either uniformly distributed over $\{0, 1\}^{2\lambda}$ in the case of $sh_\star$ or is pseudorandom over $\{0, 1\}^{2\lambda}$ by the PRF pseudorandomness. For any unauthorized set of size at least 2 all shares must not be the center node, so the key is not given to the adversary. Therefore, the pseudorandomness of the PRF implies all shares in this set are pseudorandom, immediately implying computational U-Anonymity. This same proof implies computational secrecy. $\square$

7.2 Bilinear Maps for Bipartite Graph Access Structures

In this section, we show how to use pairings to construct share-size efficient fully anonymous secret sharing schemes for bipartite graph access structures. The type of bipartite graph access structures we consider are presented as follows.

Definition 7.3 (Bipartite Graph Access Structure). *A $((k_1, n_1), (k_2, n_2))$ -bipartite graph access structure consists of $n_1 + n_2$ parties that can be partitioned into sets L and R with n_1 and n_2 parties respectively. The minimal authorized sets consist of any k_1 parties from L and k_2 parties from R .*

We will provide a solution for the $((1, n), (1, n))$ -bipartite graph access structure. Here, an minimal authorized set consists of any share from the left partition and any share from the right partition.

Theorem 7.4. *Let λ be a security parameter. Assuming the hardness of DLIN (See Def. 7.7), there exists a U-FASS scheme for sharing a 1-bit secret with $(1 - \mu)$ -correctness for a negligible function μ , computational secrecy, and computational U-Anonymity for the $((1, n), (1, n))$ -bipartite graph access structure with a per-party share size of $O(\lambda)$.*

Remark 7.1. The construction immediately extends to $((1, n_1), (1, n_2))$ -bipartite graph access structures for $n_1 \neq n_2$.

We will use the following construction to prove the above theorem.

Construction 7.5 (Computational U-FASS for Complete Bipartite Graph). *Let $p = \text{poly}(\lambda) \ll q$.*

- $\text{Share}(1^\lambda, s \in \{1, \dots, p\} \subseteq \mathbb{Z}_q^*)$:
 1. Randomly sample a dual vectorspace pair $(\mathbb{B}, \mathbb{B}^*) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$ and randomly sample $\alpha, \beta \in \mathbb{Z}_q$.
 2. Multiplicatively share s into s_1 and s_2 by randomly choosing s_1 from $\mathbb{Z}_q$.
 3. For every party $i \in L$, set party i 's share to be $(g^{s_1 b_1 + r_{i,2} b_2 + r_{i,3} b_3}, g^{b_1 + \tilde{r}_{i,2} b_2 + \tilde{r}_{i,3} b_3})$ for freshly sampled uniform random $r_{i,2}, r_{i,3}, \tilde{r}_{i,2}, \tilde{r}_{i,3} \in \mathbb{Z}_q$.
 4. For every party $i \in R$, set party i 's share to be $(g^{s_2 \alpha b_1^* + r_{i,4} b_4^* + r_{i,5} b_5^*}, g^{\alpha b_1^* + \tilde{r}_{i,4} b_4^* + \tilde{r}_{i,5} b_5^*})$ for freshly sampled uniform random $r_{i,4}, r_{i,5}, \tilde{r}_{i,4}, \tilde{r}_{i,5} \in \mathbb{Z}_q$.
 5. Output the party shares.

• $\text{Recon}(1^\lambda, \vec{sh} = (sh_1, \dots, sh_t))$:

1. For every $i \in [t]$, parse $sh_i = (sh_{i,1}, sh_{i,2})$
2. For all $\binom{t}{2}$ many pairs of shares, say sh_i and sh_j , compute $x = e(sh_{i,1}, sh_{j,1})$ and $y = e(sh_{i,2}, sh_{j,2})$ and brute force the value of s from $y^s = x$ for values of s up to $p(\lambda)$ where p is the polynomial bound on the domain of the secret. Output $\perp$ if no such value of s is found.

Proof of Theorem 7.4. The proof follows from Lemma 7.6, Lemma 7.14, and Lemma 7.15. $\square$

Lemma 7.6 (Correctness). *There exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$, such that the above secret sharing scheme above satisfies $1 - \mu$ correctness.*

Proof. With $1 - \frac{1}{q} = 1 - \text{negl}(\lambda)$ probability the multiplicative sharing succeeds, i.e. s_1 is not 0 and $s_1 s_2 = s$. For any authorized set of shares, there exists a pair of shares such that

$$\begin{aligned} x &= e(sh_{i,1}, sh_{j,1}) = g^{s_1 s_2 \alpha} = g^{s\alpha} \in G \\ y &= e(sh_{i,2}, sh_{j,2}) = g^\alpha. \end{aligned}$$

Since the secret s comes from $[0, p(\lambda)] \subseteq \mathbb{Z}_q$, brute forcing s such that $y^s = x$ will recover s in polynomial time. For an unauthorized set, all shares come from only one of the two partitions. It suffices to consider what occurs when pairing between two left partition shares as a symmetric argument can be about pairings between two right partition shares. Pairing two left partition shares sh_i and sh_j will result in two group elements of the form $(g^{s_1 s_1 + r_{i,2} r_{j,2} + r_{i,3} r_{j,3}}, g^{1 + \tilde{r}_{i,2} \tilde{r}_{j,2} + \tilde{r}_{i,3} \tilde{r}_{j,3}})$ which are both distributed uniformly and independently of each other. The probability that the first value is a polynomial bounded power of the second is therefore negligible, and the reconstruction algorithm will output $\perp$ with all but negligible probability. The total number of pairs $\binom{t}{2} \leq \binom{n}{2} = O(n^2)$, so by a union bound, the probability that there exists a pair of left-left or right-right pairing the returns a non-bot value is negligible. $\square$

To show the pseudorandom unauthorized share property, we follow the framework provided in [Lew12] and define a simple subspace hiding assumption from which we can reduce the decisional linear problem. Our subspace hiding assumption can be seen a special and simpler case of the subspace hiding assumption provided by [Lew12]. We first recall the standard definition of the decisional linear assumption.

Definition 7.7 (Decisional Linear Assumption (DLIN)). *Given a group generator $\mathcal{G}$, we define the following distribution:*

$$\begin{aligned} \mathbb{G} &:= (q, G, G_T, e) \xleftarrow{\$} \mathcal{G}, \\ g &\xleftarrow{\$} G, a, b, c, d, r \xleftarrow{\$} \mathbb{Z}_q, \\ D &:= (g, g^a, g^b, g^{ac}, g^{bd}). \end{aligned}$$

We assume that for any PPT algorithm cA with output in $\{0, 1\}$, there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$, such that

$$\left| \Pr \left[\mathcal{A}(1^\lambda, D, g^{c+d}) = 1 \right] - \Pr \left[\mathcal{A}(1^\lambda, D, g^r) = 1 \right] \right| \leq \mu(\lambda)$$

The first subspace hiding assumption we introduce allows us to switch out the secret hidden in the $\mathbf{b}_1^*$ dimension of a share even in the presence of other share values that also contain the secret in the $\mathbf{b}_1^*$ dimension. Observe below that $\mathbb{D}$ allows adversary can obtain an arbitrary polynomial number of shares all containing the secret s in the $\mathbf{b}_1^*$ dimension. Namely, $\mathcal{A}$ can rerandomize the $\mathbf{b}_2^*$ and the $\mathbf{b}_3^*$ dimensions of the vector $g^{sb_1^* + \alpha b_2^* + \beta b_3^*}$. The assumption stipulates that nevertheless, the adversary is unable to tell what the secret is. This assumption will gives us computational secrecy.

Assumption 7.8 (Dual Vectorspace Subspace Hiding Assumption 1). *We will consider the following subspace assumption. Let $(\mathbb{B} = (\mathbf{b}_1, \dots, \mathbf{b}_5), \mathbb{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_5^*))$ denote a dual vector space over $\mathbb{F}_q$ where for all $i \in [5]$, $\mathbf{b}_i, \mathbf{b}_i^* \in \mathbb{F}_q^5$, we have $\langle \mathbf{b}_i, \mathbf{b}_i^* \rangle = 1$ and for all $i \neq j$, we have $\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle = 0$. Define the distributions:*

$$\begin{aligned} (\mathbb{B}, \mathbb{B}^*) &\xleftarrow{\$} \text{Dual}(\mathbb{F}_q^5) \\ s, \alpha, \beta, \gamma, \delta, \hat{r}_2, \hat{r}_3, \theta &\xleftarrow{\$} \mathbb{F}_q \\ \mathbb{D} &:= (g, g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*}, g^{\hat{r}_2\alpha\mathbf{b}_2^*}, g^{\hat{r}_3\beta\mathbf{b}_3^*}) \\ T_0 &:= g^{s\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^*} \\ T_1 &:= g^{\theta\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^*} \end{aligned}$$

Then we assume that for any PPT algorithm $\mathcal{A}$ that

$$|\Pr[\mathcal{A}(\mathbb{D}, T_0) = 1] - \Pr[\mathcal{A}(\mathbb{D}, T_1) = 1]| \leq \text{negl}(\lambda).$$

The next two subspace hiding assumptions are introduced to obtain computational U-Anonymity. By using the previous assumption, we can argue that the secret is hidden, but we cannot argue that share is pseudorandom. To do so, we need that both vectors in a share must be of the form $g^{\sum_{i=1}^5 r_i \mathbf{b}_i^*}$ where r_i are uniformly chosen. That is the vector in the exponent must be a uniformly chosen random vector over the entirety of $\mathbb{Z}_q^5$. Assumption 7.9 allows us to argue that a share of the form $g^{\sum_{i=1}^3 r_i \mathbf{b}_i^*}$ is indistinguishable to a share of the form $g^{\sum_{i=1}^4 r_i \mathbf{b}_i^*}$.

Assumption 7.9 (Dual Vectorspace Subspace Hiding Assumption 2). *We will consider the following subspace assumption. Let $(\mathbb{B} = (\mathbf{b}_1, \dots, \mathbf{b}_5), \mathbb{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_5^*))$ denote a dual vector space over $\mathbb{F}_q$ where for all $i \in [5]$, $\mathbf{b}_i, \mathbf{b}_i^* \in \mathbb{F}_q^5$, we have $\langle \mathbf{b}_i, \mathbf{b}_i^* \rangle = 1$ and for all $i \neq j$, we have $\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle = 0$. Define the distributions:*

$$\begin{aligned} (\mathbb{B}, \mathbb{B}^*) &\xleftarrow{\$} \text{Dual}(\mathbb{F}_q^5) \\ s, \theta, \alpha, \beta, \gamma, \delta, \hat{r}_2, \hat{r}_3, \hat{r}_4, r &\xleftarrow{\$} \mathbb{F}_q \\ \mathbb{D} &:= (g, g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*}, g^{\hat{r}_2\alpha\mathbf{b}_2^*}, g^{\hat{r}_3\beta\mathbf{b}_3^*}, g^{\hat{r}_4\mathbf{b}_4^*}) \\ T_0 &:= g^{\theta\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^*} \\ T_1 &:= g^{\theta\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^* + r\mathbf{b}_4^*} \end{aligned}$$

Then we assume that for any PPT algorithm $\mathcal{A}$ that

$$|\Pr[\mathcal{A}(\mathbb{D}, T_0) = 1] - \Pr[\mathcal{A}(\mathbb{D}, T_1) = 1]| \leq \text{negl}(\lambda).$$

Assumption 7.10 allows us to argue that a share of the form $g^{\sum_{i=1}^4 r_i \mathbf{b}_i^*}$ is indistinguishable to a share of the form $g^{\sum_{i=1}^5 r_i \mathbf{b}_i^*}$. Thus the next assumption allows us to complete the computational U-Anonymity argument.

Assumption 7.10 (Dual Vectorspace Subspace Hiding Assumption 3). *We will consider the following subspace assumption. Let $(\mathbb{B} = (\mathbf{b}_1, \dots, \mathbf{b}_5), \mathbb{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_5^*))$ denote a dual vector space over $\mathbb{F}_q$ where for all*

$i \in [5]$, $\mathbf{b}_i, \mathbf{b}_i^* \in \mathbb{F}_q^5$ we have $\langle \mathbf{b}_i, \mathbf{b}_i^* \rangle = 1$ and for all $i \neq j$, we have $\langle \mathbf{b}_i, \mathbf{b}_j^* \rangle = 0$. Define the distributions:

$$\begin{aligned} (\mathbb{B}, \mathbb{B}^*) &\stackrel{\$}{\leftarrow} \text{Dual}(\mathbb{F}_q^5) \\ s, \theta, \alpha, \beta, \gamma, \delta, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, r_4, r_5 &\stackrel{\$}{\leftarrow} \mathbb{F}_q \\ \mathbb{D} &:= (g, g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*}, g^{\hat{r}_2\alpha\mathbf{b}_2^*}, g^{\hat{r}_3\beta\mathbf{b}_3^*}, g^{\hat{r}_4\mathbf{b}_4^*}, g^{\hat{r}_5\mathbf{b}_5^*}) \\ T_0 &:= g^{\theta\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\gamma\mathbf{b}_3^* + r_4\mathbf{b}_4^*} \\ T_1 &:= g^{\theta\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\gamma\mathbf{b}_3^* + r_4\mathbf{b}_4^* + r_5\mathbf{b}_5^*} \end{aligned}$$

Then we assume that for any PPT algorithm $\mathcal{A}$ that

$$|\Pr[\mathcal{A}(\mathbb{D}, T_0) = 1] - \Pr[\mathcal{A}(\mathbb{D}, T_1) = 1]| \leq \text{negl}(\lambda).$$

Remark 7.2 (Symmetry of Assumptions). All these assumptions are symmetric, i.e. they are true for any permutation/ordering of $\mathbf{b}_1^*, \mathbf{b}_2^*, \dots, \mathbf{b}_5^*$. Similarly, these assumptions are symmetric and hold for $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_5$ as well since $\mathbb{B}$ and $\mathbb{B}^*$ are identically distributed. Therefore, it suffices to state these generic assumptions.

We now show that DLIN reduces to every one of three assumptions. Therefore, these three assumptions are at least as hard as DLIN, and our construction can simply be based on the hardness of DLIN.

Lemma 7.11. *If there exists a PPT algorithm $\mathcal{A}$ that breaks Assumption 7.8 above, then there exists a PPT algorithm $\mathcal{B}$ that breaks the DLIN assumption.*

Proof. We construct $\mathcal{B}$ as follows. On input $(g, g^\alpha, g^\beta, g^{\alpha\gamma}, g^{\beta\delta}, T)$, $\mathcal{B}$ will

1. Compute uniform dual basis $(\mathbb{V} := (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5), \mathbb{V}^* := (\mathbf{v}_1^*, \mathbf{v}_2^*, \mathbf{v}_3^*, \mathbf{v}_4^*, \mathbf{v}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$. That is,

$$\langle \mathbf{v}_i, \mathbf{v}_j^* \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}.$$

2. Define implicitly:

$$\begin{aligned} \mathbf{b}_1 &:= \mathbf{v}_1 - \alpha^{-1}\mathbf{v}_2 - \beta^{-1}\mathbf{v}_3 & \mathbf{b}_1^* &:= \mathbf{v}_1^* \\ \mathbf{b}_2 &:= \mathbf{v}_2 & \alpha\mathbf{b}_2^* &:= \alpha\mathbf{v}_2^* + \mathbf{v}_1^* \\ \mathbf{b}_3 &:= \mathbf{v}_3 & \beta\mathbf{b}_3^* &:= \beta\mathbf{v}_3^* + \mathbf{v}_1^* \\ \mathbf{b}_4 &:= \mathbf{v}_4 & \mathbf{b}_4^* &:= \mathbf{v}_4^* \\ \mathbf{b}_5 &:= \mathbf{v}_5 & \mathbf{b}_5^* &:= \mathbf{v}_5^* \end{aligned}$$

3. Sample a uniform random scalar $s, \hat{r}_2, \hat{r}_3 \in \mathbb{Z}_q$.

4. Compute

$$\begin{aligned} g_1 &:= (g)^{s\mathbf{v}_1^*} \cdot (g^\alpha)^{\mathbf{v}_2^*} (g)^{\mathbf{v}_1^*} \cdot (g^\beta)^{\mathbf{v}_3^*} (g)^{\mathbf{v}_1^*} = g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*} \\ g_2 &:= \left((g^\alpha)^{\mathbf{v}_2^*} (g)^{\mathbf{v}_1^*} \right)^{\hat{r}_2} = g^{\hat{r}_2\alpha\mathbf{b}_2^*} \\ g_3 &:= \left((g^\beta)^{\mathbf{v}_3^*} (g)^{\mathbf{v}_1^*} \right)^{\hat{r}_3} = g^{\hat{r}_3\beta\mathbf{b}_3^*} \\ g_T &:= (g)^{s\mathbf{v}_1^*} \cdot (g^{\alpha\gamma})^{\mathbf{v}_2^*} \cdot (g^{\beta\delta})^{\mathbf{v}_3^*} \cdot (T)^{\mathbf{v}_1^*} \end{aligned}$$

5. Output $\mathcal{A}(g, g_1, g_2, g_3, g_T)$.

First observe that $(\mathbf{b}_1, \dots, \mathbf{b}_5)$ and $(\mathbf{b}_1^*, \dots, \mathbf{b}_5^*)$ form an orthonormal basis of $\mathbb{Z}_q^5$. Moreover, they are uniformly distributed over the set of all dual vectorspace pairs because given any dual vectorspace pair $(\mathbb{B} = (\mathbf{b}_1, \dots, \mathbf{b}_5), \mathbb{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_5^*))$ satisfying the above constraints and any $\alpha \in \mathbb{Z}_q$ and $\beta \in \mathbb{Z}_q$, it is always possible to uniquely solve for vectors $\mathbf{v}_1, \dots, \mathbf{v}_5, \mathbf{v}_1^*, \dots, \mathbf{v}_5^* \in \mathbb{Z}_q^5$ that satisfy the dual basis properties. Explicitly, $\mathbf{v}_1, \dots, \mathbf{v}_5$ are related to $\mathbf{b}_1, \dots, \mathbf{b}_5$ by the following invertible change-of-basis matrix

$$[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_5] \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\alpha^{-1} & 1 & 0 & 0 & 0 \\ -\beta^{-1} & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \dots \quad \mathbf{b}_5],$$

which is non-singular for all values of $\alpha, \beta \in \mathbb{F}_q^*$. An analogous invertible change-of-basis matrix exists between $\mathbf{v}_1^*, \dots, \mathbf{v}_5^*$ and $\mathbf{b}_1^*, \dots, \mathbf{b}_5^*$.

As computed the above, the values g_1, g_2, g_3 computed by the algorithm $\mathcal{B}$ are exactly of the form when included with the group generator g , match exactly the form of the common input $\mathbb{D} = (g, g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*}, g^{\hat{r}_2\alpha\mathbf{b}_2^*}, g^{\hat{r}_3\beta\mathbf{b}_3^*})$ in the definition of Assumption 7.8. Then observe that if T is of the form $g^{\gamma+\delta}$, then

$$g_T = g^{s\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^*}.$$

Observe that this value is distributed identically to T_0 in Definition 7.8. If T is instead of the form $g^{\gamma+\delta+r}$, then observe that

$$g_T = g^{(s+r)\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^*} = g^{\theta\mathbf{b}_1^* + \alpha\gamma\mathbf{b}_2^* + \beta\delta\mathbf{b}_3^*},$$

where $\theta = s + r$ is uniformly distributed in $\mathbb{F}_q$ since r is uniformly distributed in $\mathbb{F}_q$. Observe that this value is distributed identically to T_1 in Definition 7.8. Therefore, the advantage of the reduction $\mathcal{B}$ in distinguishing the DLIN assumption is exactly given by advantage of $\mathcal{A}$ in distinguishing the subspace hiding problem defined in Assumption 7.8. $\square$

Lemma 7.12. *If there exists a PPT algorithm $\mathcal{A}$ that breaks Assumption 7.9 above, then there exists a PPT algorithm $\mathcal{B}$ that breaks the DLIN assumption.*

Proof. We construct $\mathcal{B}$ as follows. On input $(g, g^\alpha, g^\beta, g^{\alpha\gamma}, g^{\beta\delta}, T)$, $\mathcal{B}$ will

1. Compute uniform dual basis $(\mathbb{V} := (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5), \mathbb{V}^* := (\mathbf{v}_1^*, \mathbf{v}_2^*, \mathbf{v}_3^*, \mathbf{v}_4^*, \mathbf{v}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$. That is,

$$\langle \mathbf{v}_i, \mathbf{v}_j^* \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}.$$

2. Define implicitly:

$$\begin{array}{ll} \mathbf{b}_1 := \mathbf{v}_1 & \mathbf{b}_1^* := \mathbf{v}_1^* \\ \mathbf{b}_2 := \mathbf{v}_2 & \alpha\mathbf{b}_2^* := \alpha\mathbf{v}_2^* + \mathbf{v}_4^* \\ \mathbf{b}_3 := \mathbf{v}_3 & \beta\mathbf{b}_3^* := \beta\mathbf{v}_3^* + \mathbf{v}_4^* \\ \mathbf{b}_4 := \mathbf{v}_4 - \alpha^{-1}\mathbf{v}_2 - \beta^{-1}\mathbf{v}_3 & \mathbf{b}_4^* := \mathbf{v}_4^* \\ \mathbf{b}_5 := \mathbf{v}_5 & \mathbf{b}_5^* := \mathbf{v}_5^* \end{array}$$

3. Sample uniform random $s, \hat{r}_2, \hat{r}_3, \hat{r}_4 \theta \leftarrow \mathbb{Z}_q$.

4. Compute

$$\begin{aligned} g_1 &:= (g)^{s\mathbf{v}_1^*} \cdot (g^\alpha)^{\mathbf{v}_2^*} (g)^{\mathbf{v}_4^*} \cdot (g^\beta)^{\mathbf{v}_3^*} (g)^{\mathbf{v}_4^*} = g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*} \\ g_2 &:= \left((g^\alpha)^{\mathbf{v}_2^*} (g)^{\mathbf{v}_4^*} \right)^{\hat{r}_2} = g^{\hat{r}_2 \alpha \mathbf{b}_2^*} \\ g_3 &:= \left((g^\beta)^{\mathbf{v}_3^*} (g)^{\mathbf{v}_4^*} \right)^{\hat{r}_3} = g^{\hat{r}_3 \beta \mathbf{b}_3^*} \\ g_4 &:= (g)^{\hat{r}_4 \mathbf{v}_4^*} \\ g_T &:= (g)^{\theta \mathbf{v}_1^*} \cdot (g^{\alpha\gamma})^{\mathbf{v}_2^*} \cdot (g^{\beta\delta})^{\mathbf{v}_3^*} \cdot (T)^{\mathbf{v}_4^*} \end{aligned}$$

5. Output $\mathcal{A}(g, g_1, g_2, g_3, g_4, g_T)$.

The remainder of the proof follows in the exact same reasoning as the proof of Lemma 7.11. $\square$

Lemma 7.13. *If there exists a PPT algorithm $\mathcal{A}$ that breaks Assumption 7.10 above, then there exists a PPT algorithm $\mathcal{B}$ that breaks the DLIN assumption.*

Proof. On input $(g, g^\alpha, g^\beta, g^{\alpha\gamma}, g^{\beta\delta}, T)$, $\mathcal{B}$ will

1. Compute uniform dual basis $(\mathbb{V} := (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5), \mathbb{V}^* := (\mathbf{v}_1^*, \mathbf{v}_2^*, \mathbf{v}_3^*, \mathbf{v}_4^*, \mathbf{v}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$. That is,

$$\langle \mathbf{v}_i, \mathbf{v}_j^* \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}.$$

2. Define implicitly:

$$\begin{aligned} \mathbf{b}_1 &:= \mathbf{v}_1 & \mathbf{b}_1^* &:= \mathbf{v}_1^* \\ \mathbf{b}_2 &:= \mathbf{v}_2 & \alpha \mathbf{b}_2^* &:= \alpha \mathbf{v}_2^* + \mathbf{v}_5^* \\ \mathbf{b}_3 &:= \mathbf{v}_3 & \beta \mathbf{b}_3^* &:= \beta \mathbf{v}_3^* + \mathbf{v}_5^* \\ \mathbf{b}_4 &:= \mathbf{v}_4 & \mathbf{b}_4^* &:= \mathbf{v}_4^* \\ \mathbf{b}_5 &:= \mathbf{v}_5 - \alpha^{-1} \mathbf{v}_2 - \beta^{-1} \mathbf{v}_3 & \mathbf{b}_5^* &:= \mathbf{v}_5^* \end{aligned}$$

3. Sample uniform random $s, \theta, \hat{r}_2, \hat{r}_3, r_4 \in \mathbb{Z}_q$.

4. Compute

$$\begin{aligned} g_1 &:= (g)^{s\mathbf{v}_1^*} \cdot (g^\alpha)^{\mathbf{v}_2^*} (g)^{\mathbf{v}_5^*} \cdot (g^\beta)^{\mathbf{v}_3^*} (g)^{\mathbf{v}_5^*} = g^{s\mathbf{b}_1^* + \alpha\mathbf{b}_2^* + \beta\mathbf{b}_3^*} \\ g_2 &:= \left((g^\alpha)^{\mathbf{v}_2^*} (g)^{\mathbf{v}_5^*} \right)^{\hat{r}_2} = g^{\hat{r}_2 \alpha \mathbf{b}_2^*} \\ g_3 &:= \left((g^\beta)^{\mathbf{v}_3^*} (g)^{\mathbf{v}_5^*} \right)^{\hat{r}_3} = g^{\hat{r}_3 \beta \mathbf{b}_3^*} \\ g_4 &:= (g)^{\hat{r}_4 \mathbf{v}_4^*} \\ g_5 &:= (g)^{\hat{r}_5 \mathbf{v}_5^*} \\ g_T &:= (g)^{\theta \mathbf{v}_1^*} \cdot (g^{\alpha\gamma})^{\mathbf{v}_2^*} \cdot (g^{\beta\delta})^{\mathbf{v}_3^*} \cdot (g)^{r_4 \mathbf{v}_4^*} \cdot (T)^{\mathbf{v}_5^*} \end{aligned}$$

5. Output $\mathcal{A}(g, g_1, g_2, g_3, g_4, g_5, g_T)$.

The remainder of the proof follows in the exact same reasoning as the proof of Lemma 7.11. $\square$

Lemma 7.14 (Computational Secrecy). *Construction 7.5 satisfies computational secrecy.*

Proof. Let X be any unauthorized set of shares for a sharing of the secret s . Let g be a generator for the group G . Let $|X| = t$ and let enumerate the shares in X to be $x_1, \dots, x_t$. Since X is unauthorized, we can assume without loss of generality that X consists of only shares from the right partition, namely that each share x_i is of the form

$$x_i = (x_{i,1}, x_{i,2}) = (g^{s_2 \alpha b_1^* + r_{i,4} b_4^* + r_{i,5} b_5^*}, g^{\alpha b_1^* + \tilde{r}_{i,4} b_4^* + \tilde{r}_{i,5} b_5^*}).$$

We now define the following hybrids:

1. Define $H_0^{\mathcal{A}}(1^\lambda, g)$:

- (a) Sample uniform $s_2 \leftarrow \mathbb{Z}_q^*$, sample uniform $\alpha, \{r_{i,4}\}_{i \in [t]}, \{r_{i,5}\}_{i \in [t]} \leftarrow \mathbb{Z}_q$.
- (b) Sample $(\mathbb{B} := (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5), \mathbb{B}^* := (\mathbf{b}_1^*, \mathbf{b}_2^*, \mathbf{b}_3^*, \mathbf{b}_4^*, \mathbf{b}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$.
- (c) Set for $i \in [t]$,

$$x_i = (x_{i,1}, x_{i,2}) = (g^{s_2 \alpha b_1^* + r_{i,4} b_4^* + r_{i,5} b_5^*}, g^{\alpha b_1^* + \tilde{r}_{i,4} b_4^* + \tilde{r}_{i,5} b_5^*}).$$

- (d) Output $\mathcal{A}(1^\lambda, g, x_1, \dots, x_t)$.

2. For $i \in [t]$, define $H_{i,1}^{\mathcal{A}}(1^\lambda, g)$:

- (a) Sample uniform $s_2 \leftarrow \mathbb{Z}_q^*$, sample uniform $\alpha \leftarrow \mathbb{Z}_q, \{r_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [t]}, \{r_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [t]}, \{\tilde{r}_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [t]}, \{\tilde{r}_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [t]}$.
- (b) Sample $(\mathbb{B} := (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5), \mathbb{B}^* := (\mathbf{b}_1^*, \mathbf{b}_2^*, \mathbf{b}_3^*, \mathbf{b}_4^*, \mathbf{b}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$.
- (c) Set for $j \in [t]$, sample uniform scalars $a_j, a'_j, \hat{r}_{j,4}, \hat{r}_{j,5}, \tilde{r}_{j,4}, \tilde{r}_{j,5} \leftarrow \mathbb{Z}_q$,

$$x_j = (x_{j,1}, x_{j,2}) = (g^{s_2 \alpha b_1^* + r_{j,4} b_4^* + r_{j,5} b_5^*}, g^{\alpha b_1^* + \tilde{r}_{j,4} b_4^* + \tilde{r}_{j,5} b_5^*}).$$

- (d) Sample uniform $\{a_j \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{c_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{c_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{\tilde{c}_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{\tilde{c}_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [i]}$.

- (e) For each $j \in [i]$, set

$$y_i = (y_{i,1}, y_{i,2}) = (g^{a_j b_1^* + c_{j,4} b_4^* + c_{j,5} b_5^*}, g^{a'_j b_1^* + \tilde{c}_{j,4} b_4^* + \tilde{c}_{j,5} b_5^*}).$$

- (f) Output $\mathcal{A}(1^\lambda, g, y_1, \dots, y_{i-1}, (y_{i,1}, x_{i,2}), x_{i+1}, \dots, x_t)$.

3. For $i \in [t]$, define $H_{i,2}^{\mathcal{A}}(1^\lambda, g)$:

- (a) Sample uniform $s_2 \leftarrow \mathbb{Z}_q^*$, sample uniform $\alpha \leftarrow \mathbb{Z}_q, \{r_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [t]}, \{r_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [t]}, \{\tilde{r}_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [t]}, \{\tilde{r}_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [t]}$.
- (b) Sample $(\mathbb{B} := (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5), \mathbb{B}^* := (\mathbf{b}_1^*, \mathbf{b}_2^*, \mathbf{b}_3^*, \mathbf{b}_4^*, \mathbf{b}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$.
- (c) Set for $j \in [t]$, sample uniform scalars $a_j, a'_j, \hat{r}_{j,4}, \hat{r}_{j,5}, \tilde{r}_{j,4}, \tilde{r}_{j,5} \leftarrow \mathbb{Z}_q$,

$$x_j = (x_{j,1}, x_{j,2}) = (g^{s_2 \alpha b_1^* + r_{j,4} b_4^* + r_{j,5} b_5^*}, g^{\alpha b_1^* + \tilde{r}_{j,4} b_4^* + \tilde{r}_{j,5} b_5^*}).$$

- (d) Sample uniform $\{a_j \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{c_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{c_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{\tilde{c}_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{\tilde{c}_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [i]}$.
- (e) For each $j \in [i]$, set y_j randomly over the $\mathbf{b}_1^*, \mathbf{b}_4^*, \mathbf{b}_5^*$ dimensions, i.e.

$$y_j = (y_{j,1}, y_{j,2}) = (g^{a_j \mathbf{b}_1^* + c_{j,4} \mathbf{b}_4^* + c_{j,5} \mathbf{b}_5^*}, g^{a'_j \mathbf{b}_1^* + \tilde{c}_{j,4} \mathbf{b}_4^* + \tilde{c}_{j,5} \mathbf{b}_5^*}).$$

- (f) Output $\mathcal{A}(1^\lambda, g, y_1, \dots, y_{i-1}, (y_{i,1}, \mathbf{y}_{i,2}), x_{i+1}, \dots, x_t)$.

Observe that the distribution of the input to $\mathcal{A}$ in the hybrid $H_0^\mathcal{A}$ is exactly that produced by the Share algorithm in Construction 7.5. Observe that the distribution of the input to $\mathcal{A}$ in the hybrid $H_{t,2}^\mathcal{A}$ is exactly t many pairs of vectors of groups, none of which have the secret s_2 . For every step of this hybrid sequence, we will use Assumption 7.8. Per Remark 7.2, this assumption is symmetric with respect to $\mathbf{b}_1^*, \mathbf{b}_4^*, \mathbf{b}_5^*$.

1. For convenience, define $H_{0,2}^\mathcal{A} := H_0^\mathcal{A}$.
2. For all $i \in [t]$, $H_{i,1}^\mathcal{A} \approx_c H_{i,2}^\mathcal{A}$.
3. For all $i \in [t]$, $H_{i,2}^\mathcal{A} \approx_c H_{i+1,1}^\mathcal{A}$.

We first argue that $H_{i,2}^\mathcal{A} \approx_c H_{i+1,1}^\mathcal{A}$ for $i \in \{0, \dots, t-1\}$ by construction a reduction $\mathcal{B}$ that distinguishes Assumption 7.8. $\mathcal{B}$ is given as input (g, g_1, g_2, g_3, T) where

$$\begin{aligned} g_1 &= g^{s_2 \mathbf{b}_1^* + \alpha \mathbf{b}_4^* + \beta \mathbf{b}_5^*} \\ g_2 &= g^{\hat{r}_4 \alpha \mathbf{b}_4^*} \\ g_3 &= g^{\hat{r}_5 \beta \mathbf{b}_5^*} \end{aligned}$$

and where T either has a $\mathbf{b}_1^*$ coefficient of s_2 or a random unrelated coefficient. The reduction $\mathcal{B}$ sets

$$x_{i,1} \leftarrow g_1 g_2^{r_i} g_3^{r'_i}$$

for uniform random $r_i, r'_i \leftarrow \mathbb{Z}_q$ for all $i \in [t]$. It will sample $\alpha^* \leftarrow \mathbb{Z}_q$ and set

$$x_{i,2} \leftarrow g_1^{\alpha^*} g_2^{\tilde{r}_i} g_3^{\tilde{r}'_i}$$

for uniform random $\tilde{r}_i, \tilde{r}'_i \leftarrow \mathbb{Z}_q$ for all $i \in [t]$. Observe that this ensures that the coefficient for the $\mathbf{b}_1^*$ dimension in $x_{i,1}$ is equivalent to the coefficient for the $\mathbf{b}_1^*$ dimension in $x_{j,1}$ for all $i \neq j$. Moreover this coefficient is uniformly distributed over $\mathbb{Z}_q$. Similarly, the coefficient for the $\mathbf{b}_1^*$ dimension for $x_{i,2}$ is equivalent to the coefficient for the $\mathbf{b}_1^*$ dimension in $x_{j,2}$ for all $i \neq j$ and is uniformly distributed over $\mathbb{Z}_q$. The reduction $\mathcal{B}$ then samples uniform $\{a_j \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{c_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{c_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{\tilde{c}_{j,4} \leftarrow \mathbb{Z}_q\}_{j \in [i]}, \{\tilde{c}_{j,5} \leftarrow \mathbb{Z}_q\}_{j \in [i]}$. For each $j \in [i]$, set y_j randomly over the $\mathbf{b}_1^*, \mathbf{b}_4^*, \mathbf{b}_5^*$ dimensions, i.e.

$$y_i = (y_{i,1}, y_{i,2}) = (g^{a_j \mathbf{b}_1^* + c_{j,4} \mathbf{b}_4^* + c_{j,5} \mathbf{b}_5^*}, g^{a'_j \mathbf{b}_1^* + \tilde{c}_{j,4} \mathbf{b}_4^* + \tilde{c}_{j,5} \mathbf{b}_5^*}).$$

$\mathcal{B}$ outputs

$$\mathcal{A}(g, y_1, \dots, (T, x_{i,2}), x_{i+1}, \dots, x_t).$$

Observe that if T has s_2 in the $\mathbf{b}_1^*$ dimension, then T is exactly of the form $x_{i,1}$ and if T has instead a uniform random value in the $\mathbf{b}_1^*$ dimension, then T is exactly distributed identically to $y_{i,2}$.

To argue $H_{i,2}^\mathcal{A} \approx_c H_{i+1,1}^\mathcal{A}$ for $i \in \{0, \dots, t-1\}$, the exact same argument works with a slight twist. We instead sample $\alpha^* \leftarrow \mathbb{Z}_q$ and set

$$x_{i,1} \leftarrow g_1^{\alpha^*} g_2^{\tilde{r}_i} g_3^{\tilde{r}'_i}$$

for uniform random $\tilde{r}_i, \tilde{r}'_i \leftarrow \mathbb{Z}_q$ for all $i \in [t]$. And we instead sample

$$x_{i,2} \leftarrow g_1 g_2^{r_i} g_3^{r'_i}$$

for uniform random $r_i, r'_i \leftarrow \mathbb{Z}_q$ for all $i \in [t]$. In other words, the relationship of $x_{i,1}$ and $x_{i,2}$ with respect to T is exactly flipped from the previous reduction. Finally, $\mathcal{B}$ outputs

$$\mathcal{A}(g, y_1, \dots, (y_{i,1}, T), x_{i+1}, \dots, x_t),$$

and the T is either distributed identically to $x_{i,2}$ or to $y_{i,2}$.

This concludes the proof of secrecy. $\square$

Lemma 7.15 (Computational U-Anonymity). *Construction 7.5 satisfies computational U-Anonymity.*

Proof. To argue computational U-Anonymity, we extend the hybrid argument in the proof of Lemma 7.14. We import and simplify hybrid $H_{t,2}^{\mathcal{A}}$ from the proof of Lemma 7.14:

- $H_{t,2}^{\mathcal{A}}(1^\lambda, g)$:
 1. Sample $(\mathbb{B} := (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5), \mathbb{B}^* := (\mathbf{b}_1^*, \mathbf{b}_2^*, \mathbf{b}_3^*, \mathbf{b}_4^*, \mathbf{b}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$.
 2. For each $j \in [2t]$, set y_j randomly over the $\mathbf{b}_1^*, \mathbf{b}_4^*, \mathbf{b}_5^*$ dimensions, i.e. for uniform random $r_{j,1}, r_{j,4}, r_{j,5} \leftarrow \mathbb{Z}_q$,

$$y_j = g^{r_{j,1}\mathbf{b}_1^* + r_{j,4}\mathbf{b}_4^* + r_{j,5}\mathbf{b}_5^*}.$$
 3. Output $\mathcal{A}(1^\lambda, g, y_1, \dots, y_t)$.

This notational simplification and the use of $[2t]$ instead of $[t]$ is possible because all the exponents are uniform vectors over the span of $\{\mathbf{b}_1^*, \mathbf{b}_4^*, \mathbf{b}_5^*\}$, so there is no longer any point in using the notation $(y_{i,1}, y_{i,2})$ used previously in Lemma 7.14 since both vectors are distributed identically in $H_{t,2}^{\mathcal{A}}$.

To attain U-Anonymity, we first use Assumption 7.9 to replace every y_j with $\tilde{y}_j = g^{r_{j,1}\mathbf{b}_1^* + r_{j,2}\mathbf{b}_2^* + r_{j,4}\mathbf{b}_4^* + r_{j,5}\mathbf{b}_5^*}$. In other words, we grow the exponent randomly in one more dimension, namely the $\mathbf{b}_2^*$ dimension. Then, we use Assumption 7.10 to grow the exponent randomly in one more dimension, namely the $\mathbf{b}_3^*$ dimension. Namely, we replace every $\tilde{y}_j$ with $\tilde{\tilde{y}}_j = g^{r_{j,1}\mathbf{b}_1^* + r_{j,2}\mathbf{b}_2^* + r_{j,3}\mathbf{b}_3^* + r_{j,4}\mathbf{b}_4^* + r_{j,5}\mathbf{b}_5^*}$. In the final hybrid, all the $2t$ vectors of length 5 are uniform randomly distributed over $G^5 := \underbrace{G \times \dots \times G}_{5 \text{ times}}$. More formally, we define another

sequence of hybrids

- $\tilde{H}_0^{\mathcal{A}}(1^\lambda, g) := H_{t,2}^{\mathcal{A}}(1^\lambda, g)$.
- For $i \in [2t]$, define $\tilde{H}_i^{\mathcal{A}}(1^\lambda, g)$:
 1. Sample $(\mathbb{B} := (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5), \mathbb{B}^* := (\mathbf{b}_1^*, \mathbf{b}_2^*, \mathbf{b}_3^*, \mathbf{b}_4^*, \mathbf{b}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$.
 2. For each $j \in [2t]$, sample uniform random $r_{j,1}, r_{j,2}, r_{j,4}, r_{j,5} \leftarrow \mathbb{Z}_q$, and define $y_j, \tilde{y}_j$ as defined above.
 3. Output $\mathcal{A}(1^\lambda, g, \tilde{y}_1, \dots, \tilde{y}_i, y_{i+1}, \dots, y_{2t})$.
- $\tilde{\tilde{H}}_0^{\mathcal{A}}(1^\lambda, g) := \tilde{H}_i^{\mathcal{A}}(1^\lambda, g)$.
- For $i \in [2t]$, define $\tilde{\tilde{H}}_i^{\mathcal{A}}(1^\lambda, g)$:
 1. Sample $(\mathbb{B} := (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4, \mathbf{b}_5), \mathbb{B}^* := (\mathbf{b}_1^*, \mathbf{b}_2^*, \mathbf{b}_3^*, \mathbf{b}_4^*, \mathbf{b}_5^*)) \leftarrow \text{Dual}(\mathbb{Z}_q^5)$.

2. For each $j \in [2t]$, sample uniform random $r_{j,1}, r_{j,2}, r_{j,3}, r_{j,4}, r_{j,5} \leftarrow \mathbb{Z}_q$, and define $\tilde{y}_j, \tilde{\tilde{y}}_j$ as defined above.
3. Output $\mathcal{A}(1^\lambda, g, \tilde{y}_1, \dots, \tilde{\tilde{y}}_i, \tilde{y}_{i+1}, \dots, \tilde{y}_{2t})$.

For $i \in \{0, \dots, 2t - 1\}$, the reduction idea for $\tilde{H}_i^{\mathcal{A}} \approx_c \tilde{H}_{i+1}^{\mathcal{A}}$ from Assumption 7.9 is identical for arguing $\tilde{H}_i^{\mathcal{A}} \approx_c \tilde{H}_{i+1}^{\mathcal{A}}$ from Assumption 7.10. We construct a reduction $\mathcal{B}$ from Assumption 7.9 to the hybrid indistinguishability. This reduction receives the input $(g, g_1, g_2, g_3, g_4, T)$ where

$$\begin{aligned} g_1 &= g^{\text{sb}_1^* + \alpha \text{b}_4^* + \beta \text{b}_5^*} \\ g_2 &= g^{\hat{r}_4 \alpha \text{b}_4^*} \\ g_3 &= g^{\hat{r}_5 \beta \text{b}_5^*} \\ g_4 &= g^{\hat{r}_2 \text{b}_2^*} \end{aligned}$$

and either T is uniformly distributed over the span of $\{\text{b}_1^*, \text{b}_4^*, \text{b}_5^*\}$, i.e. $T = g^{\theta \text{b}_1^* + \alpha \gamma \text{b}_4^* + \beta \delta \text{b}_5^*}$ or T is uniformly distributed over the span of $\{\text{b}_1^*, \text{b}_2^*, \text{b}_4^*, \text{b}_5^*\}$, i.e. $T = g^{\theta \text{b}_1^* + \alpha \gamma \text{b}_4^* + \beta \delta \text{b}_5^* + r \text{b}_2^*}$. Then observe that $\mathcal{B}$ can produce generate random vectors in the span of $\{\text{b}_1^*, \text{b}_4^*, \text{b}_5^*\}$, i.e. of the form of the form y_i . Similarly, $\mathcal{B}$ can generate random vectors in the span of $\{\text{b}_1^*, \text{b}_2^*, \text{b}_4^*, \text{b}_5^*\}$, i.e. of the form $\tilde{y}_i$. The former is done by taking random powers of g_1, g_2 , and g_3 and multiplying them, and the latter is done by taking random powers of g_1, g_2, g_3 , and g_4 and multiplying them. Therefore $\mathcal{B}$ can run $\mathcal{A}(1^\lambda, g, \tilde{y}_1, \dots, \tilde{y}_i, T, y_{i+2}, \dots, y_{2t})$, where $\mathcal{A}$'s inputs are distributed either exactly as in $\tilde{H}_i^{\mathcal{A}}$ when T is in the span of $\{\text{b}_1^*, \text{b}_4^*, \text{b}_5^*\}$ or $\tilde{H}_{i+1}^{\mathcal{A}}$ when T is in the span of $\{\text{b}_1^*, \text{b}_2^*, \text{b}_4^*, \text{b}_5^*\}$.

This concludes the proof of U-Anonymity. $\square$

7.3 A General Compiler from Standard Secret Sharing to FASS

We now describe a transformation from any (information-theoretic or computational) standard secret sharing scheme $\text{SS} = (\text{Share}, \text{Recon})$ (see Def. 3.2) for any monotone access structure Γ to a computational multi-dealer fully anonymous secret sharing (M-FASS) scheme $\text{MFASS} = (\text{MFASS.Share}, \text{MFASS.Recon})$ for Γ . If the per-party share size is bounded by S and a Turing machine representation of Recon runs in time bounded by T , then the resulting M-FASS scheme has per-party share size $S \text{ poly}(\lambda, \log T)$.

Theorem 7.16. *For any access structure $\Gamma = \{\Gamma_\lambda\}$ where Γ_λ involves $n = n(\lambda)$ parties, if there exists a (information-theoretically or computationally secure) standard secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ (see Def. 3.2), where Recon can be represented by a TM that runs in time bounded by $T : \mathbb{N} \rightarrow \mathbb{N}$ and where the per-party share size is bounded by S bits, then assuming the existence of the following primitives:*

1. Secret key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$.
2. Multi-bit compute-and-compare obfuscation for any distribution $\mathcal{P}_{\text{MBCC}}^{\text{TM}}$ that satisfies α -pseudo-entropy for any polynomial $\alpha = \alpha(\lambda)$.
3. Any injective PRG $(\text{Gen}, \mathcal{G})$ in NC^1 with polynomial stretch $m = m(\lambda) \geq \alpha(\lambda)$.

there exists a computationally secure M-FASS scheme for Γ with per-party share size $S \cdot \text{poly}(\lambda, \log T)$ bits.

We prove Theorem 7.16 using Construction 7.17.

Construction 7.17 (Compiler to Computational M-FASS). *M-FASS Scheme $(\text{MFASS.Share}, \text{MFASS.Recon})$*
 $\text{MFASS.Share}(1^\lambda, \mathbf{s} \in \{0, 1\}^\lambda)$:

1. Sample a uniform random string $\mathbf{r} \in \{0,1\}^\lambda$ and compute the shares $\widetilde{\mathbf{sh}} = (\widetilde{\mathbf{sh}}_1, \dots, \widetilde{\mathbf{sh}}_n) \leftarrow \text{Share}(1^\lambda, \mathbf{r})$. Let $M := \max_{i \in [n]} |\widetilde{\mathbf{sh}}_i|$.
2. Sample PRG $G \leftarrow \text{Gen}(1^\lambda, 1^{n \cdot m})$, and let g_i be the i th block of the output of G (equivalently, one can view G as a concatenation of λ -many PRGs $g_1, \dots, g_n : \{0,1\}^\lambda \rightarrow \{0,1\}^{m(\lambda)}$). Then, compute for all $i \in [n]$, $\mathbf{y}_i \leftarrow g_i(\mathbf{r})$.
3. Generate a secret key: $\text{sk} \leftarrow \text{Enc.Gen}(1^\lambda)$.
4. Encrypt the padded shares: For $i \in [n]$, compute $\text{ct}_i \leftarrow \text{Enc}\left(1^\lambda, \text{sk}, \text{Pad}_M(\mathbf{e}_i \parallel \widetilde{\mathbf{sh}}_i)\right)$ where $\mathbf{e}_i \in \{0,1\}^n$ is the i th elementary vector denoting the party identity i , and where $\text{Pad}_M(x) = (\ell_x \parallel x \parallel \bar{0})$ for $\ell_x \in \{0,1\}^{\lceil \log(n+M) \rceil}$ that describes the length of x , and $\bar{0} = 0^{n+M-|x|}$.
5. For each $j \in [n]$, generate the function that has sk hardcoded that takes as input the string description of a circuit $\langle F \rangle$ and an input of arbitrary length x :

$f_{\text{sk},j}(\langle F \rangle, x)$:

1. Parse x as a tuple of ciphertexts $(\text{ct}_1, \dots, \text{ct}_k)$.
2. Decrypt the ciphertexts: For $i \in [k]$, let $\text{Pad}_M(\mathbf{e}'_i \parallel \widetilde{\mathbf{sh}}'_i) \leftarrow \text{Dec}(1^\lambda, \text{sk}, \text{ct}_i)$ and strip off the padding.
3. Run F for at most T steps: $F\left(1^\lambda, \mathbf{x}, (\widetilde{\mathbf{sh}}'_i)_{i \in [k]}\right)$ where $\mathbf{x} = \sum_i \mathbf{e}'_i$ and let $\mathbf{r}'$ be the output of F . Output $g_j(\mathbf{r}')$.

6. For $j \in [n]$, generate a fresh obfuscation $O_j = \text{MBCC}[f_{\text{sk},j}, s, y_j]$.
7. Output the shares $(\text{sh}_j = (O_j, \text{ct}_j))_{j \in [n]}$.

$\text{MFASS.Recon}(1^\lambda, \text{sh}')$:

1. Parse the shares as obfuscations and ciphertexts $((O_i, \text{ct}_i))_{i \in [k]}$.
2. Output $O_1(\langle \text{Recon} \rangle, \text{ct}_1, \dots, \text{ct}_k)$.

Proof of Theorem 7.16.

Correctness: Consider any authorized set of share $\text{sh}_A = \{(O_i, \text{ct}_i)\}_{i \in [k]}$ for some k . By the correctness of the MBCC obfuscation, $O_1(\langle \text{Recon} \rangle, (\text{ct}_i)_{i \in [k]})$ will output s with overwhelming probability if $f_{\text{sk}}(\text{ct}_1, \dots, \text{ct}_k) = y_1$. Decryption correctness guarantees that $\text{Dec}(1^\lambda, \text{sk}, \text{ct}_i) = \widetilde{\mathbf{sh}}_i$ and the correctness of the reconstruction algorithm for authorized sets guarantees that $\text{Recon}\left(1^\lambda, (\text{sh}_i)_{i \in [k]}\right) = \mathbf{r}$. Therefore, the output of $f_{\text{sk}}(\text{ct}_1, \dots, \text{ct}_k)$ will be $y_1 = g_1(\mathbf{r})$.

For any unauthorized set of shares, the reconstruction algorithm will not output $\mathbf{r}$ with overwhelming probability, so by the injectivity of g_i , the output of $f_{\text{sk},i}$ will not be equal to $g_i(\mathbf{r})$.

Secrecy: Consider any ensemble $\mathcal{D} = \{D_\lambda\}$ that satisfies α -pseudo-entropy where $(f_{\text{sk}}, s, \text{aux}) \leftarrow D_\lambda$ is such that f_{sk} is a function sampled as defined the construction and s is the fixed secret. That is, we allow the adversary any reasonable auxiliary information that maintains the α -pseudo-entropy of f_{sk} , which when

considered alone reveals zero information about the target point y . Choose the PRG length parameter $m(\lambda) \geq \alpha(\lambda)$. Consider any unauthorized set of shares $\text{sh}_B = \{(O_i, \text{ct}_i)\}_{i \in [k]}$ for some k where O_i are MBCC obfuscations with respect to f_{sk} where $(f_{\text{sk}}, \mathbf{s}, \text{aux}) \leftarrow D_\lambda$ that satisfies α -pseudo-entropy. We now show that

$$\mathbf{H}_{\text{HILL}}\left(y_1 \mid (f_{\text{sk},i})_{i \in [k]}, \mathbf{s}, \text{aux}, (O_i)_{i \in \{2, \dots, k\}}, (\text{ct}_i)_{i \in [k]}\right) \geq m(\lambda) \geq \alpha(\lambda) \quad (25)$$

where $m(\lambda)$ is the length of the output of g_i . First we will observe that $\mathbf{s}$ is a fixed vector completely independent of the choice of y_1 , and we will remove $\mathbf{s}$ from subsequent indistinguishability arguments. Define the random variables

$$\begin{aligned} X &:= y_1 \\ Y &:= ((f_{\text{sk},i})_{i \in [k]}, \text{aux}, (O_i)_{i \in \{2, \dots, k\}}, (\text{ct}_i)_{i \in [k]}). \end{aligned}$$

We will show that $(X, Y) \approx_c (U_m, Y)$ where U_m is the uniform distribution over $\{0, 1\}^{m(\lambda)}$.

First, the statistical (resp. computational) standard secret sharing secrecy implies the statistical (resp. computational) indistinguishability

$$(X, Y) \approx_{s/c} \left(X, \left((f_{\text{sk},i})_{i \in [k]}, \text{aux}, (O_i)_{i \in \{2, \dots, k\}}, (\text{ct}'_i)_{i \in [k]} \right) \right)$$

where the distribution of ct'_i is given by first sampling a secret $\mathbf{r}'$ independently of $\mathbf{r}$ and shares $\tilde{\text{sh}}' \leftarrow \text{Share}(1^\lambda, \mathbf{r}')$, and let $\text{ct}'_i = \text{Enc}(1^\lambda, \text{sk}, \tilde{\text{sh}}'_i)$. If this were not true, then we could construct a reduction B breaking the standard secret sharing secrecy by having B choose the secrets $\mathbf{r}$ and $\mathbf{r}'$, computing $G(\mathbf{r})$, asking the challenger for the shares sh , and then using the hybrid distinguisher on the input $(\text{sh}, G(\mathbf{r}))$.

Then, we can appeal to the PRG pseudorandomness to swap all the y_i to uniform random values since $G(\mathbf{r}) = y_1 y_2 \dots y_n$ and X was exactly defined to be the distribution of y_1 ,

$$\left(X, \left((f_{\text{sk},i})_{i \in [k]}, \text{aux}, (O_i)_{i \in \{2, \dots, k\}}, (\text{ct}'_i)_{i \in [k]} \right) \right) \approx_c \left(U_m, \left((f_{\text{sk},i})_{i \in [k]}, \text{aux}, (\tilde{O}_i)_{i \in \{2, \dots, k\}}, (\text{ct}'_i)_{i \in [k]} \right) \right)$$

where $\tilde{O}_i = \text{Obf}(1^\lambda, \text{MBCC}[f_{\text{sk},i}, \mathbf{s}, \mathbf{u}_i])$.

Finally, we can obtain the claim that

$$(U_m, ((f_{\text{sk},i})_{i \in [k]}, \text{aux}, (\tilde{O}_i)_{i \in \{2, \dots, k\}}, (\text{ct}'_i)_{i \in [k]})) \approx_c (U_m, Y)$$

by repeating the previous hybrids in the reverse direction, however, this time with treating U_m as uniform auxiliary information. Then, since U_m is an independent uniform random variable from Y , the average conditional min-entropy satisfies $\tilde{\mathbf{H}}_\infty(U_m \mid Y) \geq m(\lambda)$. Finally, the definition of HILL entropy (Def. 3.18) implies Equation 25.

By treating $O_2, \dots, O_k, \text{ct}_1, \dots, \text{ct}_k$ as auxiliary input and using Equation 25, we can appeal to the security (Def. 3.16) of the compute-and-compare obfuscation to claim:

$$((O_1, \dots, O_k), (\text{ct}_1, \dots, \text{ct}_k)) \approx_c \left((\text{Sim}(1^\lambda, \text{params}), O_2, \dots, O_k), (\text{ct}_1, \dots, \text{ct}_k) \right)$$

where the public parameters params consists of the input and output lengths for f_{sk} in unary. Observe that we can continue this same exact argument for every i , since we need only argue that the pseudo-entropy

for each $y_i = g_i(\mathbf{r})$ is at least $\alpha(\lambda)$, and the pseudo-entropy each time is $m(\lambda) \geq \alpha(\lambda)$. Putting together this sequence of arguments gives:

$$((O_1, \dots, O_k), (ct_1, \dots, ct_k)) \approx_c \left((\text{Sim}(1^\lambda, \text{params}), \dots, \text{Sim}(1^\lambda, \text{params})), (ct_1, \dots, ct_k) \right).$$

At this point we can appeal to the secret key encryption scheme's semantic security and replace all the ciphertexts with encryptions of 0.

M-FASS: First, note that all shares are of the same size due to the padding. Then, observe that the proof of secrecy extends immediately to the setting in which the unauthorized set of shares consists of all-but-one shares of some secret s_0 and a single share from a sharing of some secret s_1 . Namely, for any secrets s_0, s_1 and any unauthorized sets B and $\{a\}$ and any subset $B' \subseteq B$ where $|B'| = |B| - 1$, define the following hybrids:

- $H_0^{\mathcal{A}}(1^\lambda, s_0, s_1, B, B', a)$:
 1. Let $\text{sh}^{(0)} \leftarrow \text{MFASS.Share}(1^\lambda, s_0)$ and $\text{sh}^{(1)} \leftarrow \text{MFASS.Share}(1^\lambda, s_1)$.
 2. Output $\mathcal{A}(1^\lambda, s_0, s_1, B, B', a, \text{sh}_{B'}^{(0)} \parallel \text{sh}_{\{a\}}^{(1)})$.
- $H_1^{\mathcal{A}}(1^\lambda, s_0, s_1, B, B', a)$:
 1. Let $\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)$ and $\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)$.
 2. Output $\mathcal{A}(1^\lambda, s_0, s_1, B, B', a, \text{sh}_{B'}^{(0)} \parallel (\text{Sim}(1^\lambda, \text{params}), \text{ct}))$.
- $H_2^{\mathcal{A}}(1^\lambda, s_0, s_1, B, B', a)$:
 1. Let $\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)$ and $\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)$.
 2. Output $\mathcal{A}(1^\lambda, s_0, s_1, B, B', a, \text{sh}_{B'}^{(0)} \parallel (\text{Sim}(1^\lambda, \text{params}), \text{Enc}(1^\lambda, 0)))$.
- $H_3^{\mathcal{A}}(1^\lambda, s_0, s_1, B, B', a)$:
 1. Let $\text{sh}^{(0)} \leftarrow \text{Share}(1^\lambda, s_0)$ and $\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)$.
 2. Output $\mathcal{A}(1^\lambda, s_0, s_1, B, B', a, \underbrace{(\text{Sim}(1^\lambda, \text{params}), \text{Enc}(1^\lambda, 0))}_{|B'| \text{ times}} \parallel (\text{Sim}(1^\lambda, \text{params}), \text{Enc}(1^\lambda, 0)))$.

We have that $H_0^{\mathcal{A}} \approx_c H_1^{\mathcal{A}}$ by the security of the MBCC obfuscation, which can be argued to apply by lower bounding the HILL entropy as done in the secrecy argument using the standard secret sharing scheme's secrecy and by observing that $\text{sh}_{B'}^{(0)}$ uses independent randomness from $\text{sh}_{\{a\}}^{(1)}$. We then have that $H_1^{\mathcal{A}} \approx_c H_2^{\mathcal{A}}$ by the semantic security of the secret key encryption scheme. Finally, we have that $H_2^{\mathcal{A}} \approx_c H_3^{\mathcal{A}}$ by the exact sequence of hybrids used in the argument made above for secrecy. $\square$

8 Robust Reconstruction

In the non-anonymous setting, shares are allowed to have a common tag that allow shares produced by a common sharing of a secret s identifying them to the reconstruction algorithm. This enables the use of the standard syntax of reconstruction for standard secret sharing in the multi-dealer setting, in which several secrets are shared. In the anonymous setting, when shares are produced by a M-FASS or U-FASS

scheme, such a common tag is not possible to include due to the anonymity requirements. Therefore, for the standard notion of reconstruction to be useful, the algorithm must somehow identify shares that come from a single sharing of a secret s among several sharings of several secrets, posing a technical challenge to simultaneously obtaining the desired anonymity. However, we recall that our notions of anonymity are limited to unauthorized sets, allowing us the possibility of obtaining a notion of “multi-dealer reconstruction” that is necessary for this type of secret sharing to be useful.

In a prior work, Eldridge et al. [EBG⁺24] already note the importance of this property and study this property in the context of the threshold access structure. They term this property “MD-Correctness” (see Section 3.1.2 of [EBG⁺24]) where the MD is an acronym for “Multiple Dealers”. In their paper, they consider the task of receiving a set of honestly generated shares from multiple dealers and recovering a list of all secrets for which an authorized set of shares exists in this set. We note that the shares corresponding to any single dealer in this set may form an authorized set or an unauthorized set. Their solution that recovers this list is technically innovative, however it is subject to several important shortcomings that we address in this work. First, their solution is based on error correcting codes and is thus subject to list-decoding bounds—therefore, their solution does not extend to an arbitrary polynomial number of dealers and shares and is in fact limited to a constant number of dealers. Moreover, their definition of MD-Correctness and their solution is critically limited to threshold access structures because it exploits error correction in the context of polynomial reconstruction. In this section, we show that all of our computational constructions—our computational U-FASS construction for the star graph access structure, our computational U-FASS construction for the bipartite graph access structure, and our computational M-FASS compiler that transforms a standard secret sharing schemes for arbitrary access structures into an M-FASS scheme for the same access structure—achieve a notion of robust reconstruction. That is, our previous computational schemes additionally support:

1. Multiple dealer reconstruction in the presence of an arbitrary polynomial number of dealers and shares, avoiding the list-decoding bounds the previous technique was subject to.
2. Multiple dealer reconstruction in the presence of an additional arbitrary polynomial amount of random insertion noise.

Definition 8.1 (Robust Reconstruction [Generalization of Definition 3.6 in [EBG⁺24]]). *A secret sharing scheme $(\text{Share}, \text{Recon})$ for an access structure Γ and secret space $\mathcal{S}$ has robust reconstruction if there exists an algorithm $\widehat{\text{Recon}}$ and there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$ such that for all polynomials $f : \mathbb{N} \rightarrow \mathbb{N}$, for all $\lambda \in \mathbb{N}$, for all secrets $s_1, \dots, s_{f(\lambda)}$ for sets $B_1, \dots, B_{f(\lambda)} \subseteq 2^{n(\lambda)}$ where $n : \mathbb{N} \rightarrow \mathbb{N}$ is the number of parties in Γ_λ , then over the coins of the sharing algorithm Share ,*

$$\Pr \left[\widehat{\text{Recon}}(\widehat{\text{sh}}) \neq \{s_i : i \text{ s.t. } B_i \in \Gamma_\lambda\} \mid \begin{array}{l} \forall i \in [f(\lambda)], \text{sh}^{(i)} \leftarrow \text{Share}(1^\lambda, s_i) \\ \widehat{\text{sh}} \leftarrow \bigcup_{i \in [f(\lambda)]} \text{sh}_{B_i}^{(i)} \end{array} \right] \leq \mu(\lambda)$$

where $\text{sh}_{B_i}^{(i)}$ is the (unordered) set of all share values corresponding to the set B_i on the sharing $\text{sh}^{(i)}$ of secret s_i .

We can augment Definition 8.1 to also include recovery in the presence of random noise by adding the additional red colored text below.

Definition 8.2 (RINT Robust Reconstruction). *We say that a secret sharing scheme $(\text{Share}, \text{Recon})$ for an access structure Γ and secret space $\mathcal{S}$ and share value space $\mathcal{V}$ has random insertion noise tolerant (RINT) robust reconstruction if there exists an algorithm $\widehat{\text{Recon}}$, such that there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$ such that for all polynomials $f, g : \mathbb{N} \rightarrow \mathbb{N}$, for all $\lambda \in \mathbb{N}$, for all secrets $s_1, \dots, s_{f(\lambda)}$ for sets*

$B_1, \dots, B_{f(\lambda)} \subseteq 2^{n(\lambda)}$ where $n : \mathbb{N} \rightarrow \mathbb{N}$ is the number of parties in Γ_λ , then over the coins of the sharing algorithm Share,

$$\Pr \left[\widehat{\text{Recon}}(\widehat{\text{sh}}) \neq \{s_i : i \text{ s.t. } B_i \in \Gamma_\lambda\} \mid \begin{array}{l} \forall i \in [f(\lambda)], \text{'sh}^{(i)} \leftarrow \text{Share}(1^\lambda, s_i) \\ \forall j \in [g(\lambda)] \text{ } r_j \leftarrow \text{Unif}(\mathcal{V}_\lambda) \\ \widehat{\text{sh}} \leftarrow \left(\bigcup_{i \in [f(\lambda)]} \text{sh}_{B_i}^{(i)} \right) \cup \{r_j\}_{j \in [g(\lambda)]} \end{array} \right] \leq \mu(\lambda)$$

where $\text{sh}_{B_i}^{(i)}$ is the (unordered) set of all share values corresponding to the set B_i on the sharing $\text{sh}^{(i)}$ of secret s_i .

Theorem 8.3. *Assuming the existence of one-way functions, there exists a computational U-FASS scheme that supports efficient RINT multiple dealer reconstruction (Def. 8.2) for sharing a 1-bit secret that satisfies $1 - \mu(\lambda)$ correctness, computational secrecy, and computational U-Anonymity, for the $(1, n(\lambda))$ -star graph access structure where $\mu : \mathbb{N} \rightarrow [0, 1]$ is a negligible function with a per-party share size of $O(\lambda)$.*

Proof. We prove that Construction 7.2 is additionally has an efficient multiple dealer reconstruction algorithm. Define $\widehat{\text{Recon}}(\widehat{\text{sh}})$ to be the algorithm that runs Recon from Construction 7.2 on all pairs of shares in $\widehat{\text{sh}}$, collects all non- $\perp$ values in a list L , and outputs the list L . This algorithm Recon iterates over all pairs of share values $(\text{sh}_1, \text{sh}_2) = ((k_1, r_1), (k_2, r_2))$ and computes $F(k_1, r_2) \oplus r_2$ for a fixed PRF $F : \{0, 1\}^\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$. If this string ends in $\lambda - 1$ many zeroes, the algorithm outputs the first bit, otherwise the algorithm outputs $\perp$.

First observe that whenever an authorized pair of shares under a single sharing of s_i for any $i \in [f(\lambda)]$ are paired, this computational step outputs s_i . This implies that $\{s_i : i \text{ s.t. } B_i \in \Gamma_\lambda\} \subseteq L$. Then observe that whenever a share from a sharing of s_i is paired with a sharing of s_j for $j \neq i$, or paired with a random share value (random noise), the value k_1 is uniformly distributed and the probability that r_2 is such that $F(k_1, r_2) \oplus r_2$ ends in $\lambda - 1$ zeroes is a negligible close to the probability that a truly random function $R : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ satisfies $R(r_2) \oplus r_2$ ends in $\lambda - 1$ zeroes, which is $2^{-\Omega(\lambda)}$, otherwise there exists a simple distinguisher between a truly random function and $F(k_1, \cdot)$ over the choice of a random key k_1 . Therefore, with all but negligible probability, a pairing of a share from a sharing of s_i with a share from a sharing of s_j OR with a uniform random share value from the domain $\{0, 1\}^{2\lambda}$ will result in $\perp$, so that $L \subseteq \{s_i : i \text{ s.t. } B_i \in \Gamma_\lambda\}$.

Finally, note that this algorithm $\widehat{\text{Recon}}$ is efficient as it iterates over a number of pairs of shares that is quadratic in the total number of shares and random noise given to it as input. $\square$

Theorem 8.4. *Assuming the hardness of DLIN (See Def. 7.7), there exists a U-FASS scheme that supports efficient RINT multi-dealer reconstruction (Def. 8.2) for sharing a 1-bit secret with $(1 - \mu)$ -correctness for a negligible function μ , computational secrecy, and computational U-Anonymity for the $((1, n), (1, n))$ -bipartite graph access structure with a per-party share size of $O(\lambda)$.*

Proof. We prove that Construction 7.5 supports efficient RINT multi-dealer reconstruction. Define $\widehat{\text{Recon}}(\widehat{\text{sh}})$ to be the algorithm that runs Recon from Construction 7.5 on all pairs of shares in $\widehat{\text{sh}}$, collects all non- $\perp$ values in a list L , and outputs the list L .

First observe that whenever an authorized pair of shares under a single sharing of s_i for any $i \in [f(\lambda)]$ are paired under the bilinear pairing e , we obtain the secret the s_i . This implies that $\{s_i : i \text{ s.t. } B_i \in \Gamma_\lambda\} \subseteq L$. Then observe that whenever a share from a sharing of s_i is paired either with a sharing of s_j for $j \neq i$ or with a random share value (random noise), then observe that in either case—any share value generated for the secret s_j or a uniform random share value—the share value for these two cases are distributed independently from the share values generated in the sharing of secret s_i . The probability—in either case—that the resulting value under the pairing e in the target group has an exponent within the bound polynomial bound p of the secret space within $\mathbb{Z}_q^*$ in Construction 7.5 is negligible.

Finally, observe that this algorithm $\widehat{\text{Recon}}$ is efficient as it iterates over a number of pairs of shares that is quadratic in the total number of shares and random noise given to it as input. $\square$

Theorem 8.5. *For any access structure $\Gamma = \{\Gamma_\lambda\}$ where Γ_λ involves $n = n(\lambda)$ parties, if there exists a (information-theoretically or computationally secure) standard secret sharing scheme $\Pi = (\text{Share}, \text{Recon})$ (see Def. 3.2), where Recon can be represented by a TM that runs in time bounded by $T : \mathbb{N} \rightarrow \mathbb{N}$ and where the per-party share size is bounded by S bits, then assuming the existence of the following primitives:*

1. Secret key encryption scheme (Gen, Enc, Dec).
2. Multi-bit compute-and-compare obfuscation for any distribution $\mathcal{P}_{\text{MBCC}}^{\text{TM}}$ that satisfies α -pseudo-entropy for any polynomial $\alpha = \alpha(\lambda)$.
3. Any injective PRG (Gen, $\mathcal{G}$) in NC^1 with polynomial stretch $m = m(\lambda) \geq \alpha(\lambda)$.

there exists a computationally secure M-FASS scheme that supports efficient RINT multi-dealer reconstruction (Def. 8.2) for Γ with per-party share size $S \cdot \text{poly}(\lambda, \log T)$ bits.

Proof. We prove that a slight modification of Construction 7.17 additionally supports efficient RINT multi-dealer reconstruction. We first modify the Share algorithm so that instead of an arbitrary SKE scheme, we use an SKE scheme that additionally supports the following property: Attempting to decrypt a ciphertext under a uniform random key results in $\perp$ with overwhelming probability. Such an SKE scheme exists assuming the existence of OWF and can be constructed exactly using the PRF construction used in Construction 7.2 where decrypted strings that do not end in some sufficient number of zeroes are stipulated as $\perp$ values. This verifiable bad-key SKE scheme otherwise satisfies the usual properties of any SKE scheme. Then, we modify the function $f_{\text{sk},i}(\langle F \rangle, x)$ in the Share algorithm of Construction 7.17 so that Step 2 of this function ignores all $\perp$ values resulting in decrypting the ciphertext, and Step 3 passes only non- $\perp$ values into the circuit F . This concludes all the modification we make for the Share algorithm. Observe that after these changes, the existing proofs of correctness, secrecy, and M-FASS still hold.

To show that this modified construction additionally supports efficient (non-RINT) multi-dealer reconstruction (Def. 8.1). We define $\widehat{\text{Recon}}(\widehat{\text{sh}})$ to be an algorithm that parses all shares as a pair $(\text{ct}, \hat{C})$ where $\hat{C}$ is a fixed length description of an obfuscated version of the circuit $f_{\text{sk},j}(\langle F \rangle, x)$ for some verifiable bad-key SKE scheme secret key sk and some index j . Then for each share, $(\text{ct}, \hat{C}) \in \widehat{\text{sh}}$, the algorithm runs $\hat{C}$ on all the set of all the $\{\text{ct}\}$. Since each sk value is uniform randomly sampled, the decryption step, Step 2, of any obfuscated circuit $\hat{C}$ in a share generated by a sharing of secret s_i for $i \in [f(\lambda)]$, where f is our polynomial bound on the number of dealers in Def. 8.2, will result in a $\perp$ value for all ct values in a share generated by the sharing of a secret s_j for $j \neq i$. Therefore, Step 2 of this algorithm identifies all the shares corresponding to a sharing of s_i . Then, Step 3 of the $\hat{C}$ applies the standard secret sharing reconstruction algorithm on the undecrypted shares corresponding to a sharing of s_i , returning s_i when this set is authorized. In other words, with all but negligible probability, at the end of attempting to run $|\widehat{\text{sh}}|$ many obfuscations, $\widehat{\text{Recon}}$ will output a list of all secrets s_i for which the set B_i of present shares in $\widehat{\text{sh}}$ is an authorized set.

To extend this modified construction to additionally support efficient RINT multi-dealer reconstruction (Def. 8.2), we need to consider the case in which certain shares in $\widehat{\text{sh}}$ are of the form (r_1, r_2) for two uniform random strings. Here r_2 may be an accidental obfuscated circuit description, whose behavior is random. To protect our reconstruction process against such strings, we can simply augment the intended secret space to be a sparse subset of a larger dummy secret domain for the standard secret sharing scheme so that the probability a random string constitutes a valid secret is exponentially small. For example, if our original secret space is $\{0, 1\}$, we can embed this space into a dummy secret domain to be in $\{0, 1\}^\lambda$ where the only two “valid” secrets are two randomly chosen strings $x_1 \in \{0, 1\}^\lambda$ and $x_2 \in \{0, 1\}^\lambda$ that are fixed

for all dealers. Then, our generated fresh obfuscations of the form $\text{MBCC}[f_{sk,j}, s', y_j]$ for $s' \in \{x_1, x_2\}$. When we attempt to interpret a random string r_2 as an obfuscated circuit to run on all the ct values of the shares, there is a negligible probability that this random circuit will hit an exponentially small random subset $\{x_1, x_2\}$ and incorrectly recover a secret value. $\square$

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A Strengthening of M-Anonymity

Definition A.1 (Strong M-Anonymity). *A secret sharing scheme (Share, Recon) for access structure Γ , and secret space $\mathcal{S}$ is δ -strong M-Anonymous if for every computationally unbounded (in the computational setting, non-uniform polynomial-time) algorithm $\mathcal{A}$ such that for all sufficiently large $\lambda \in \mathbb{N}$, for all $B, B' \in 2^{[n]} \setminus \Gamma_\lambda$ such that $|B| = |B'|$ for all partitions $(P_1, \dots, P_t)$ of B' , for all t -sequences of secrets $(s_1, s_2, \dots, s_t) \in \mathcal{S}_\lambda^t$,*

$$\left| \Pr_{\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)} \left[\mathcal{A} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t), \text{sh}_B^{(1)} \right) = 1 \right] - \Pr_{\forall i \in [t], \text{sh}^{(i)} \leftarrow \text{Share}(1^\lambda, s_i)} \left[\mathcal{A} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t), \left(\text{sh}_{P_i}^{(i)} \right)_{i \in [t]} \right) = 1 \right] \right| \leq \delta(\lambda).$$

Remark A.1. Statistical (resp. computational) Strong M-Anonymity, without a specified δ parameter, is defined so that for every unbounded adversary (resp. non-uniform polynomial-time) $\mathcal{A}$ there exists a negligible function $\delta : \mathbb{N} \rightarrow [0, 1]$ such that the above statement holds (i.e. for all sufficiently large $\lambda \in \mathbb{N} \dots$).

Remark A.2. It may be surprising at first glance that M-Anonymity (Def. 4.2) implies strong M-Anonymity. For example, consider the DNF access structure $\Phi(x_1, \dots, x_4)$ that consists of all minterms except for the minterms $\{x_1, x_2, x_3, x_3, x_1x_2, x_3x_4\}$ and observe that there are only two unauthorized sets, namely $\{1, 2\}$ and $\{3, 4\}$, which are disjoint. It doesn't seem that M-Anonymity should imply any notion of anonymity for the shares of these two sets because they are disjoint, nor does it immediately suggest a hybrid sequence from the distribution of the shares for parties in $\{1, 2\}$ to the distribution of the shares for parties in $\{3, 4\}$. The proof that M-Anonymity implies strong M-Anonymity will circumvent this obstacle by a sequence of hybrids that involve fresh sharings of fresh random secrets.

Lemma A.2 (M-Anonymity Implies Strong M-Anonymity). *Any secret sharing scheme (Share, Recon) that satisfies statistical (respectively computational) M-Anonymity (Def. 4.2) also satisfies statistical (resp. computational) Strong M-Anonymity (Def. A.1).*

Proof. Let $B, B' \in 2^{[n]} \setminus \Gamma_\lambda$ and let $\ell = |B| = |B'|$. Consider any partition of B' ,

$$(P_1 = \{P_{1,j}\}_{j \in [i_1]}, \dots, P_t = \{P_{t,j}\}_{j \in [i_t]})$$

where $\sum_{k=1}^t i_k = \ell$ and so that

$$(B'_1, \dots, B'_\ell) := (P_{1,1}, \dots, P_{1,i_1}, \dots, P_{t,1}, \dots, P_{t,i_t})$$

is an enumeration of B' and let $\Pi(B'_i)$ denote the index i^* such that $B'_i \in P_{i^*}$. Let $(s_1, \dots, s_t)$ be any t -sequence of secrets.

We define $2\ell + 1$ experiments $H_0^\mathcal{A}, \dots, H_{2\ell}^\mathcal{A}$. Our starting point is the experiment $H_0^\mathcal{A}$ that gives the adversary $\mathcal{A}$ the shares for set B from a single sharing of s_1 . Each experiment $H_i^\mathcal{A}$ for $i \in [\ell]$ is a modification of experiment $H_{i-1}^\mathcal{A}$ where we replace the share for party B_i with a fresh sharing of an independent uniformly sampled secret for the party B'_i for $i \in [\ell]$. In experiment $H_\ell^\mathcal{A}$, the adversary is given ℓ -many shares for unauthorized set of parties B' , where each share is a fresh sharing of an independently uniformly sampled secret. Then, in each experiment $H_{\ell+i}^\mathcal{A}$ for $i \in [\ell]$, we replace the share of party B'_i with a sharing of $s_{\Pi(B'_i)}$ for the party B'_i . Finally $H_{2\ell}^\mathcal{A}$ is exactly the experiment that corresponds to a mixture of shares of $(s_1, \dots, s_t)$ for partition $(P_1, \dots, P_t)$.

Throughout the following experiments, for notational convenience, let $(B_1, \dots, B_\ell)$ be a fixed but arbitrary enumeration of the set B implicitly given in the description of B as input.

$H_0^{\mathcal{A}}(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t))$:

1. Let $\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)$.
2. Output $\mathcal{A}((s_1, \dots, s_t), B, (P_1, \dots, P_t), \text{sh}_B^{(1)})$.

For $i \in [\ell]$, $H_i^{\mathcal{A}}(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t))$:

1. Let $\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)$.
2. For $j \in [i]$, sample a uniform random secret $r_j \leftarrow S_\lambda$ and compute $\widetilde{\text{sh}}^{(j)} \leftarrow \text{Share}(1^\lambda, r_j)$.
3. Let $\text{sh}_{\text{mix}} = (\widetilde{\text{sh}}_{B'_1}^{(1)}, \dots, \widetilde{\text{sh}}_{B'_i}^{(i)}, \text{sh}_{B_{i+1}}^{(1)}, \dots, \text{sh}_{B_\ell}^{(1)})$.
4. Output $\mathcal{A}((s_1, \dots, s_t), B, (P_1, \dots, P_t), \text{sh}_{\text{mix}})$.

For $i \in [\ell]$, $H_{\ell+i}^{\mathcal{A}}(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t))$:

1. For $k \in [t]$, sample $\text{sh}^{(k)} \leftarrow \text{Share}(1^\lambda, s_k)$.
2. For $j \in \{i+1, \dots, \ell\}$, sample a uniform random secret $r_j \leftarrow S_\lambda$ and compute $\widetilde{\text{sh}}^{(j)} \leftarrow \text{Share}(1^\lambda, r_j)$.
3. Let $\text{sh}_{\text{mix}} = (\text{sh}_{B'_1}^{(\Pi(B'_1))}, \dots, \text{sh}_{B'_i}^{(\Pi(B'_i))}, \widetilde{\text{sh}}_{B'_{i+1}}^{(i+1)}, \dots, \widetilde{\text{sh}}_{B'_\ell}^{(1)})$.
4. Output $\mathcal{A}((s_1, \dots, s_t), B, (P_1, \dots, P_t), \text{sh}_{\text{mix}})$.

Lemma A.3. *If (Share, Recon) satisfies statistical (respectively computational) M-Anonymity, then for all $B, B' \in 2^{[n]} \setminus \Gamma_\lambda$ such that $\ell := |B| = |B'|$, for all partitions $(P_1, \dots, P_t)$ of B' , for all $(s_1, \dots, s_t) \in S_\lambda^t$, for all $i \in [2\ell]$, for all unbounded algorithms (resp. non-uniform polynomial-time algorithms) $\mathcal{A}$, there exists a negligible function $\mu : \mathbb{N} \rightarrow [0, 1]$ such that for all sufficiently large $\lambda \in \mathbb{N}$,*

$$\left| \Pr \left[H_{i-1}^{\mathcal{A}} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t) \right) = 1 \right] - \Pr \left[H_i^{\mathcal{A}} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t) \right) = 1 \right] \right| \leq \mu(\lambda)$$

Proof. Throughout the proof, fix an arbitrary enumerations of $B = (B_1, \dots, B_\ell)$. For $i \in [\ell]$, suppose for sake of contradiction that there exists a computationally unbounded (resp. non-uniform polynomial-time) algorithm $\mathcal{A}$ and a polynomial $p : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\left| \Pr \left[H_{i-1}^{\mathcal{A}} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t) \right) = 1 \right] - \Pr \left[H_i^{\mathcal{A}} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t) \right) = 1 \right] \right| \geq \frac{1}{p(\lambda)}$$

for infinitely many values of $\lambda \in \mathbb{N}$. Then, by a standard averaging argument¹⁷, there exists a secret r^* for which we can construct a computationally unbounded (resp. non-uniform polynomial-time) algorithm $\mathcal{B}$ that contradicts the statistical (resp. computational) M-Anonymity property with respect to $s_1, r^*, \widehat{B}, \widehat{B}'$ where s_1 is as given in the Lemma statement and where $\widehat{B} = \{B_\ell, \dots, B_{i+1}, B_i\}$, $\widehat{B}' = \{B_\ell, \dots, B_{i+1}\} \subset \widehat{B}$, and $a = B' \setminus \{i\}$. Observe that $\widehat{B}$ is unauthorized because it is a subset of an unauthorized set.

We now construct the reduction $\mathcal{B}_i$ that receives as input $(s_1, r^*, \widehat{B}, \widehat{B}', a, X)$ where $X = (X_1, \dots, X_{\ell-1}, X^*)$ is either the shares $\text{sh}_{\widehat{B}}^{(1)}$ or $\text{sh}_{\widehat{B}'}^{(1)} \parallel \text{sh}_a^{(\star)}$, where $\text{sh}^{(1)} \leftarrow \text{Share}(1^\lambda, s_1)$ and $\text{sh}^{(\star)} \leftarrow \text{Share}(1^\lambda, r^*)$.

¹⁷To slightly expound upon this averaging argument, in $H_{i-1}^{\mathcal{A}}$ the i th share in sh_{mix} is a sharing of a random secret r_i and in $H_i^{\mathcal{A}}$ the i th share is always a sharing of the fixed secret s_1 given by the Lemma statement. Since $\mathcal{A}$'s output is distinguishable between $H_{i-1}^{\mathcal{A}}$ and $H_i^{\mathcal{A}}$ with at least probability $1/p(\lambda)$, there must exist a fixed choice of some of the coins for $H_{i-1}^{\mathcal{A}}$ corresponding to a fixed choice of r_i , for which this non-negligible advantage holds. Set $r^* \leftarrow r_i$.

1. $\mathcal{B}_i$ has $(s_2, \dots, s_t), B, (P_1, \dots, P_t)$ hardcoded.
2. Sample uniform random secret values $r_1, r_2, \dots, r_{i-1} \leftarrow \mathcal{S}$ and compute $\widetilde{\text{sh}}^{(j)} \leftarrow \text{Share}(1^\lambda, r_j)$.
3. Let $\text{sh}_{\text{mix}} = (\widetilde{\text{sh}}_{B'_1}^{(1)}, \dots, \widetilde{\text{sh}}_{B'_{i-1}}^{(i-1)}, X^*, X_{\ell-1}, \dots, X_1)$
4. Output $\mathcal{A}((s_1, \dots, s_t), B, (P_1, \dots, P_t), \text{sh}_{\text{mix}})$.

Then observe that the input of $\mathcal{A}$ in this reduction is exactly that of either $H_{i-1}^{\mathcal{A}}$ or $H_i^{\mathcal{A}}$ with a fixed choice of r^* that was chosen such that the two cases of X^* are distinguishable by $\mathcal{A}$ with probability at least $\frac{1}{p(\lambda)}$. Then the existence of $\mathcal{B}_i$ contradicts the statistical (resp. computational) M-Anonymity property, so no such $\mathcal{A}$ can exist.

For the case of $i \in \{\ell+1, \dots, 2\ell\}$ a slightly modification to the reduction must occur due to the multiple secret share dealings. At a high-level, we can use the exact same reduction idea above to switch each partition separately, though the formal notation is a bit tedious. For any $i \in \{\ell+1, \dots, 2\ell\}$, let $i^* = i - \ell$ and consider the index of the partition element $P_{j_i^*}$ that party B_{i^*} is in, namely $j_i^* = \Pi(B_{i^*}) \in [t]$. For notational convenience, we have chosen an enumeration of the unauthorized set B' such that all parties in the same partition are in consecutive positions. Therefore, let $\{i_L, \dots, i_R\}$ all the parties in in partition j_i^* so that $i_L \leq i^* \leq i_R$.

Suppose for sake of contradiction that there exists a computationally unbounded (resp. non-uniform polynomial-time) algorithm $\mathcal{A}$ and a polynomial $p : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\left| \Pr \left[H_{i-1}^{\mathcal{A}} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t) \right) = 1 \right] - \Pr \left[H_i^{\mathcal{A}} \left(1^\lambda, (s_1, \dots, s_t), B, (P_1, \dots, P_t) \right) = 1 \right] \right| \geq \frac{1}{p(\lambda)}$$

for infinitely many values of $\lambda \in \mathbb{N}$. Then, by the same standard averaging argument, there exists a secret r^* for which we can construct a computationally unbounded (resp. non-uniform polynomial-time) algorithm $\mathcal{B}$ that contradicts the statistical (resp. computational) M-Anonymity property with respect to $s_{j_i^*}, r^*, \widehat{B}, \widehat{B}'$ where $s_{j_i^*}$ is as given in the Lemma statement and where $\widehat{B} = \{B'_{i_L}, \dots, B'_{i^*}\}, \widehat{B}' = \{B'_{i_L}, \dots, B'_{i^*-1}\} \subset \widehat{B}$, and $a = B'_{i^*}$. Observe that $\widehat{B}$ is unauthorized because it is a subset of an unauthorized set.

We now construct the reduction $\mathcal{B}_i$ that receives as input $(s_1, r^*, \widehat{B}, \widehat{B}', a, X)$ where $X = (X_{i_L}, \dots, X_{i-1}, X^*)$ is either the shares $\text{sh}_{\widehat{B}}^{(j_i^*)}$ or $\text{sh}_{\widehat{B}'}^{(j_i^*)} \parallel \text{sh}_a^{(*)}$, where $\text{sh}^{(j_i^*)} \leftarrow \text{Share}(1^\lambda, s_{j_i^*})$ and $\text{sh}^{(*)} \leftarrow \text{Share}(1^\lambda, r^*)$.

1. $\mathcal{B}$ has $(s_1, s_2, \dots, s_t), B, (P_1, \dots, P_t)$ hardcoded.
2. For all $1 \leq j \leq j_i^*$, compute $\text{sh}^{(j)} \leftarrow \text{Share}(1^\lambda, s_j)$.
3. Sample uniform random secret values $r_{i^*+1}, r_{i^*+2}, \dots, r_\ell \leftarrow \mathcal{S}$ and compute $\widetilde{\text{sh}}^{(j)} \leftarrow \text{Share}(1^\lambda, r_j)$ for $j \in \{i^*+1, \dots, \ell\}$.
4. Let $\text{sh}_{\text{mix}} = \left(\widetilde{\text{sh}}_{B'_1}^{(\Pi(B'_1))}, \dots, \widetilde{\text{sh}}_{B'_{i_L-1}}^{(\Pi(B'_{i_L-1}))}, X_{i_L}, X_{i_L+1}, \dots, X_{i^*-1}, X^*, \widetilde{\text{sh}}_{B'_{i^*+1}}^{(i^*+1)}, \dots, \widetilde{\text{sh}}_{B'_\ell}^{(\ell)} \right)$
5. Output $\mathcal{A}((s_1, \dots, s_t), B, (P_1, \dots, P_t), \text{sh}_{\text{mix}})$.

Then observe that the input of $\mathcal{A}$ in this reduction is exactly that of either $H_{i-1}^{\mathcal{A}}$ or $H_i^{\mathcal{A}}$ with a fixed choice of r^* that was chosen such that the two cases of X^* are distinguishable by $\mathcal{A}$ with probability at least $\frac{1}{p(\lambda)}$. Then the existence of $\mathcal{B}_i$ contradicts the statistical (resp. computational) M-Anonymity property, so no such $\mathcal{A}$ can exist. $\square$

Lemma A.3 implies the desired Theorem statement $\square$